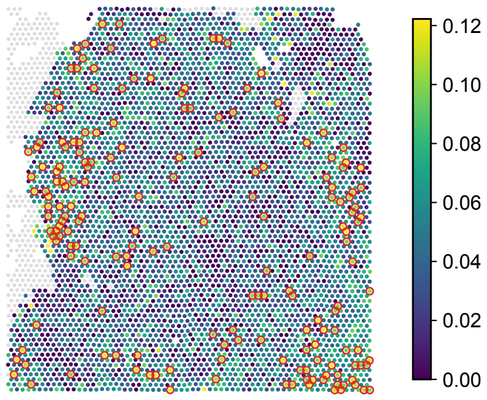
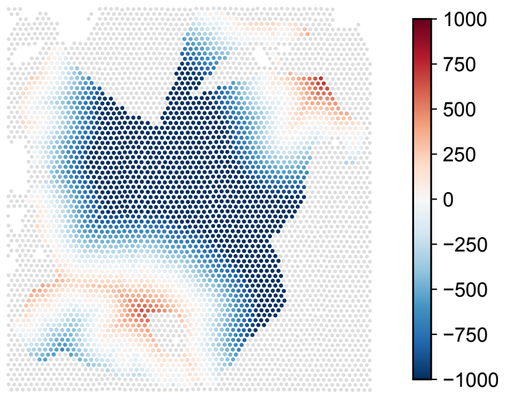
CanDis_2026_Sample1 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

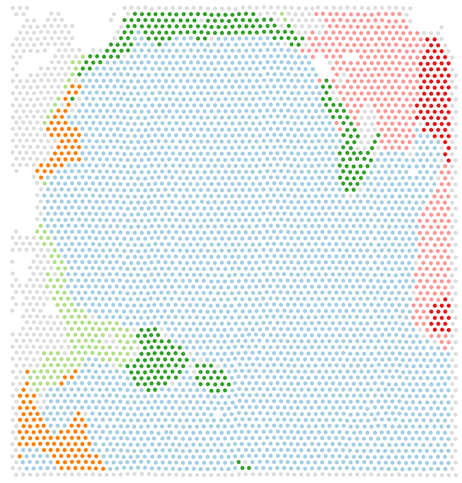
MP10 AUCell and frozen P95



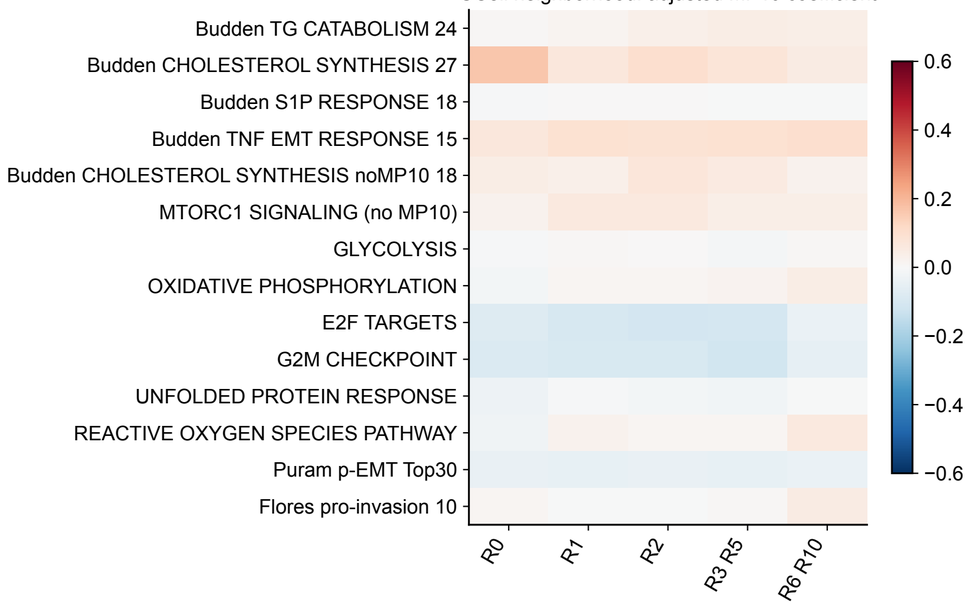
Strict v3.4 signed distance (µm)



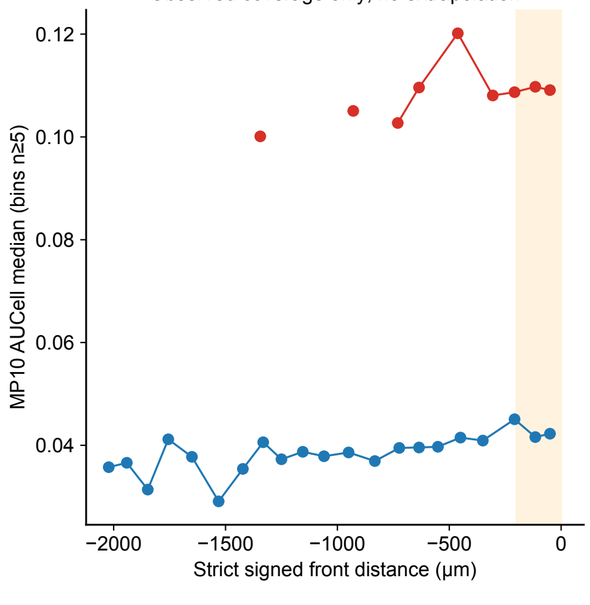
Formal R12 0–8; grey = unlabelled



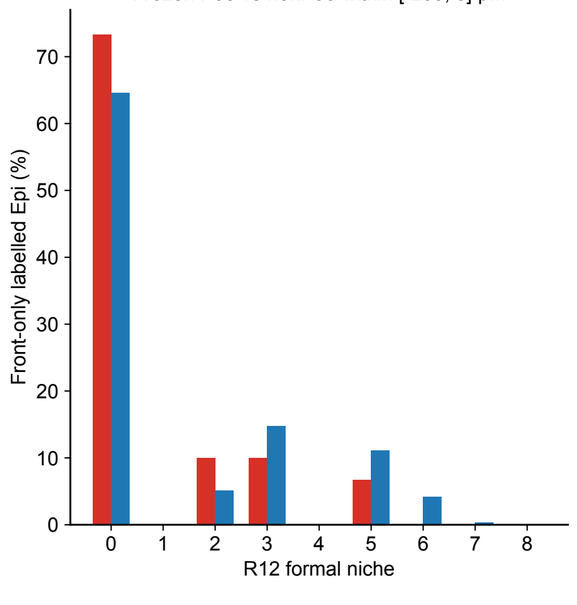
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

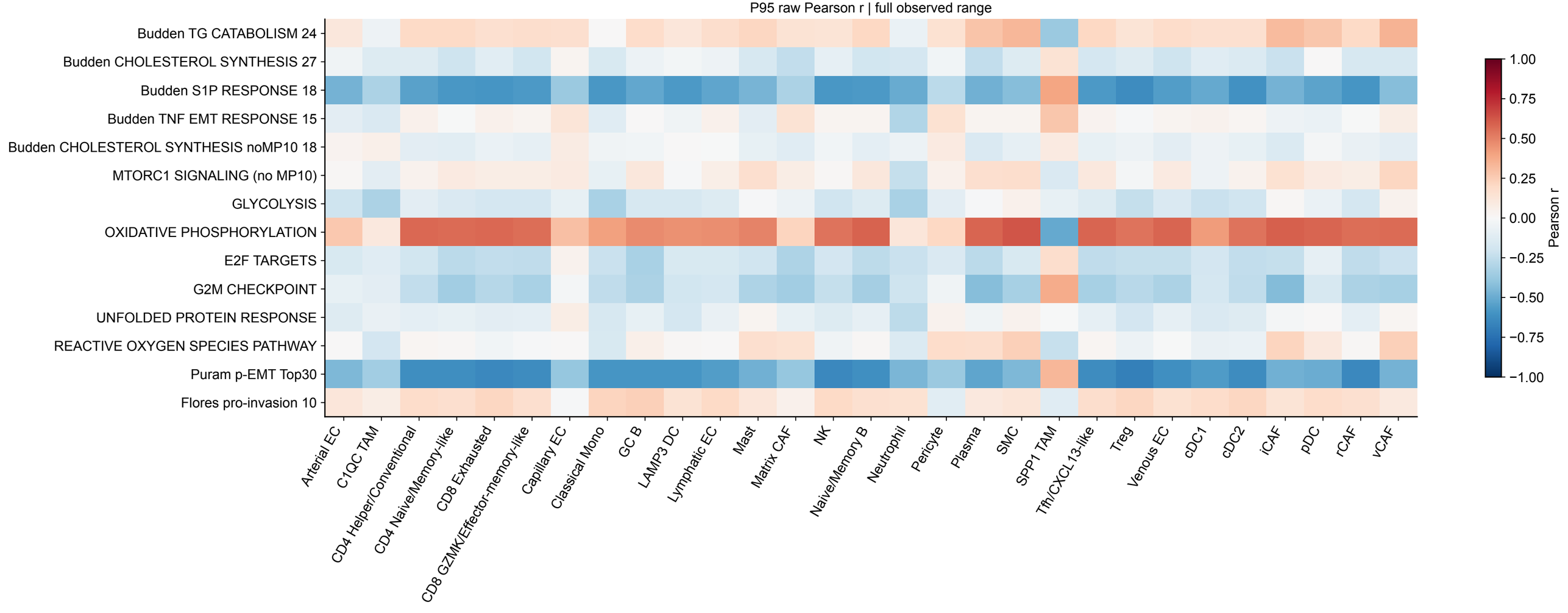


Tissue 4823 | Epi 4486 | internal 3636 | P95 183
Exact front in Epi 2759 | R12 labelled 4227

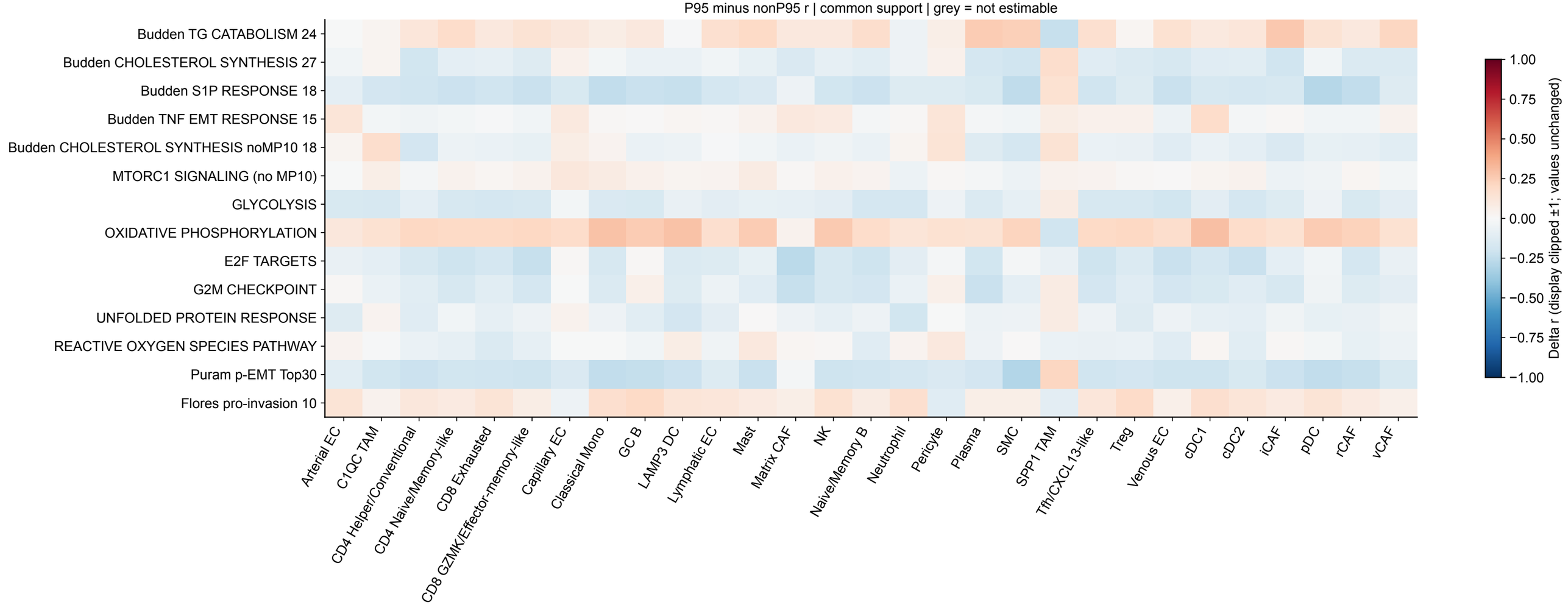
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

CanDis_2026_Sample1 | all 29 candidates; fixed core programs

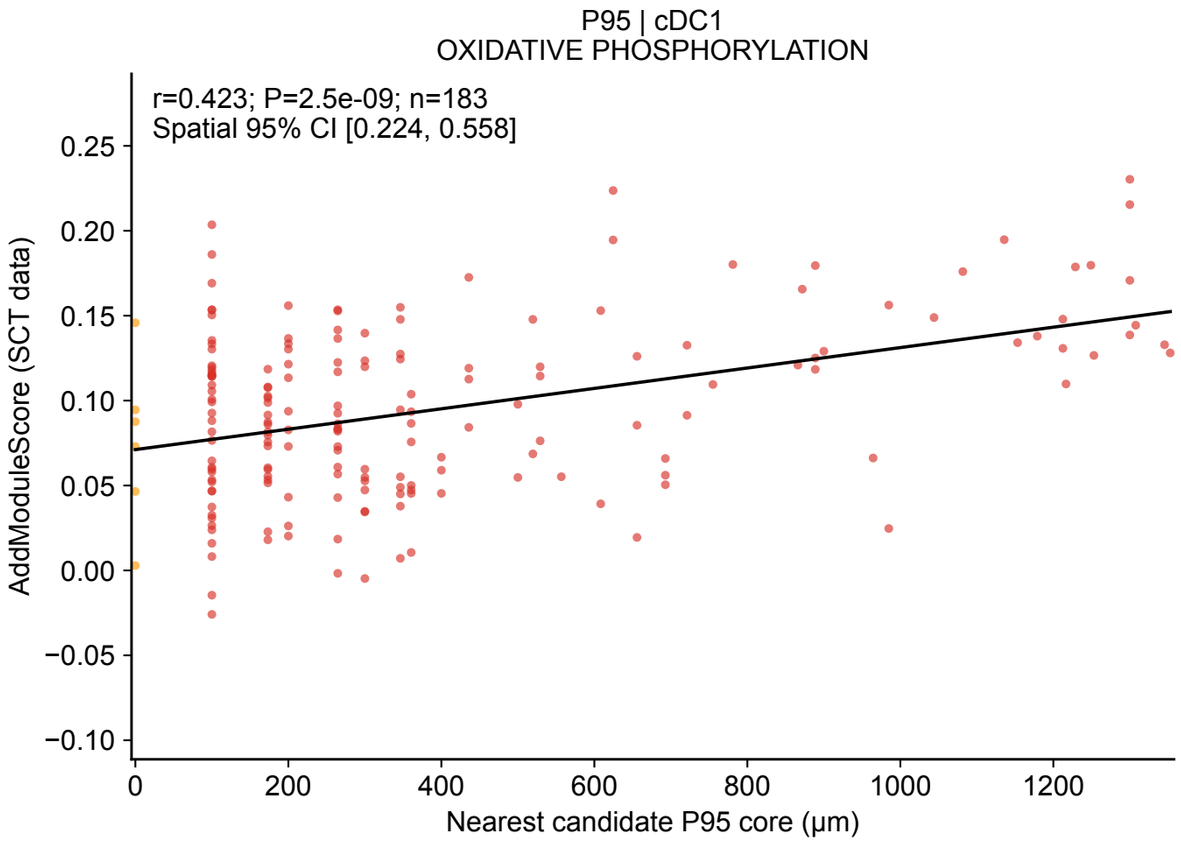
P95 raw Pearson r | full observed range

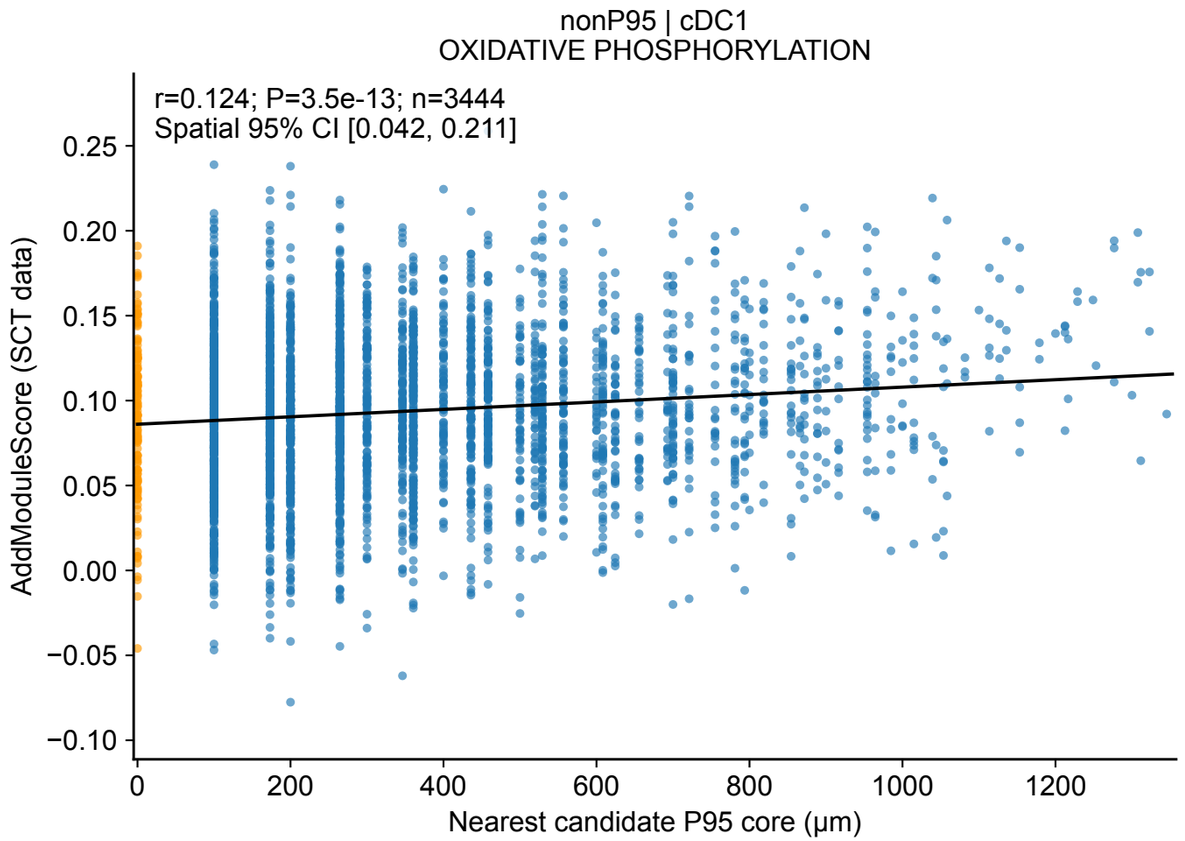


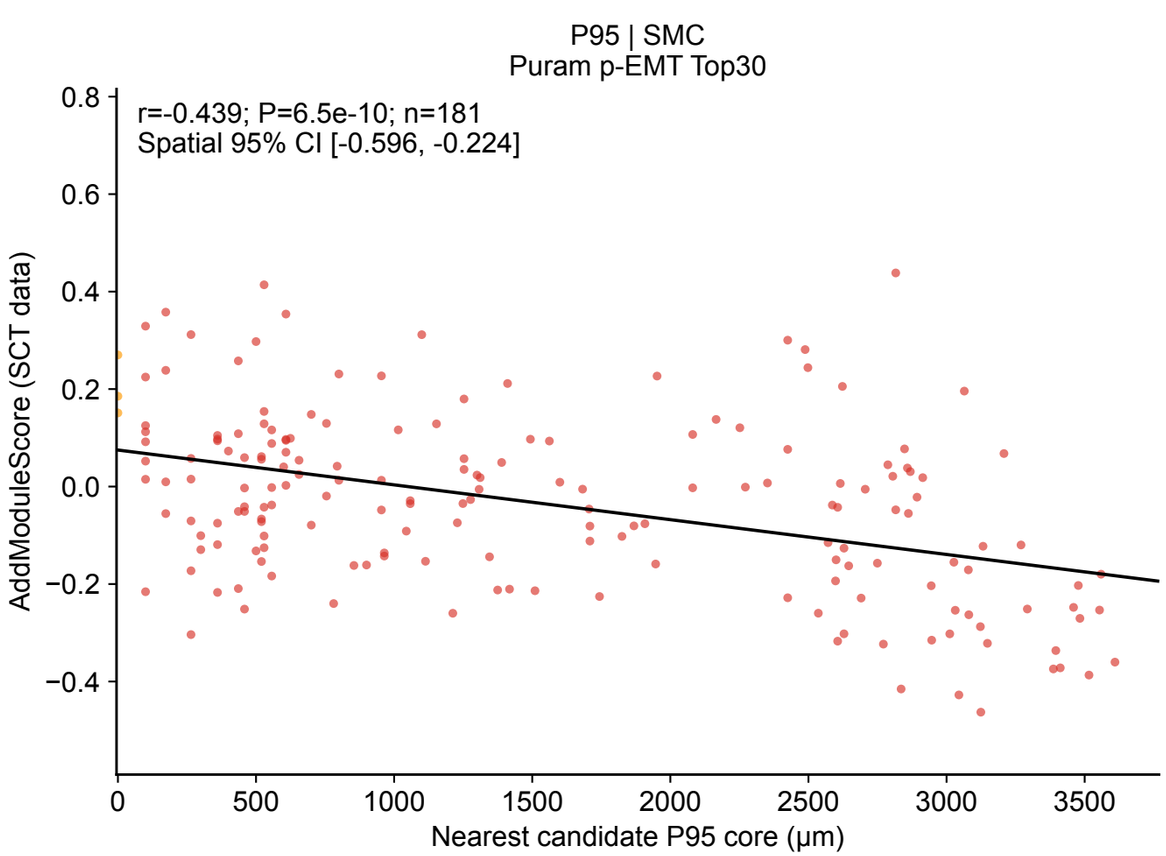
P95 minus nonP95 r | common support | grey = not estimable

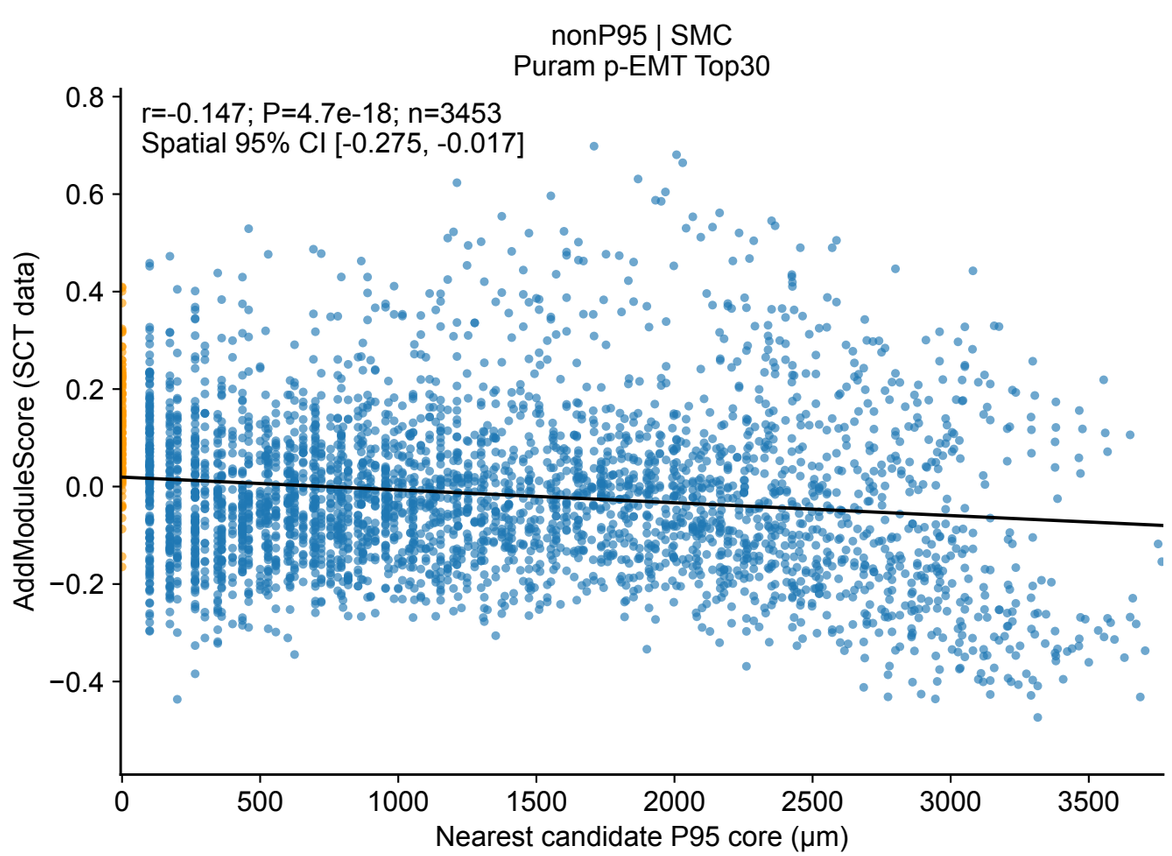


CanDis_2026_Sample1 | exploratory scatter display selected by coverage and |delta r|, not P







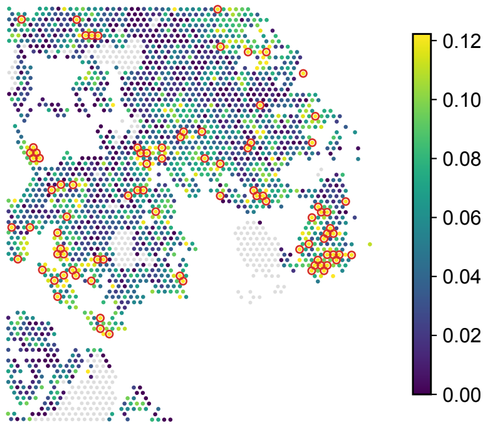


cDC1 / OXIDATIVE PHOSPHORYLATION: delta r=0.299 [0.146, 0.410]; common 0–1353 µm; spatial P=4.6e-05
SMC / Puram p-EMT Top30: delta r=-0.292 [-0.433, -0.119]; common 0–3764 µm; spatial P=0.00072
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

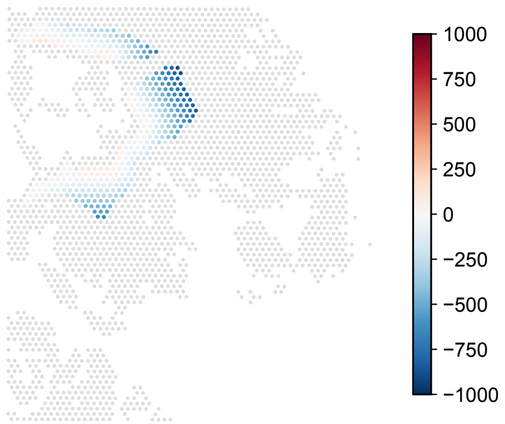
P95 nonP95 distance 0

CanDis_2026_Sample10 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

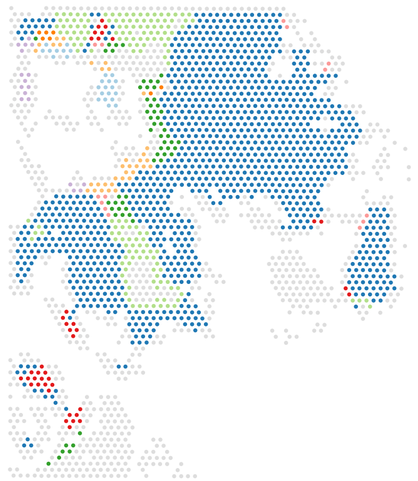
MP10 AUCell and frozen P95



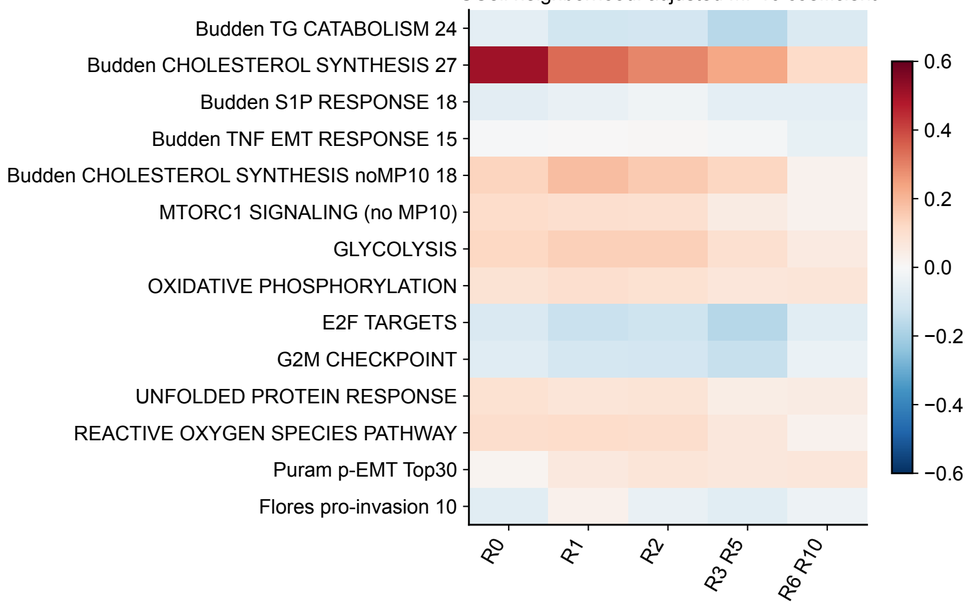
Strict v3.4 signed distance (µm)



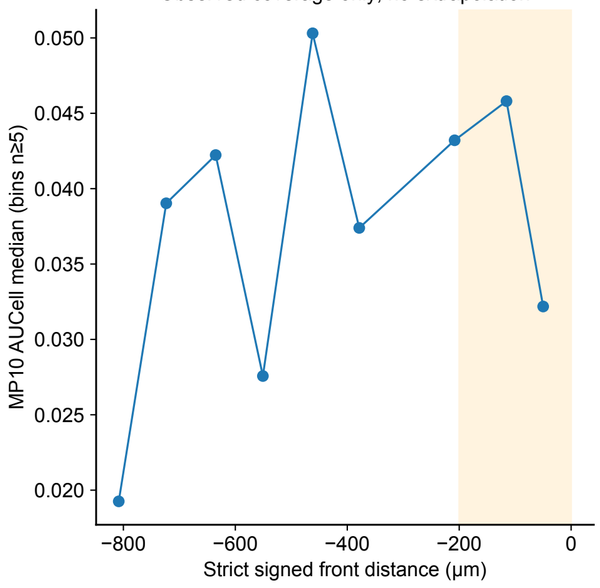
Formal R12 0–8; grey = unlabelled



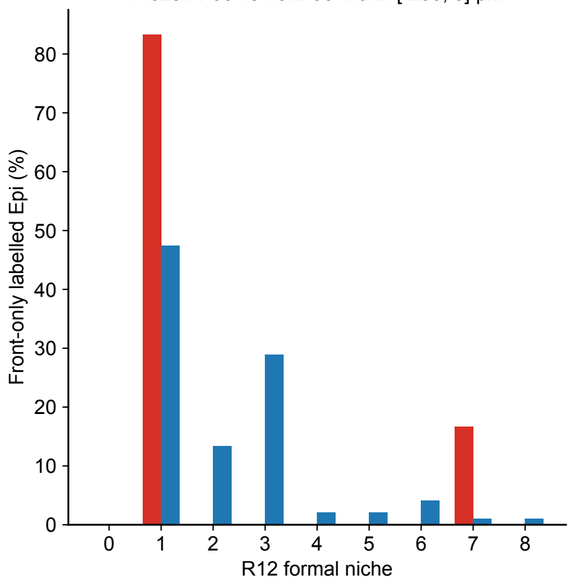
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



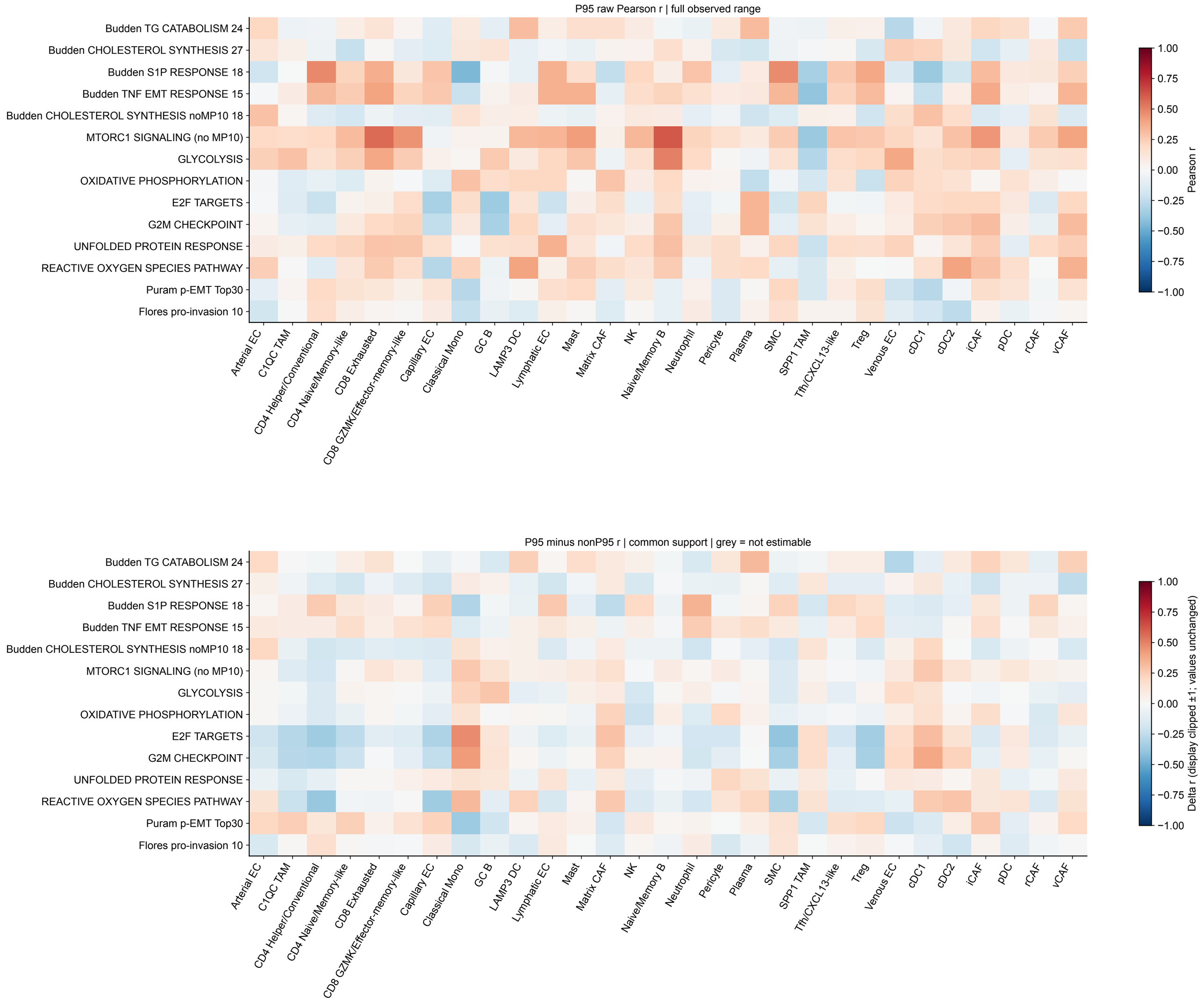
Frozen P95 vs nonP95 within [-200, 0] µm



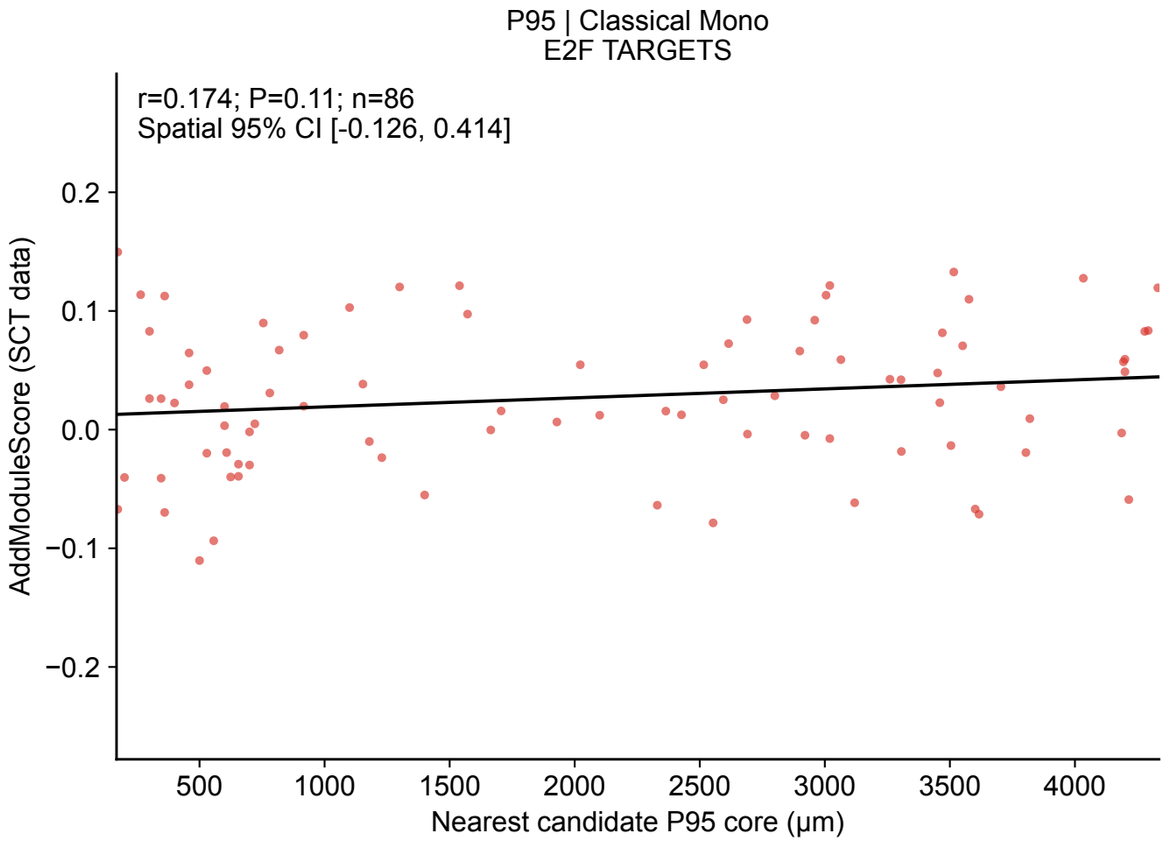
Tissue 2360 | Epi 2111 | internal 1726 | P95 87
Exact front in Epi 321 | R12 labelled 1500

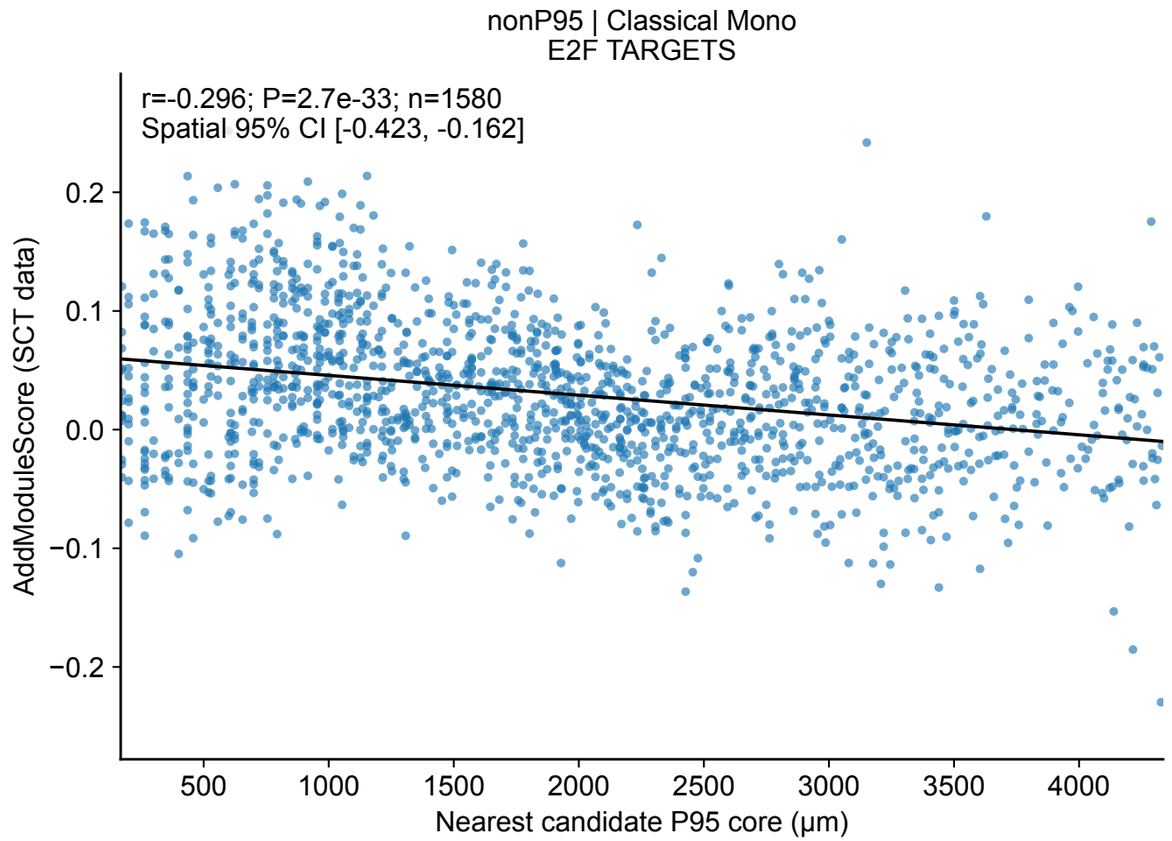
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

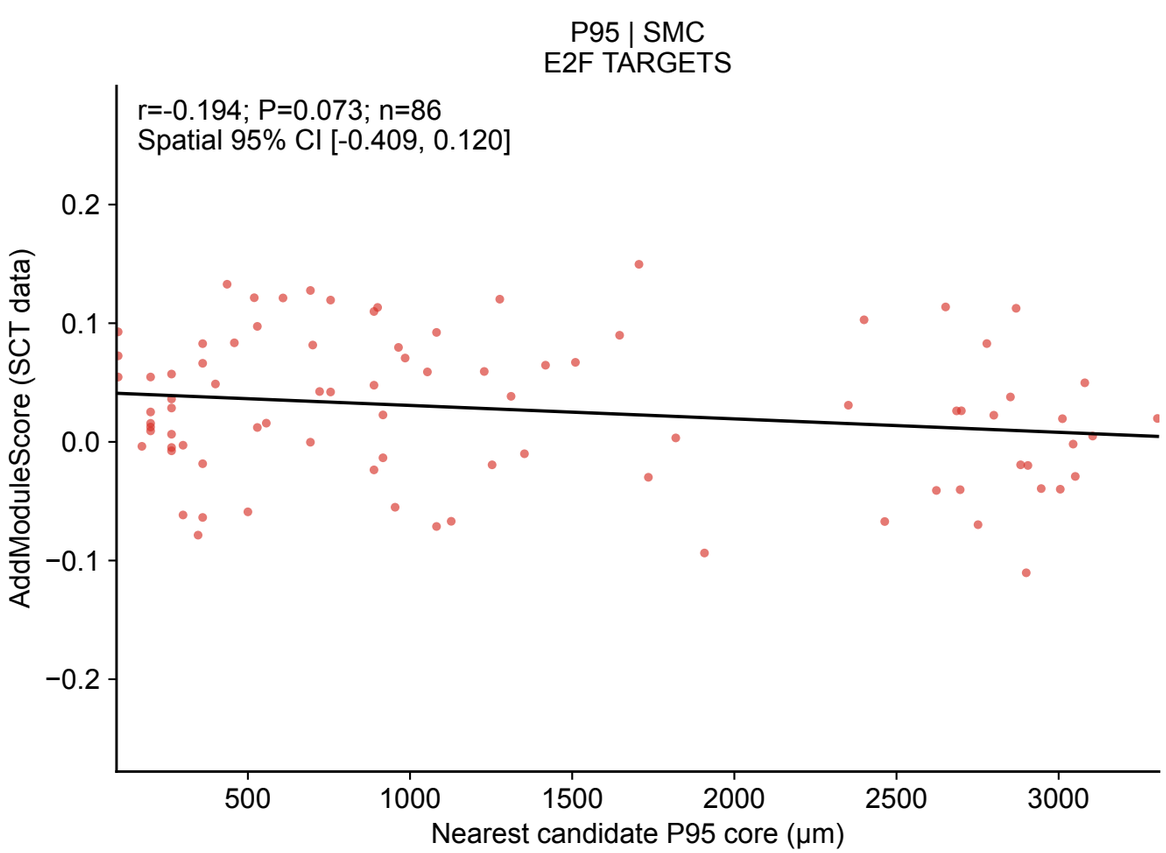
CanDis_2026_Sample10 | all 29 candidates; fixed core programs

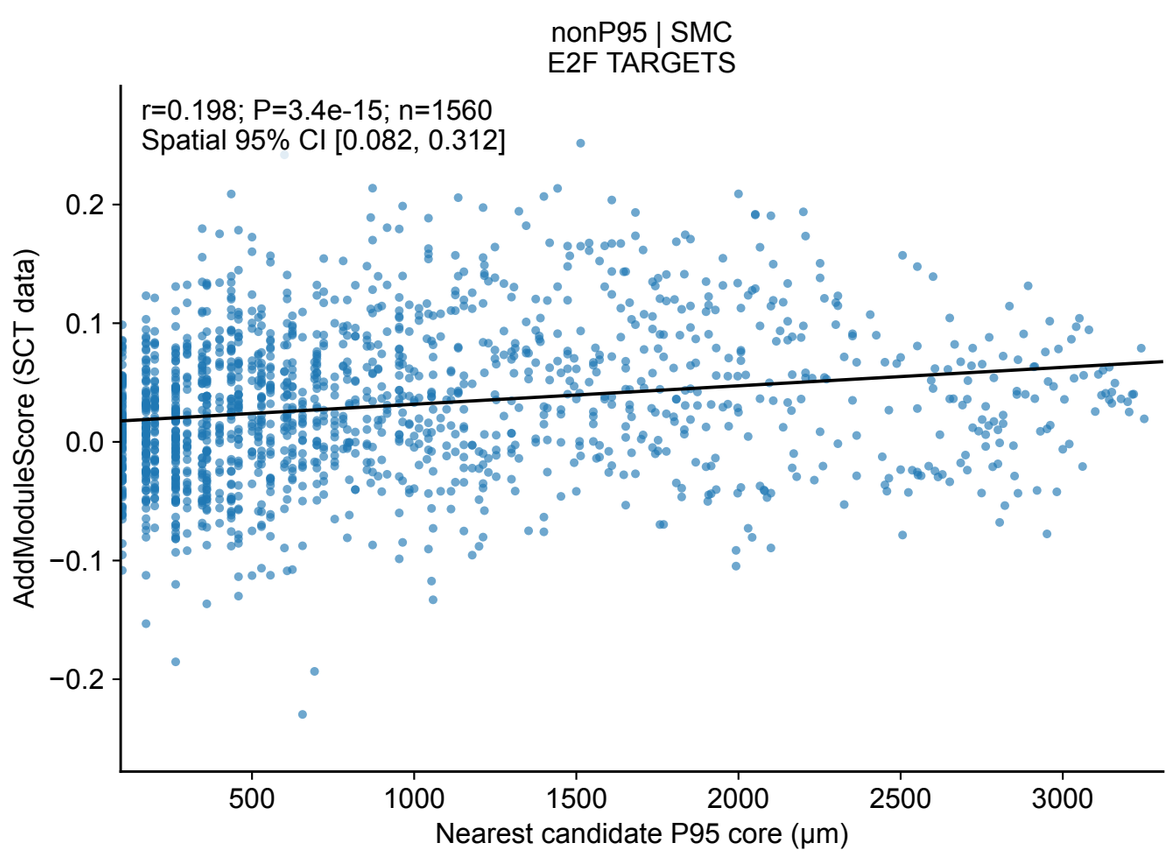


CanDis_2026_Sample10 | exploratory scatter display selected by coverage and |delta r|, not P





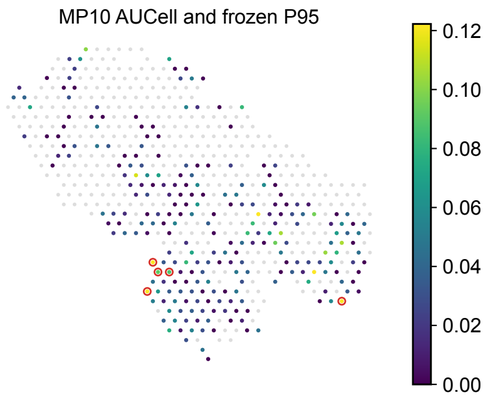


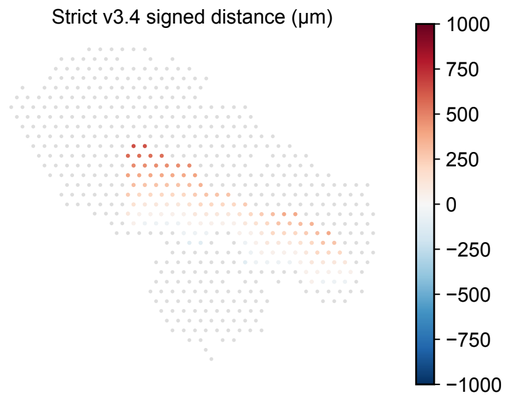


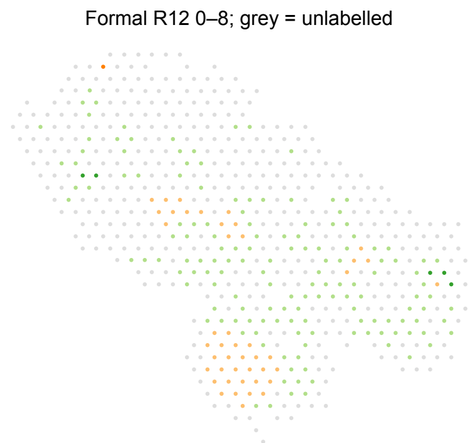
Classical Mono / E2F TARGETS: delta r=0.470 [0.195, 0.684]; common 173–4331 µm; spatial P=0.00018
SMC / E2F TARGETS: delta r=-0.392 [-0.591, -0.103]; common 100–3305 µm; spatial P=0.001
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

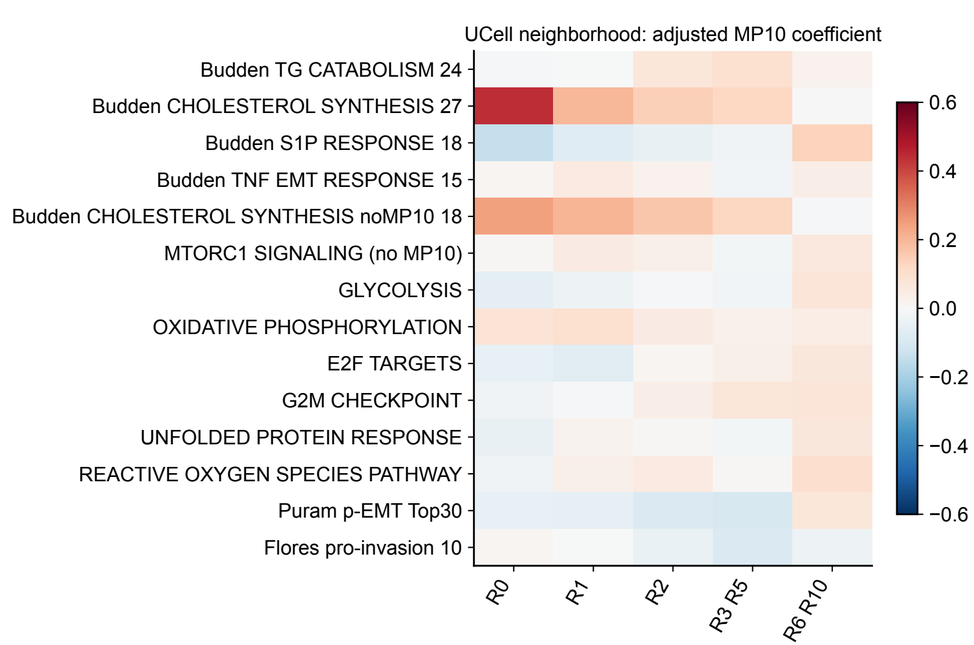
P95 nonP95 distance 0

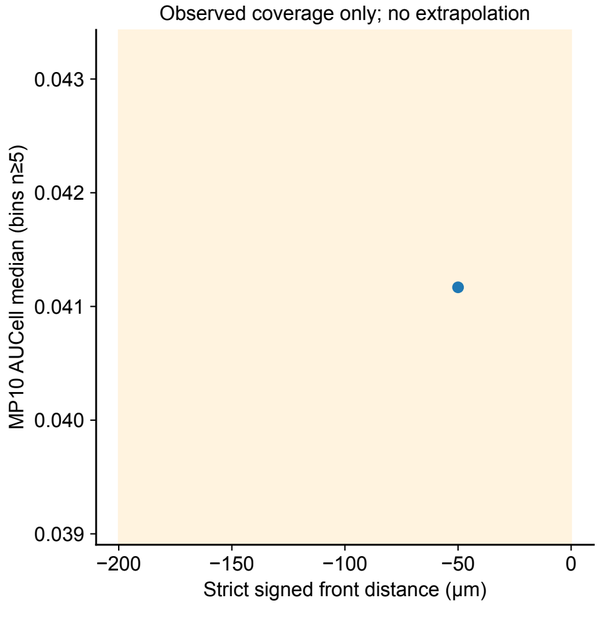
CanDis_2026_Sample11 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

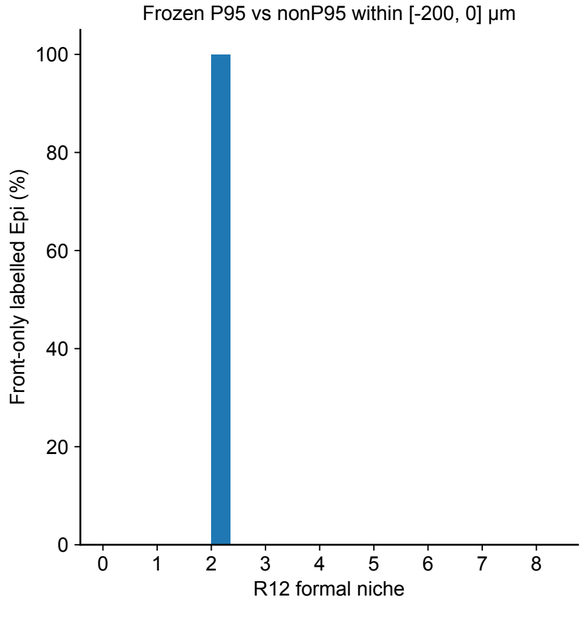








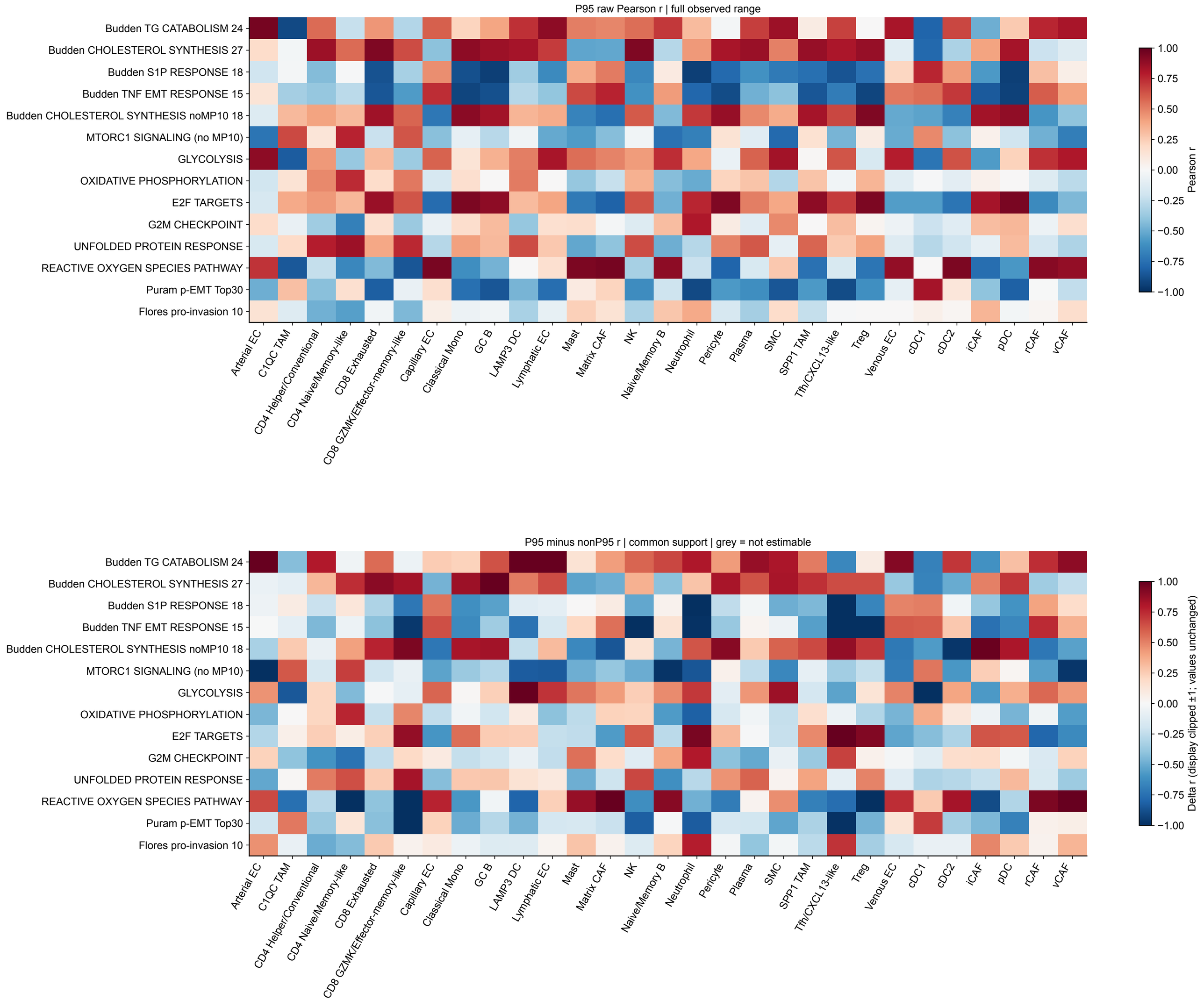




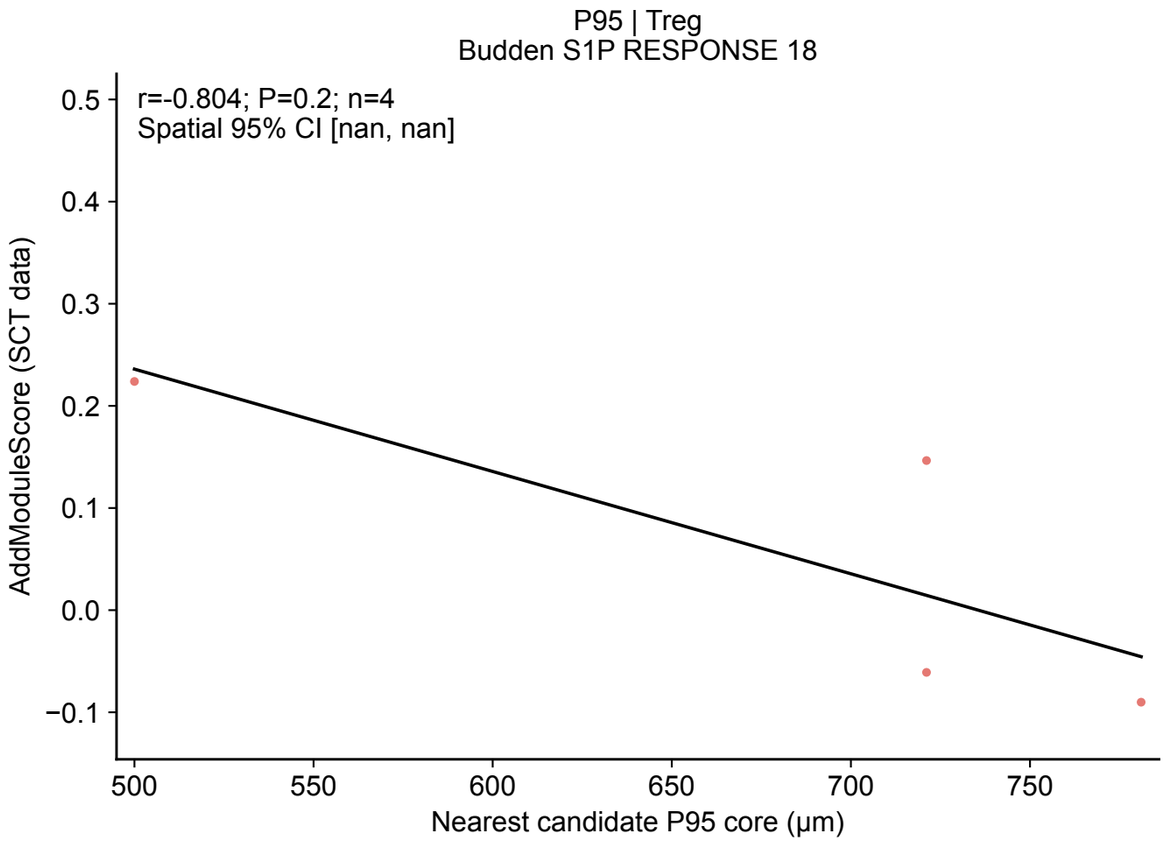
Tissue 577 | Epi 247 | internal 93 | P95 5
Exact front in Epi 77 | R12 labelled 200

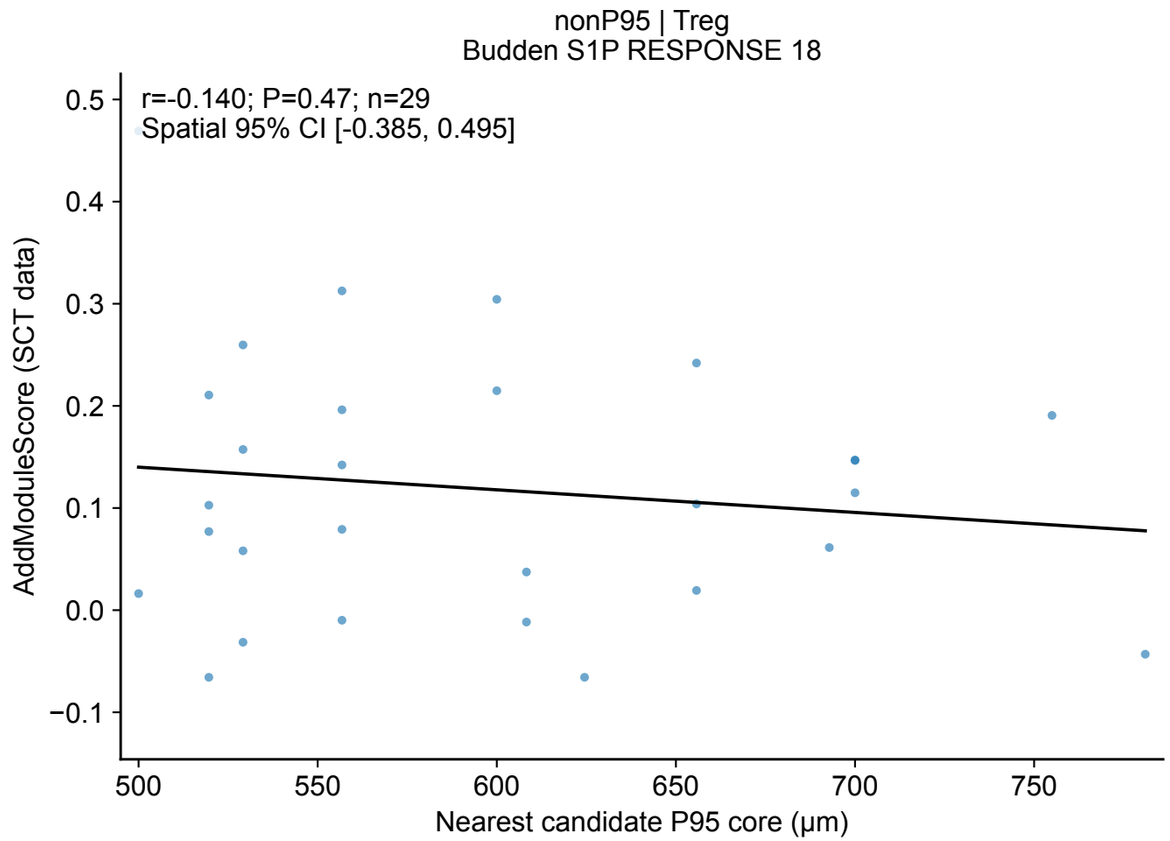
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

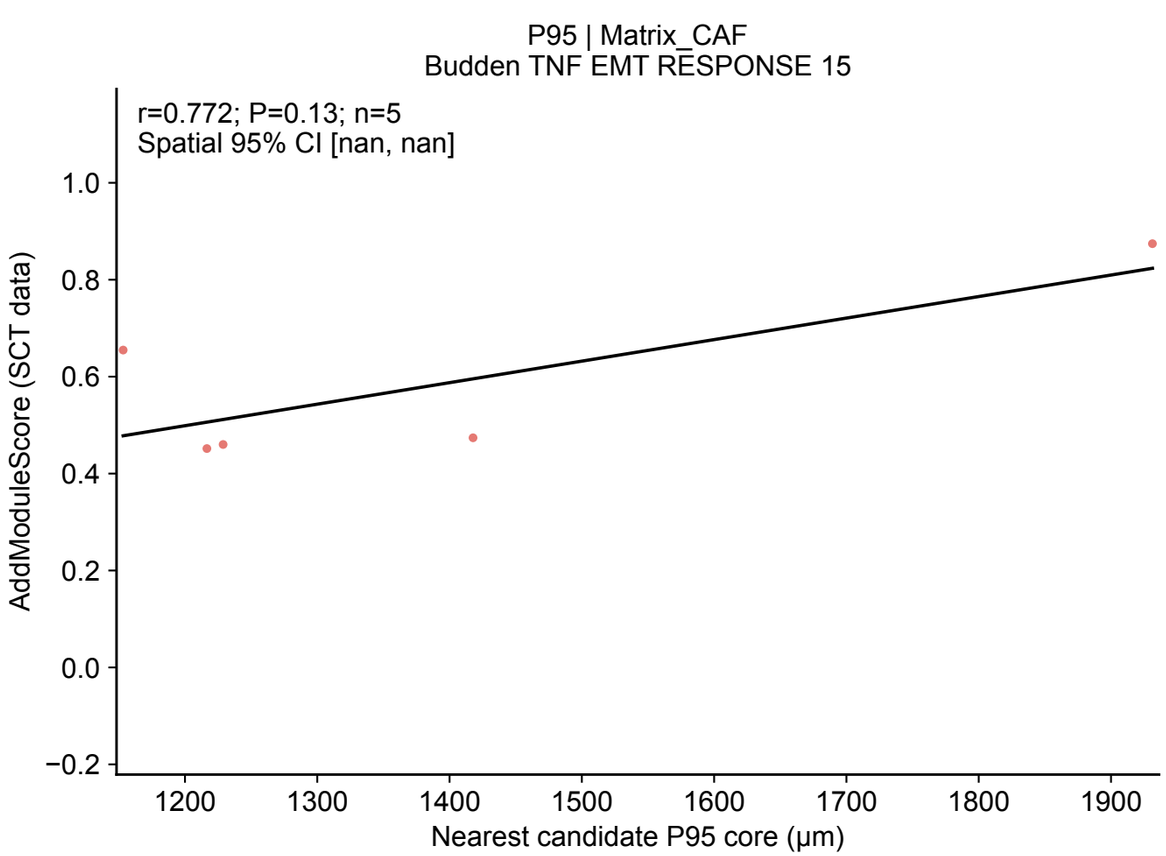
CanDis_2026_Sample11 | all 29 candidates; fixed core programs

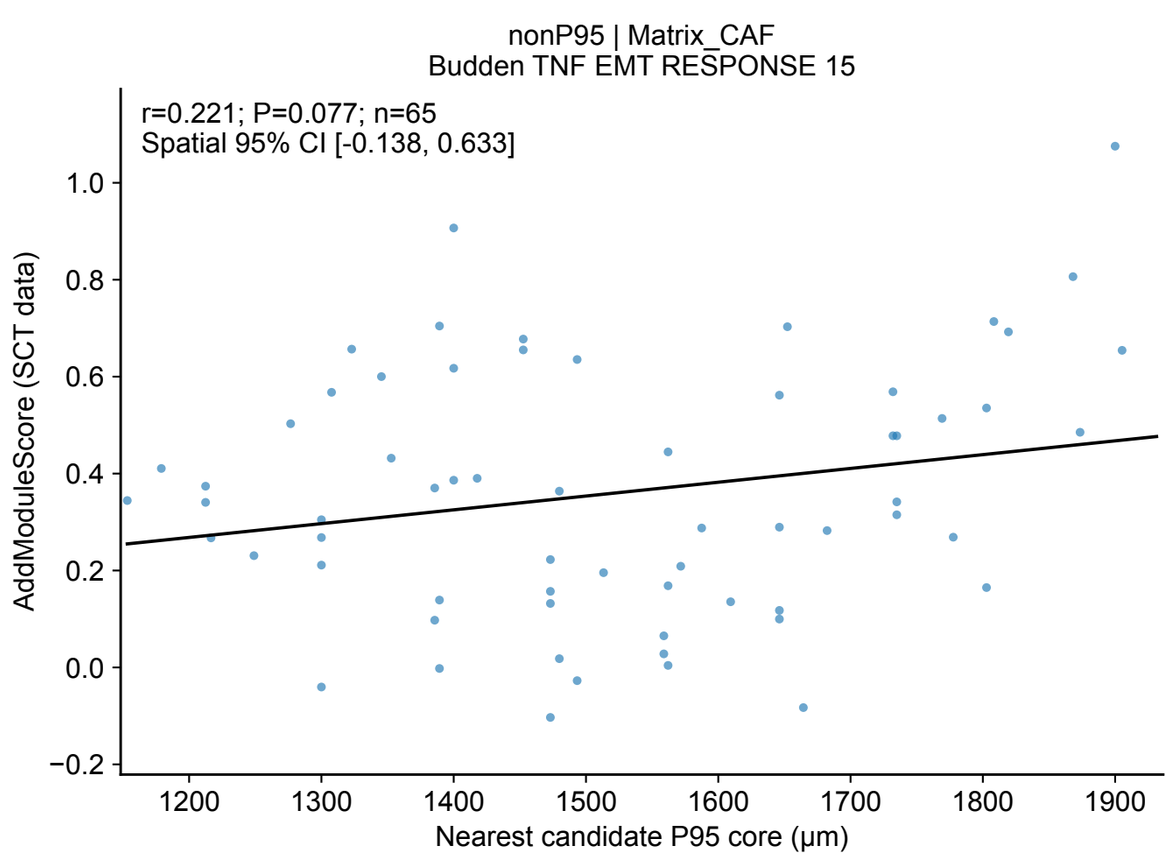


CanDis_2026_Sample11 | exploratory scatter display selected by coverage and |delta r|, not P







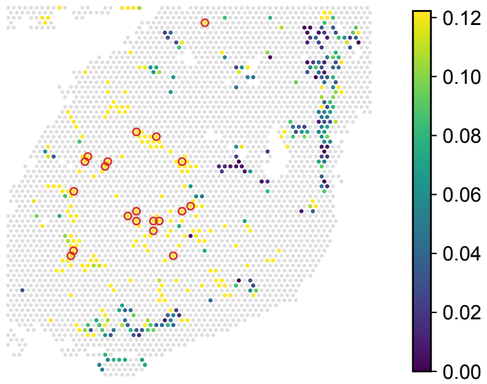


Treg / Budden S1P RESPONSE 18: delta r=-0.664 [nan, nan]; common 500–781 µm; spatial P=nan
Matrix_CAF / Budden TNF EMT RESPONSE 15: delta r=0.551 [nan, nan]; common 1153–1931 µm; spatial P=nan
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

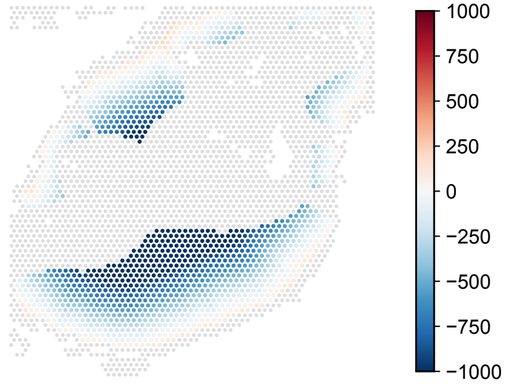
P95 nonP95 distance 0

CanDis_2026_Sample12 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

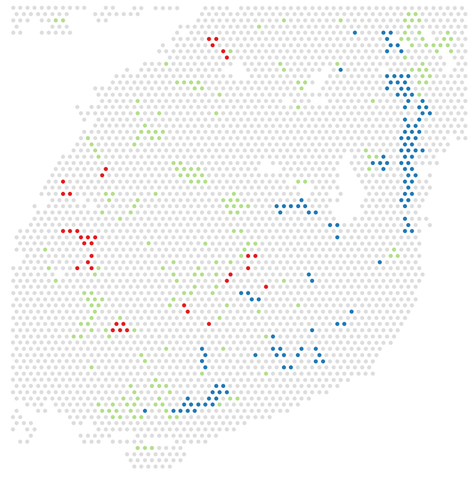
MP10 AUCell and frozen P95



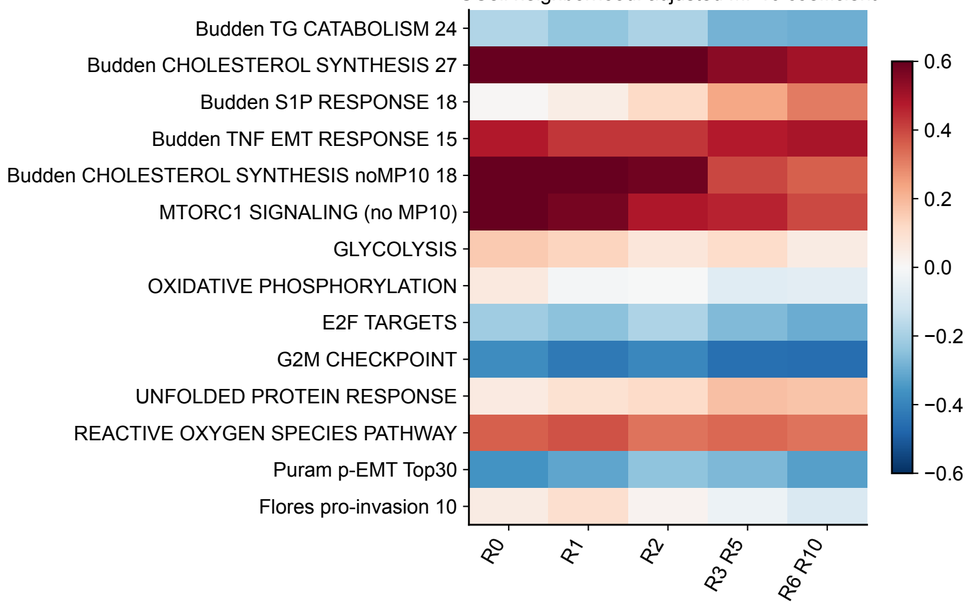
Strict v3.4 signed distance (µm)



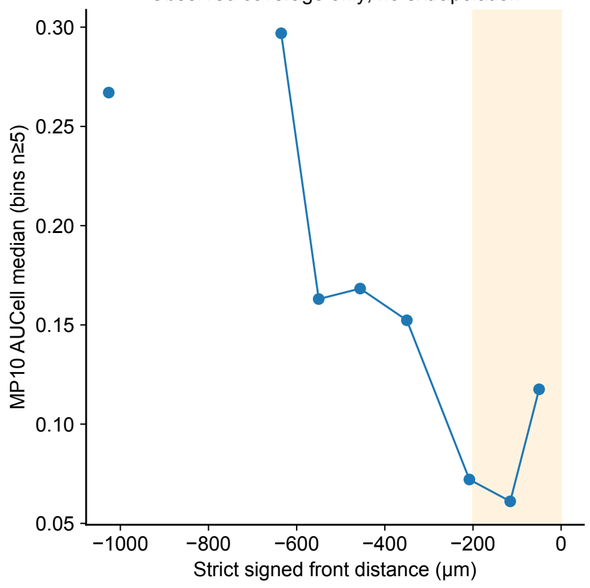
Formal R12 0–8; grey = unlabelled



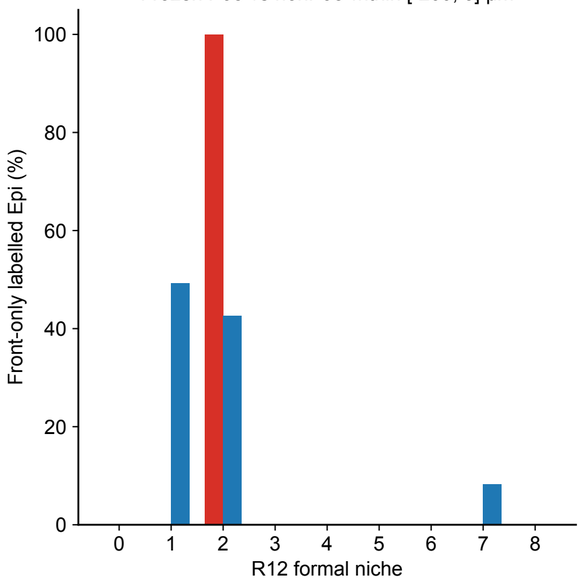
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

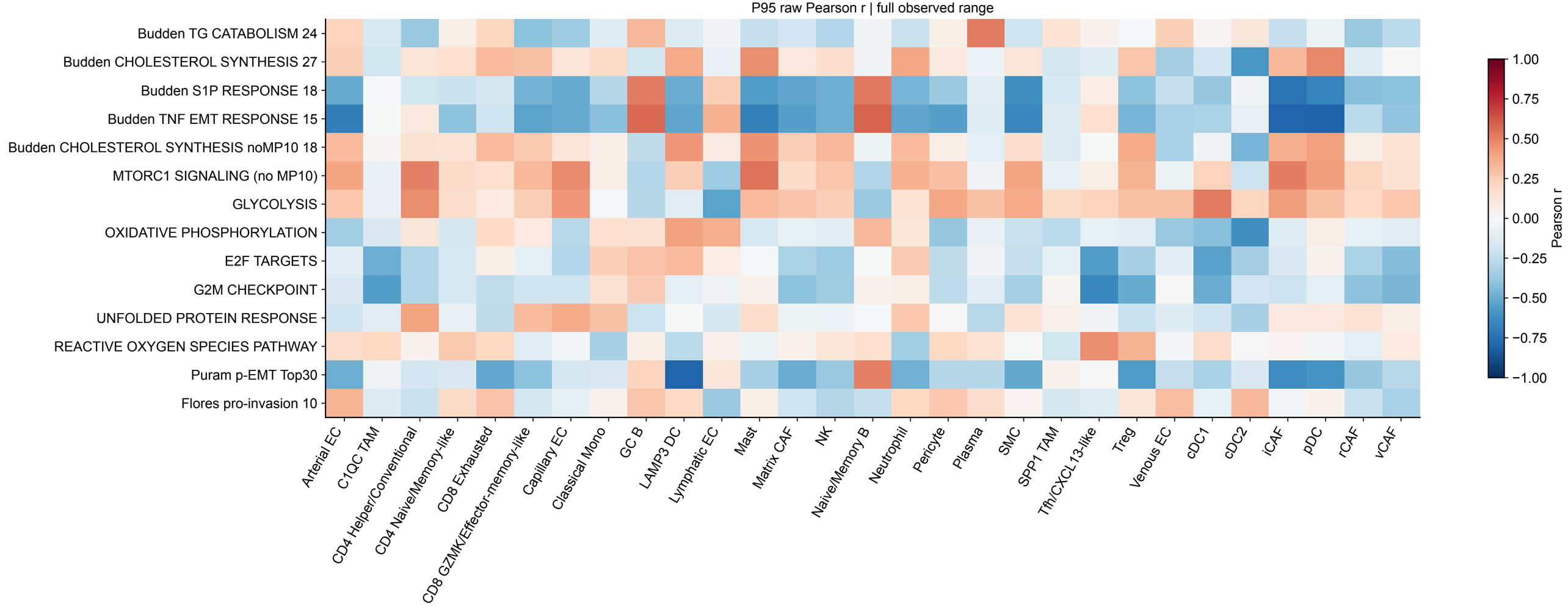


Tissue 3508 | Epi 405 | internal 390 | P95 20
Exact front in Epi 189 | R12 labelled 370

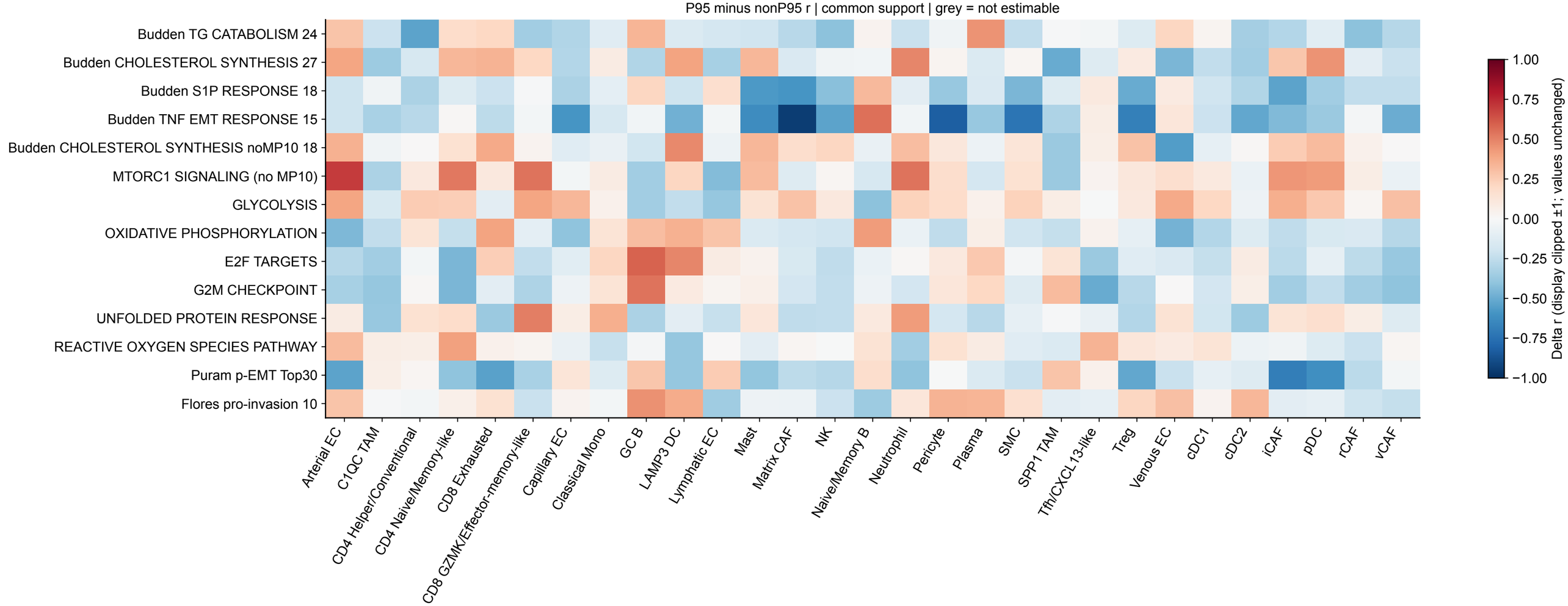
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

CanDis_2026_Sample12 | all 29 candidates; fixed core programs

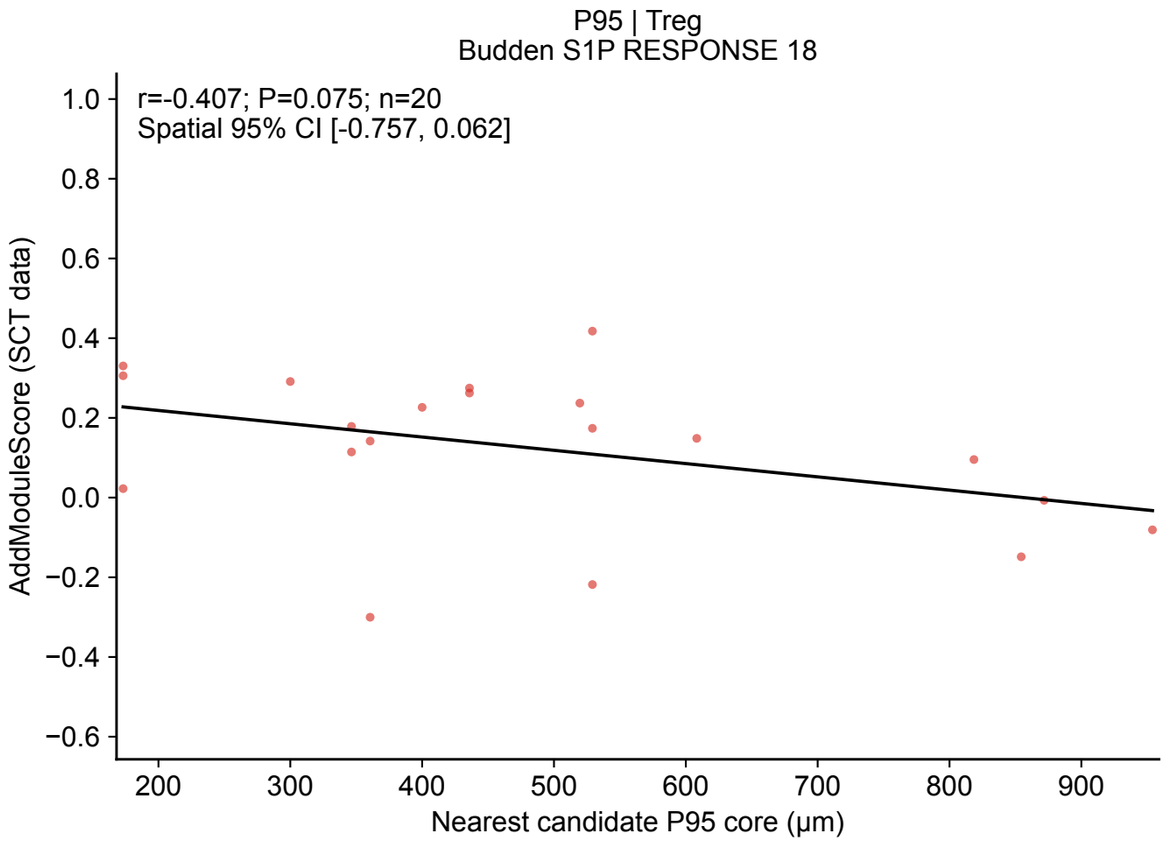
P95 raw Pearson r | full observed range

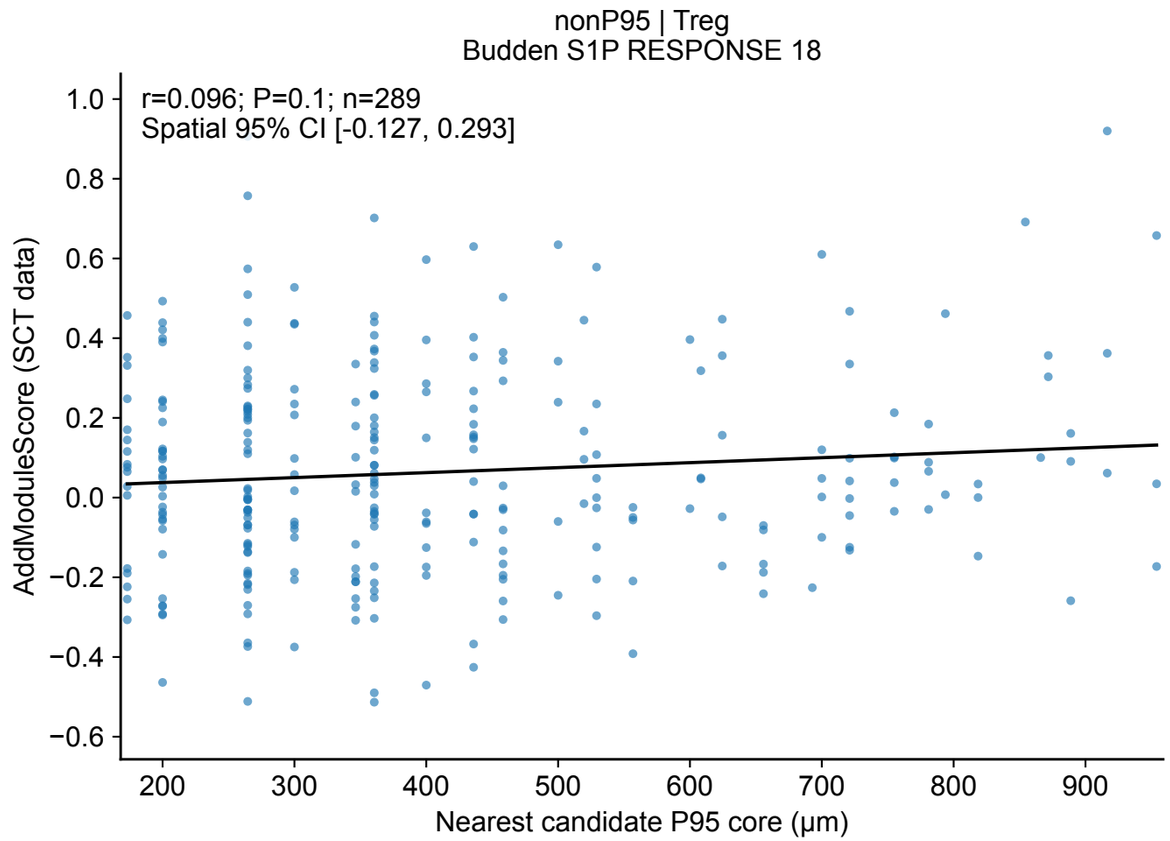


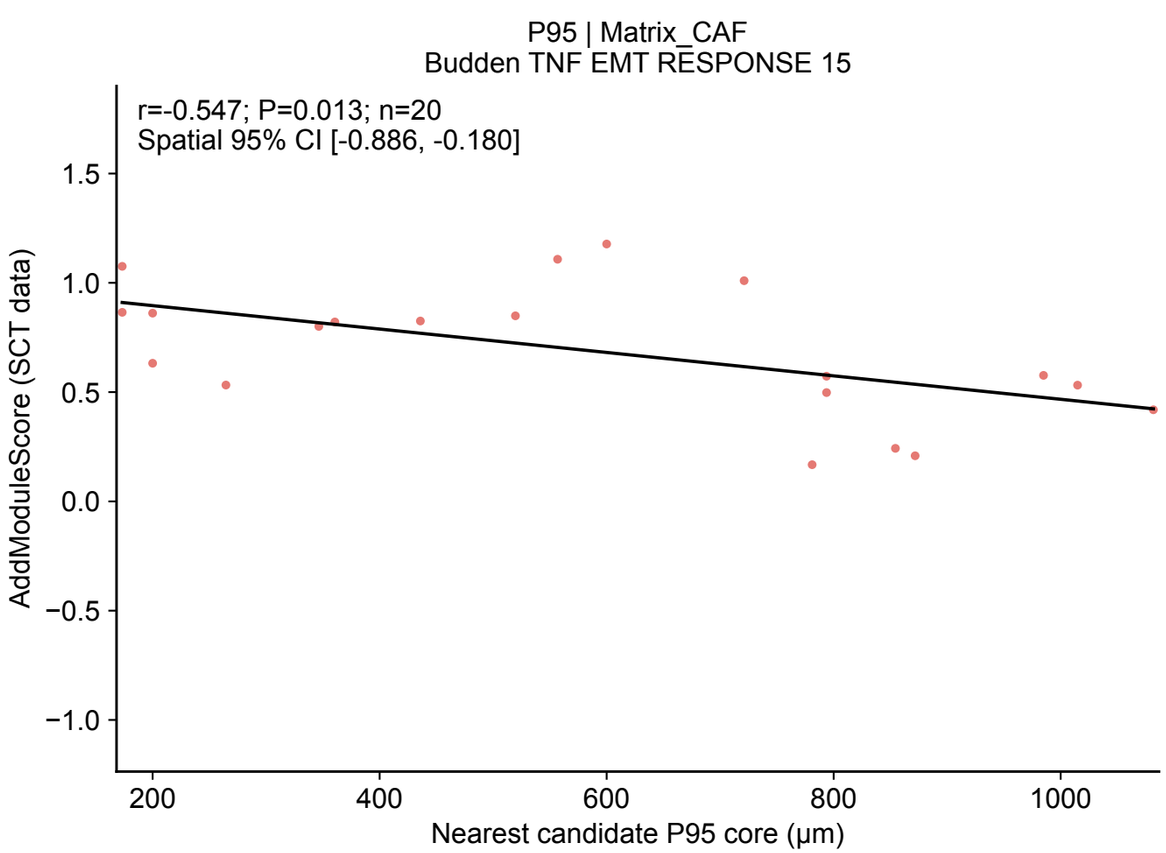
P95 minus nonP95 r | common support | grey = not estimable

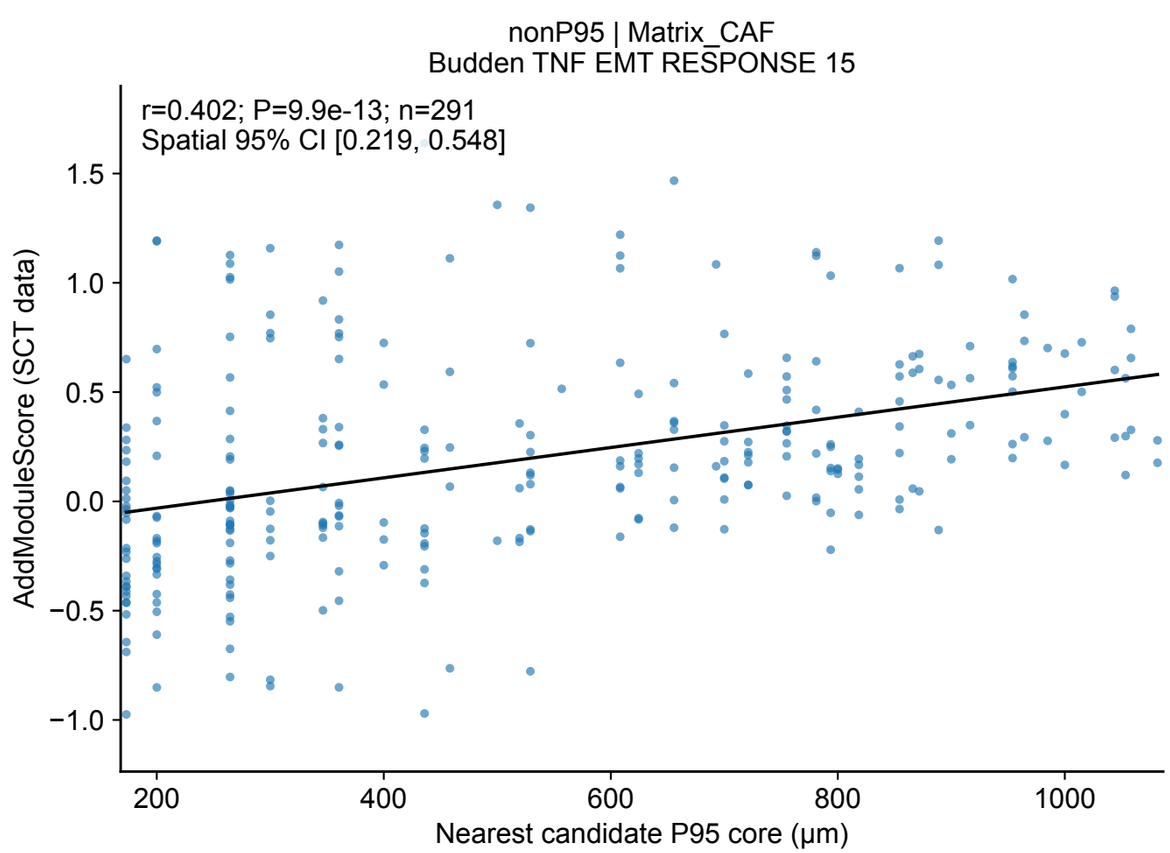


CanDis_2026_Sample12 | exploratory scatter display selected by coverage and |delta r|, not P







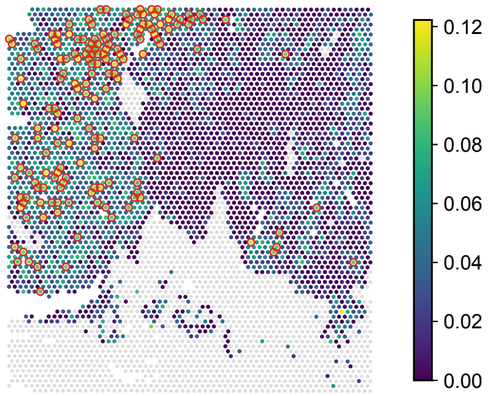


Treg / Budden S1P RESPONSE 18: delta r=-0.503 [-0.943, 0.076]; common 173–954 µm; spatial P=0.043
Matrix_CAF / Budden TNF EMT RESPONSE 15: delta r=-0.949 [-1.292, -0.547]; common 173–1082 µm; spatial P=2.5e-05
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

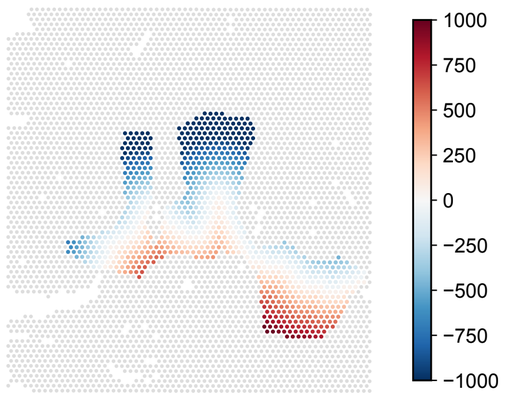
P95 nonP95 distance 0

CanDis_2026_Sample13 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

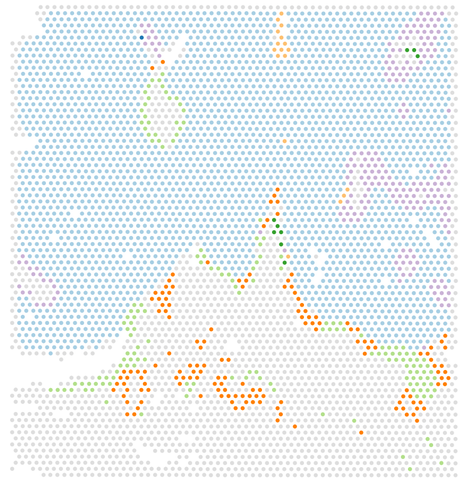
MP10 AUCell and frozen P95



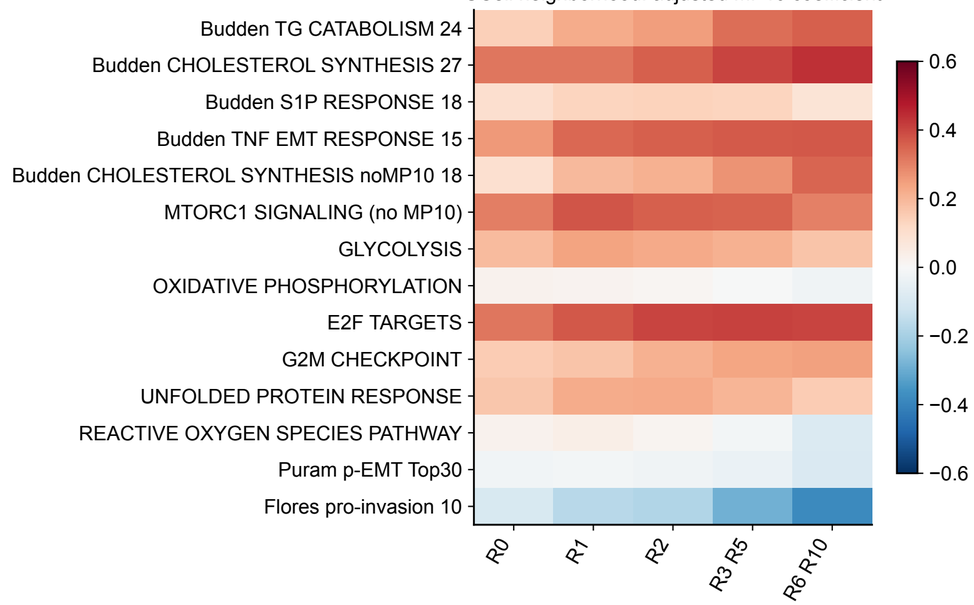
Strict v3.4 signed distance (µm)



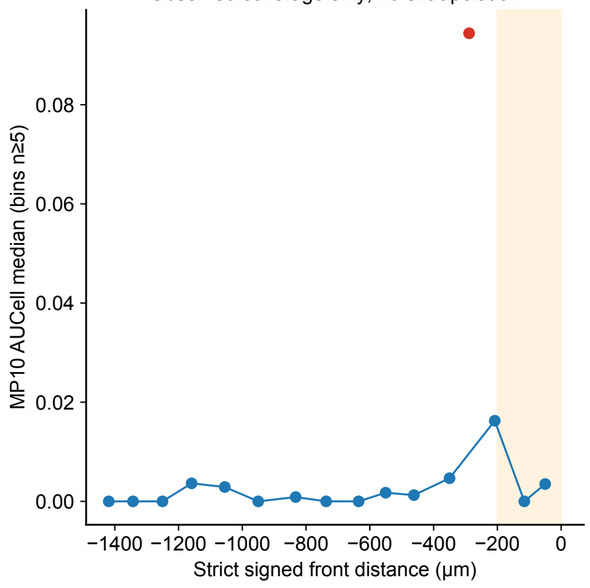
Formal R12 0–8; grey = unlabelled



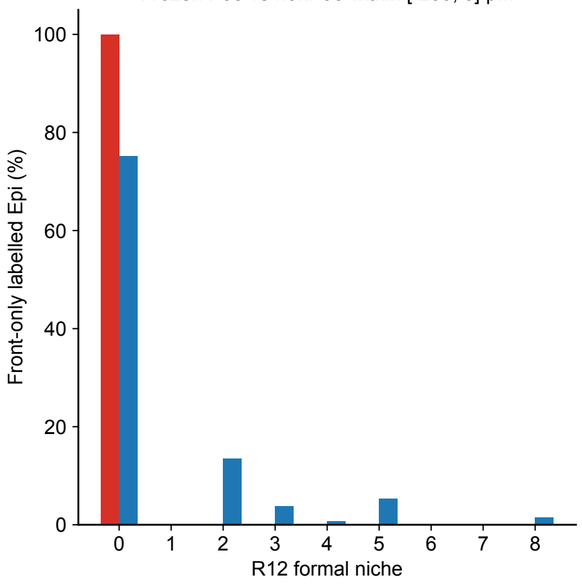
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

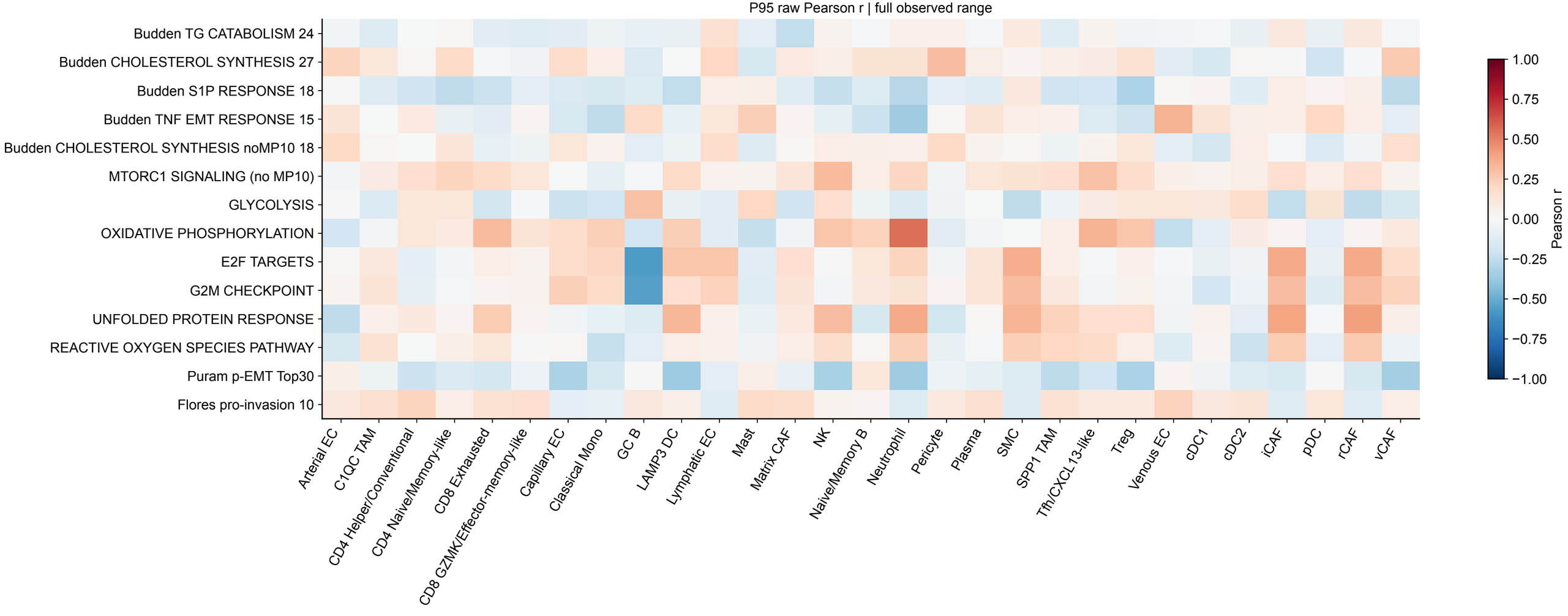


Tissue 4835 | Epi 3430 | internal 3151 | P95 158
Exact front in Epi 603 | R12 labelled 3182

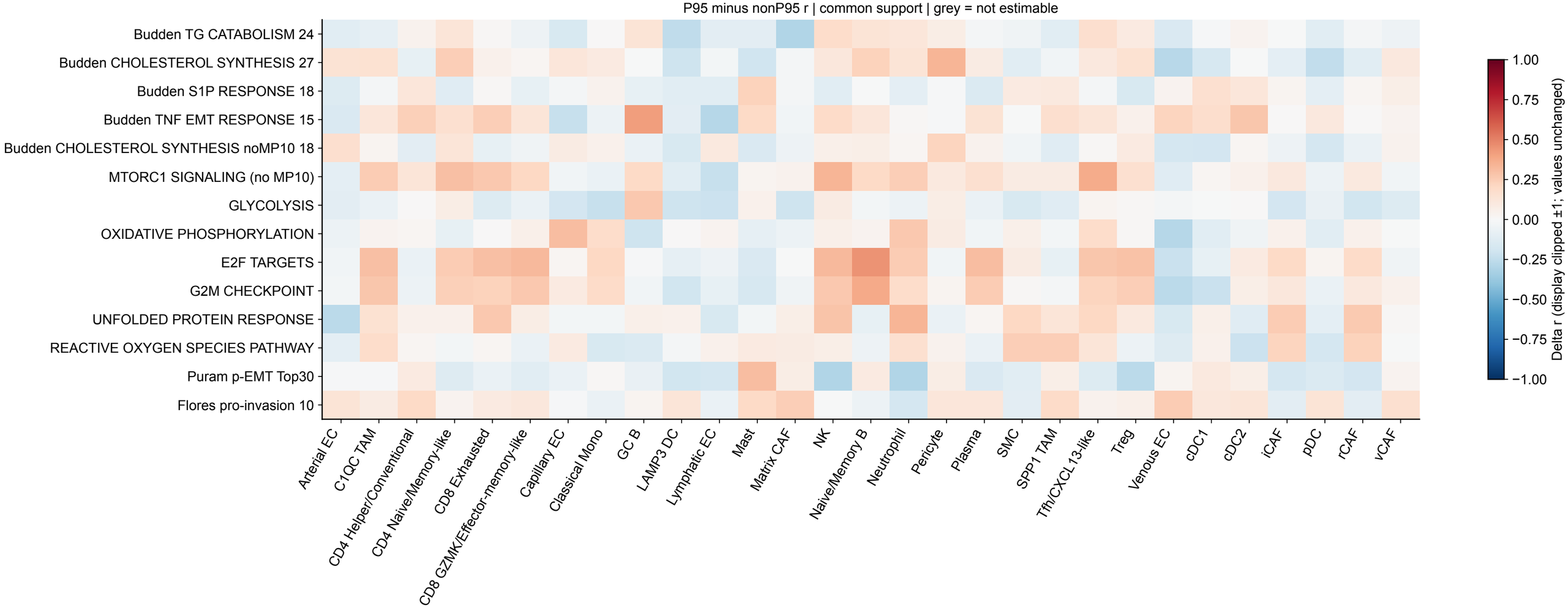
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

CanDis_2026_Sample13 | all 29 candidates; fixed core programs

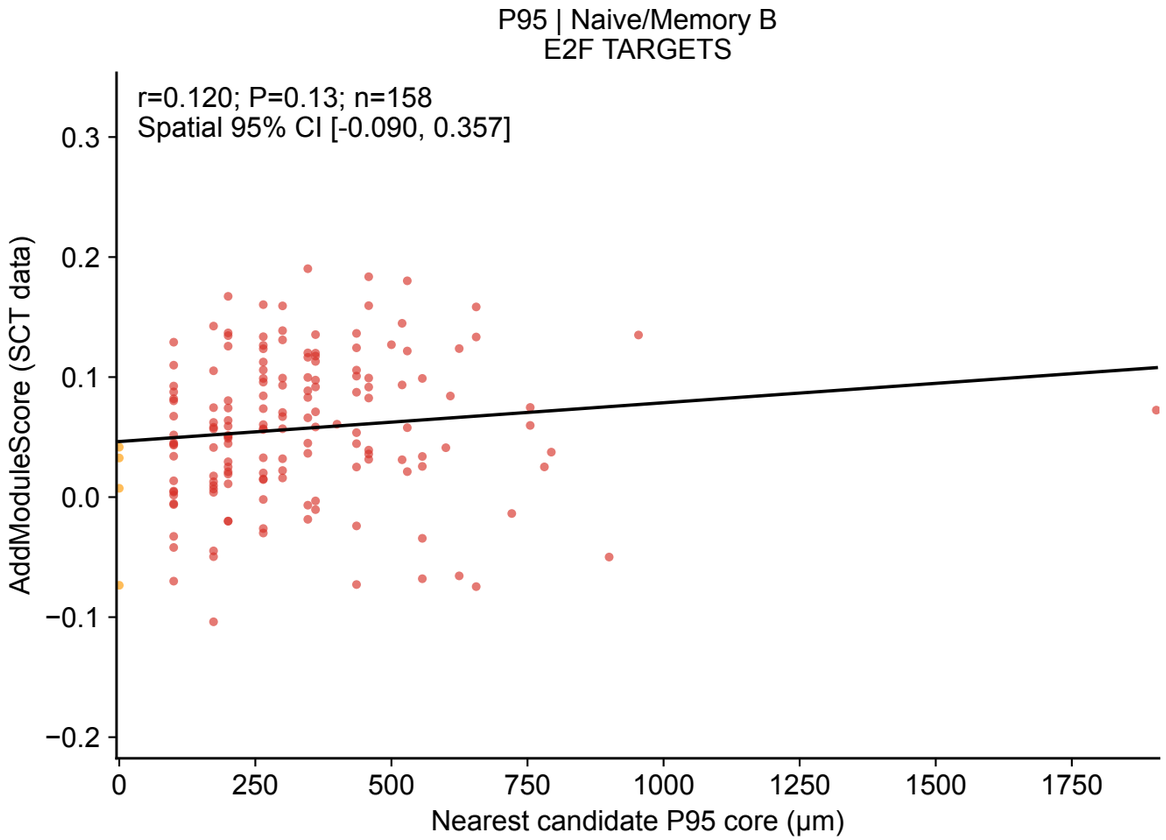
P95 raw Pearson r | full observed range

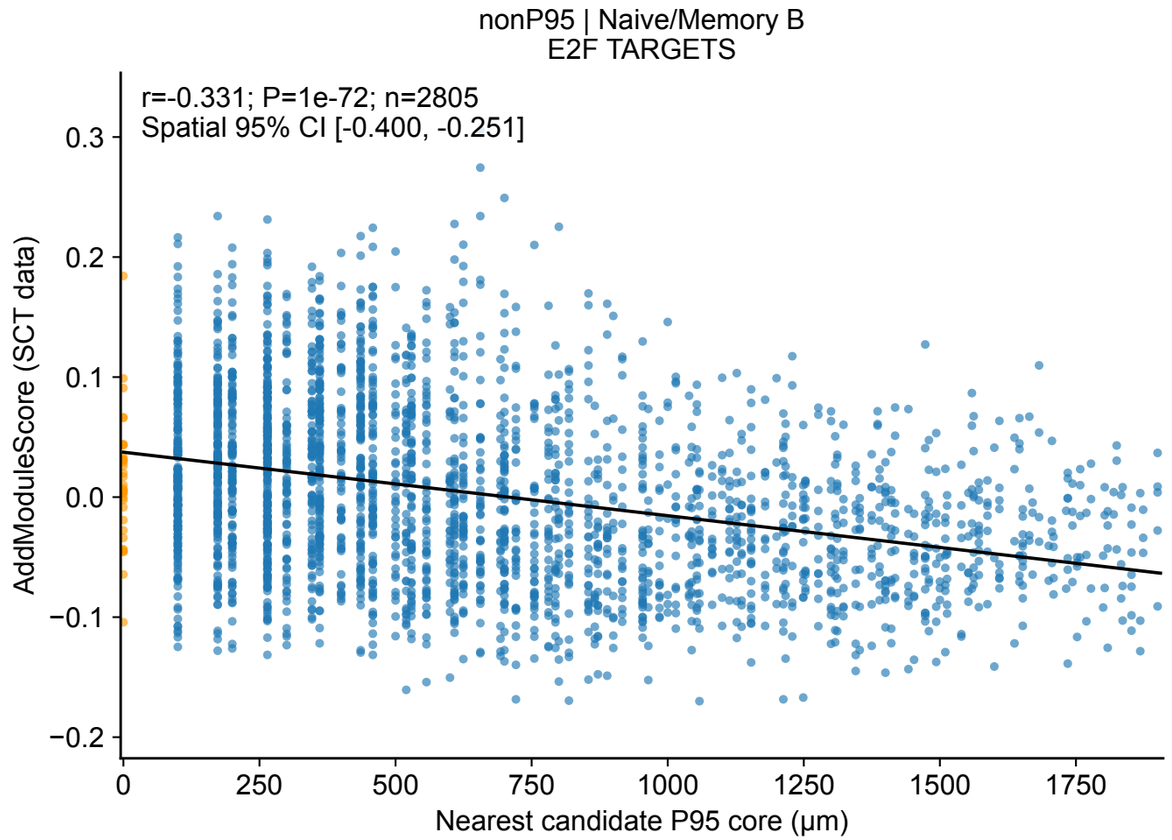


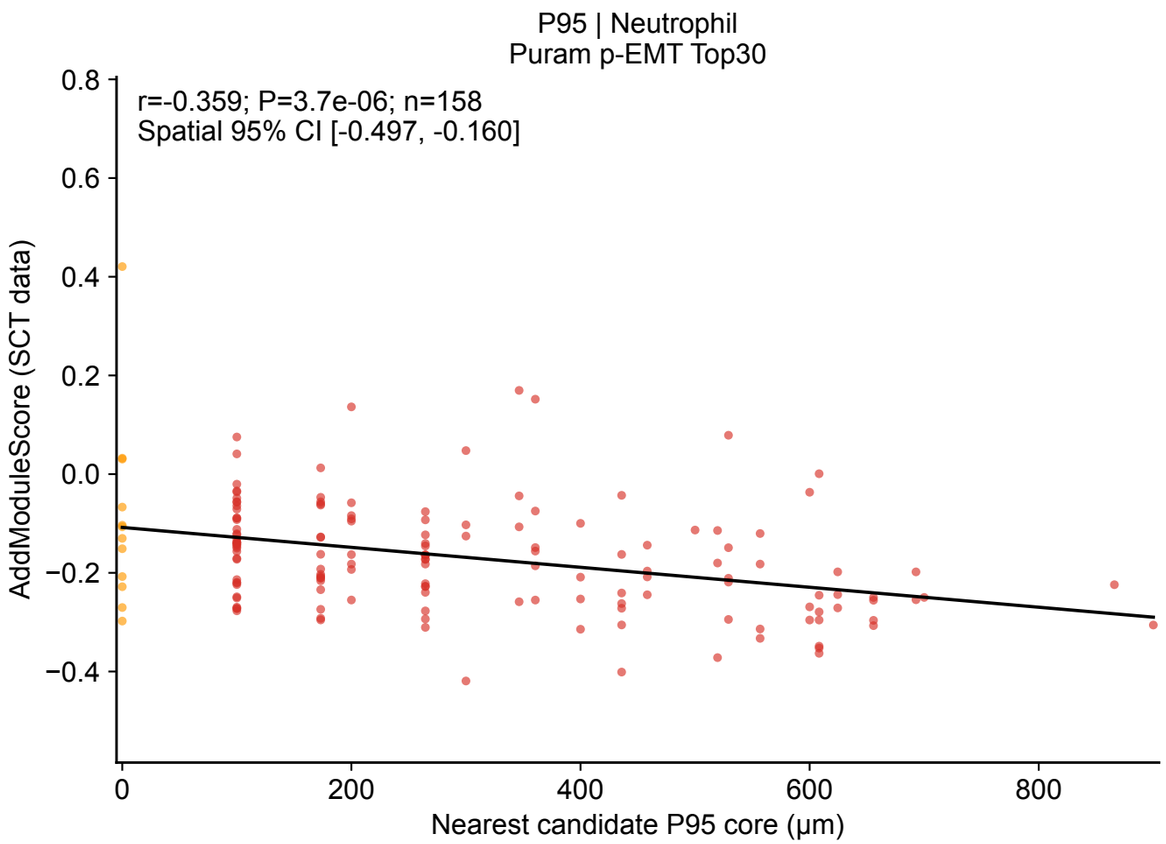
P95 minus nonP95 r | common support | grey = not estimable

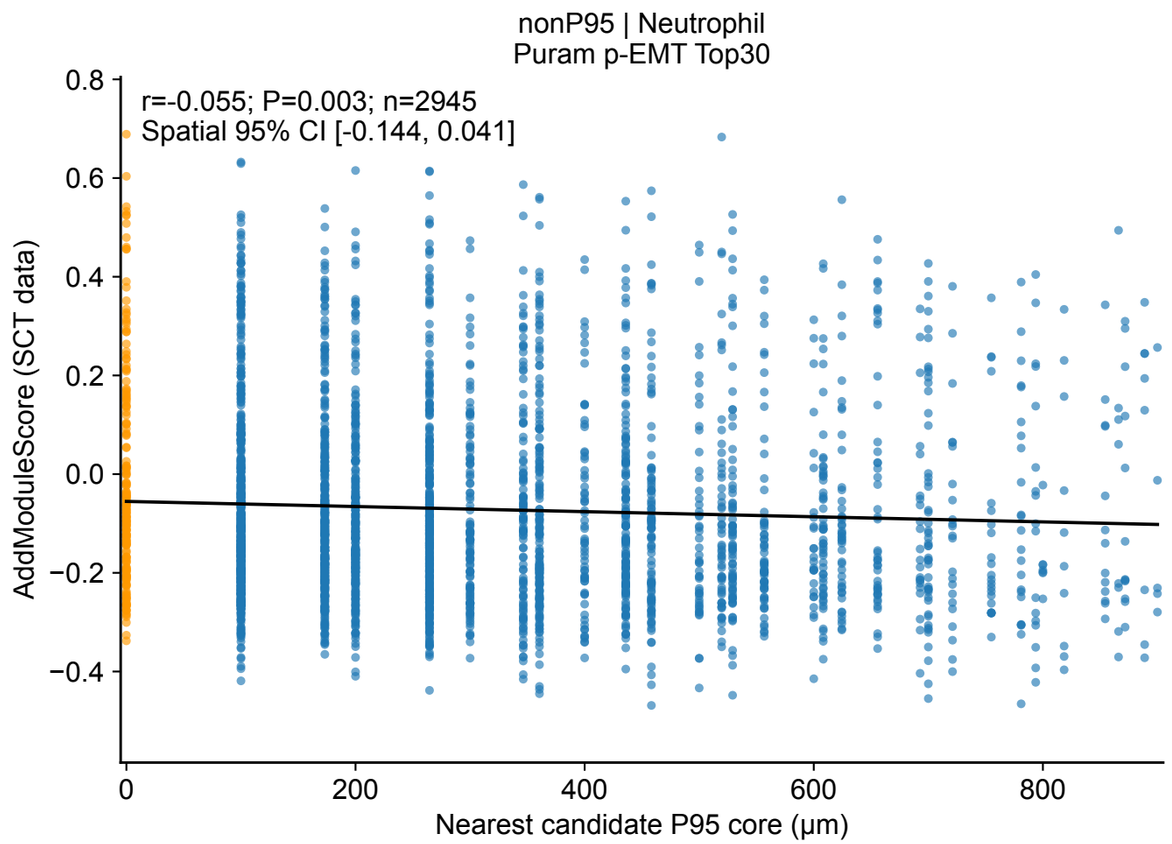


CanDis_2026_Sample13 | exploratory scatter display selected by coverage and |delta r|, not P







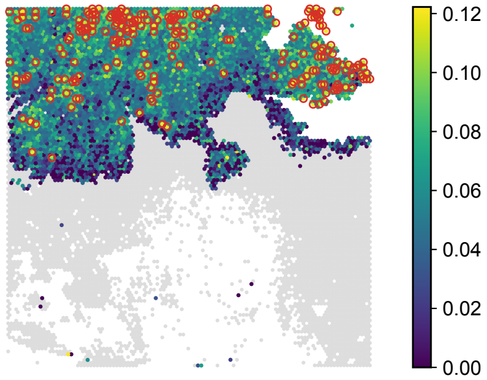


Naive/Memory B / E2F TARGETS: delta r=0.451 [0.236, 0.701]; common 0–1905 µm; spatial P=0.00011
Neutrophil / Puram p-EMT Top30: delta r=-0.304 [-0.465, -0.110]; common 0–900 µm; spatial P=0.0014
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

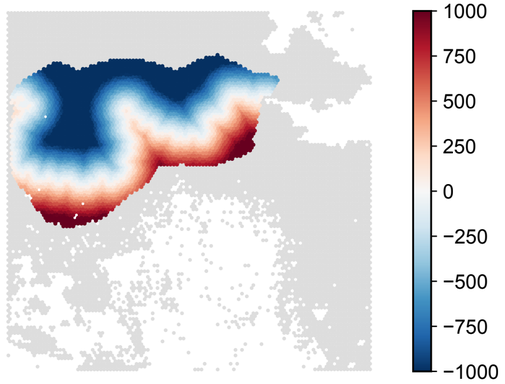
P95 nonP95 distance 0

CanDis_2026_Sample2 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

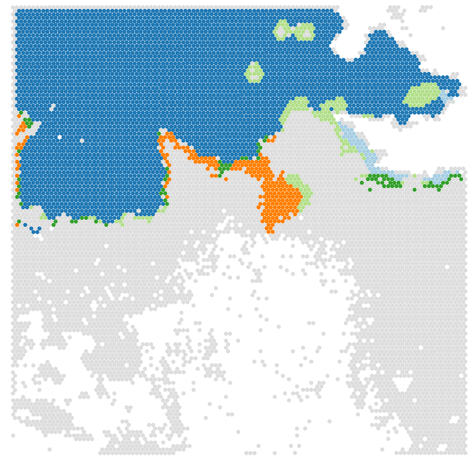
MP10 AUCell and frozen P95



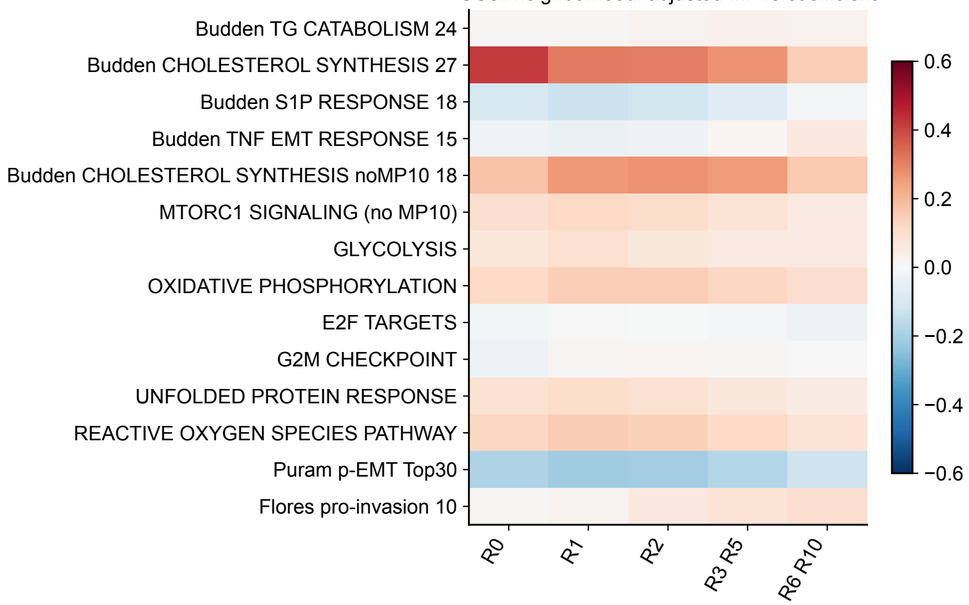
Strict v3.4 signed distance (µm)



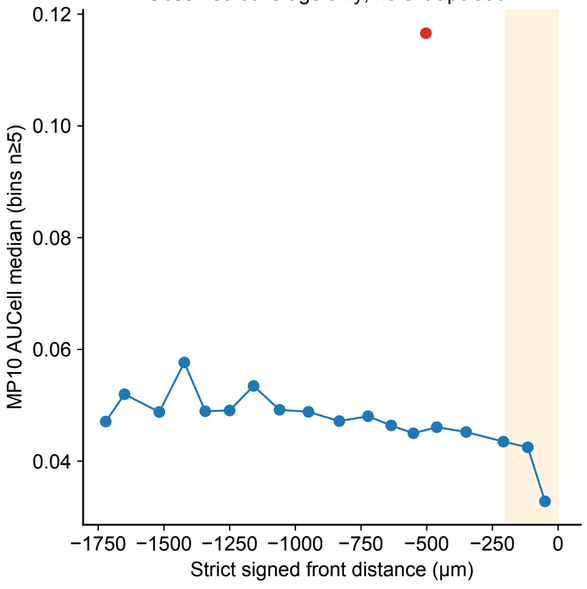
Formal R12 0–8; grey = unlabelled



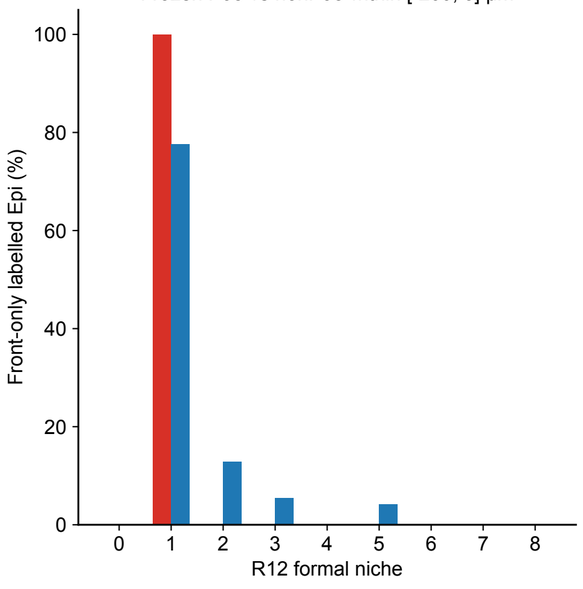
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

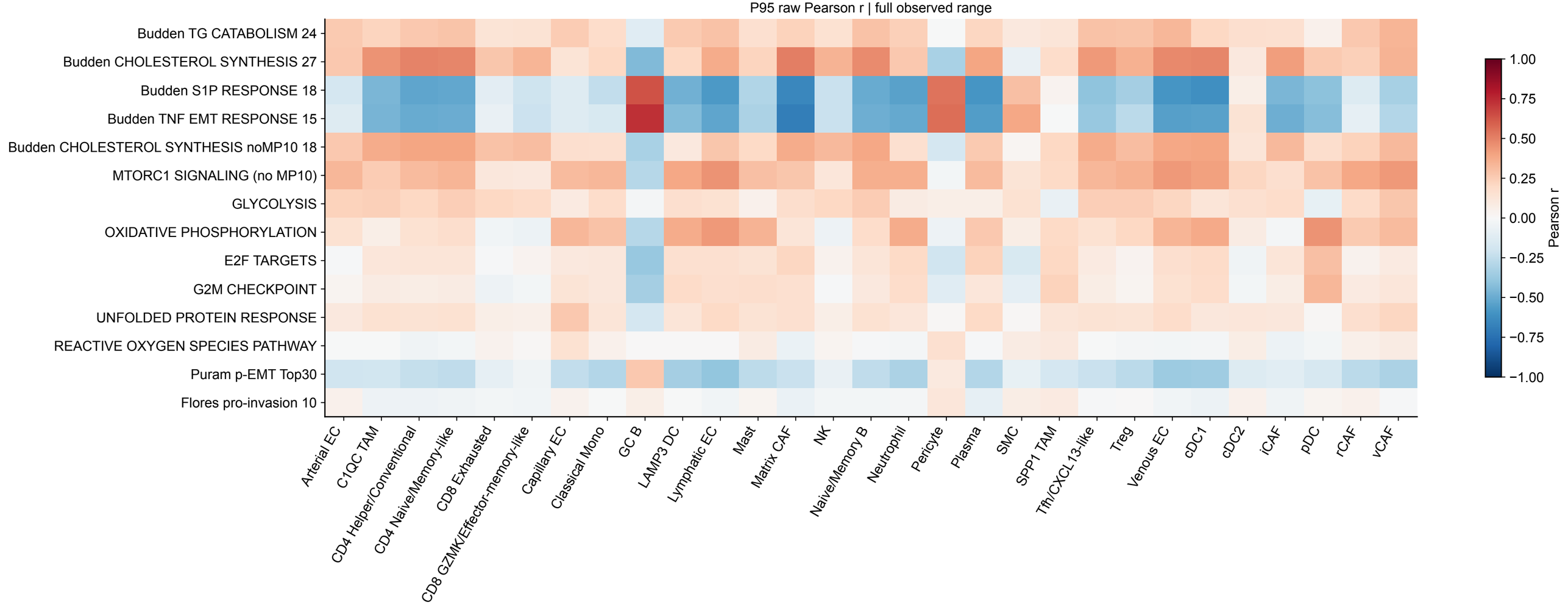


Tissue 10062 | Epi 4928 | internal 4363 | P95 219
Exact front in Epi 2460 | R12 labelled 4580

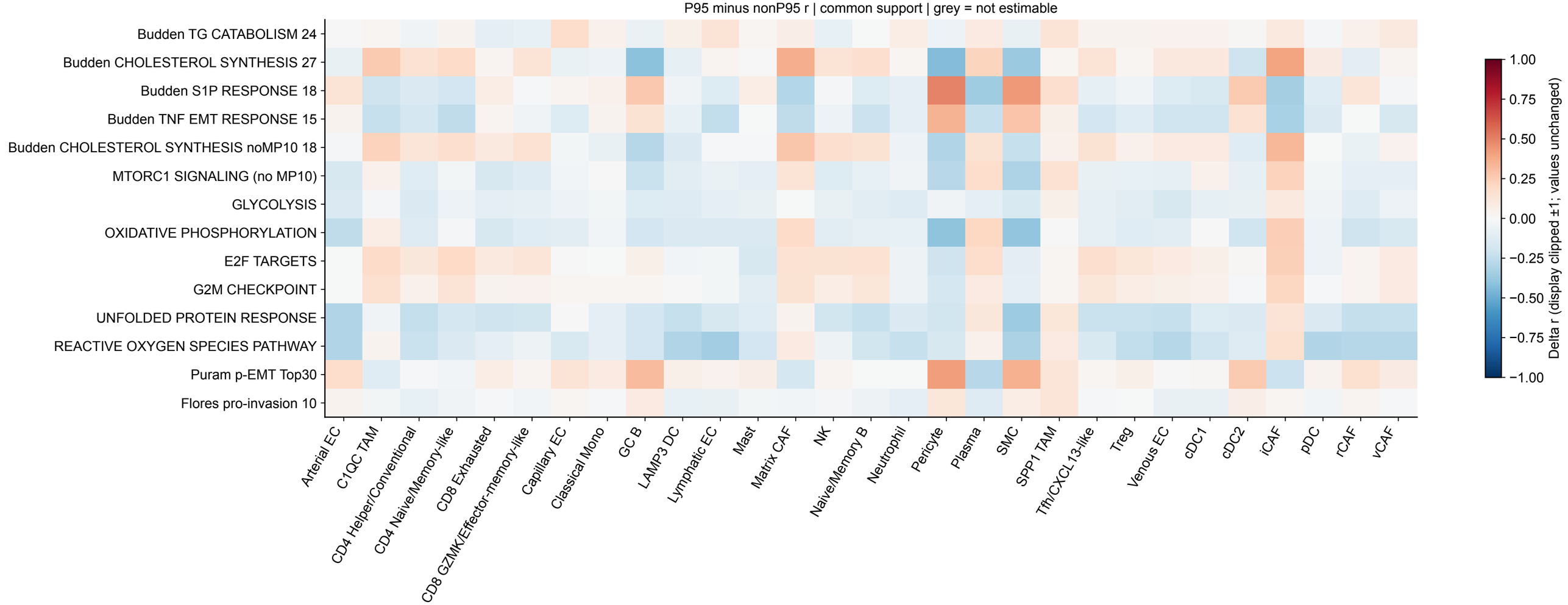


CanDis_2026_Sample2 | all 29 candidates; fixed core programs

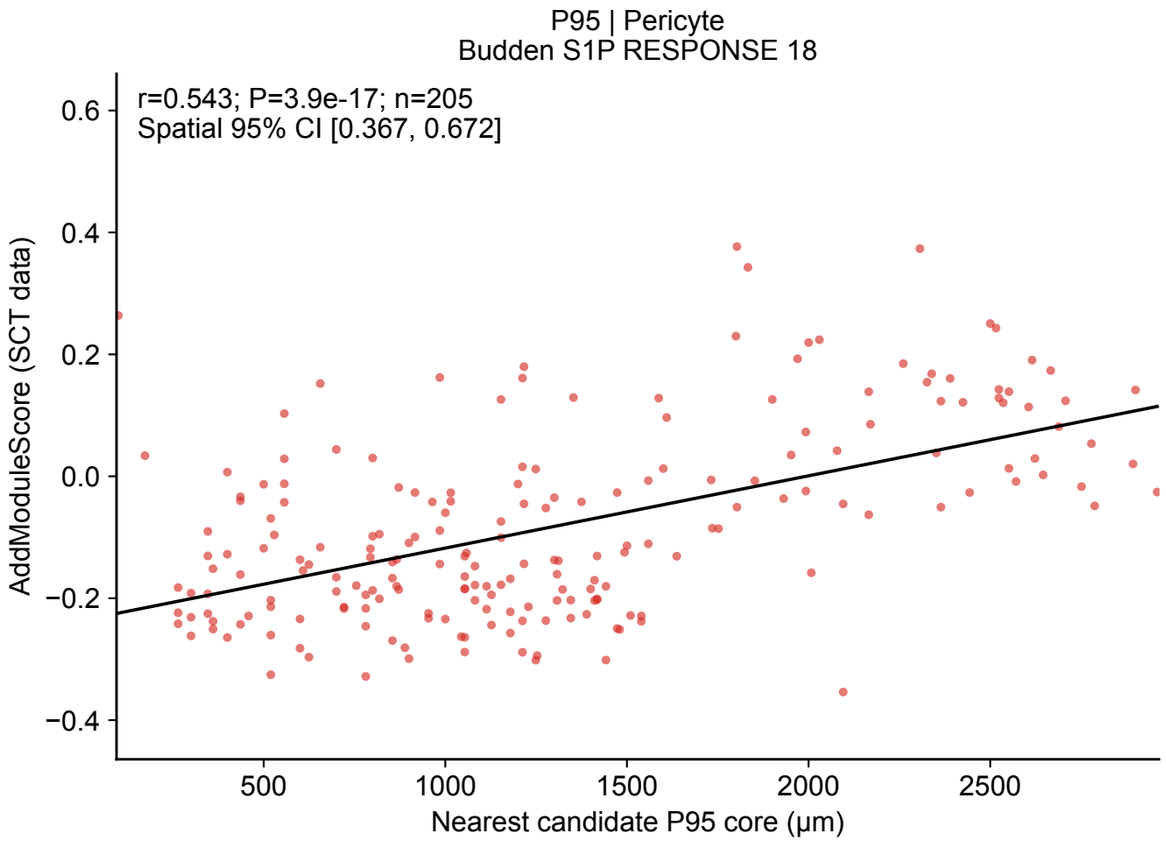
P95 raw Pearson r | full observed range

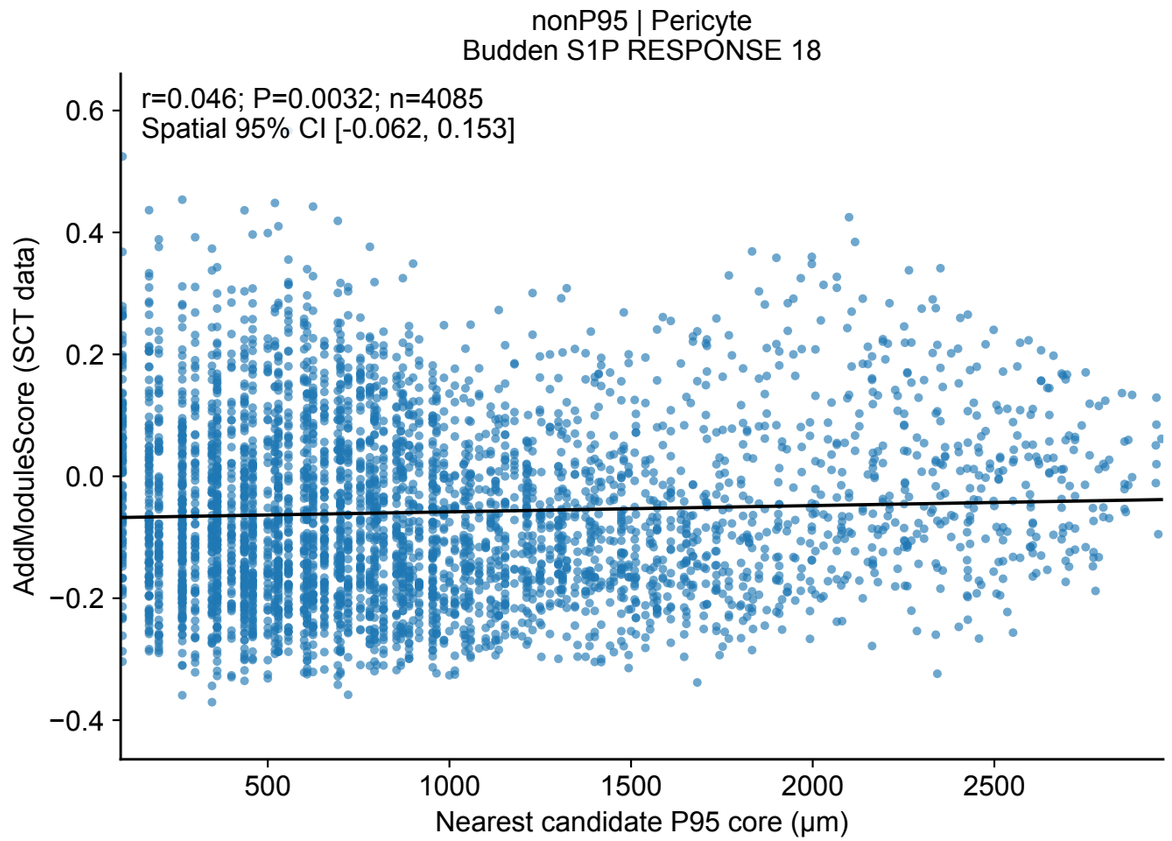


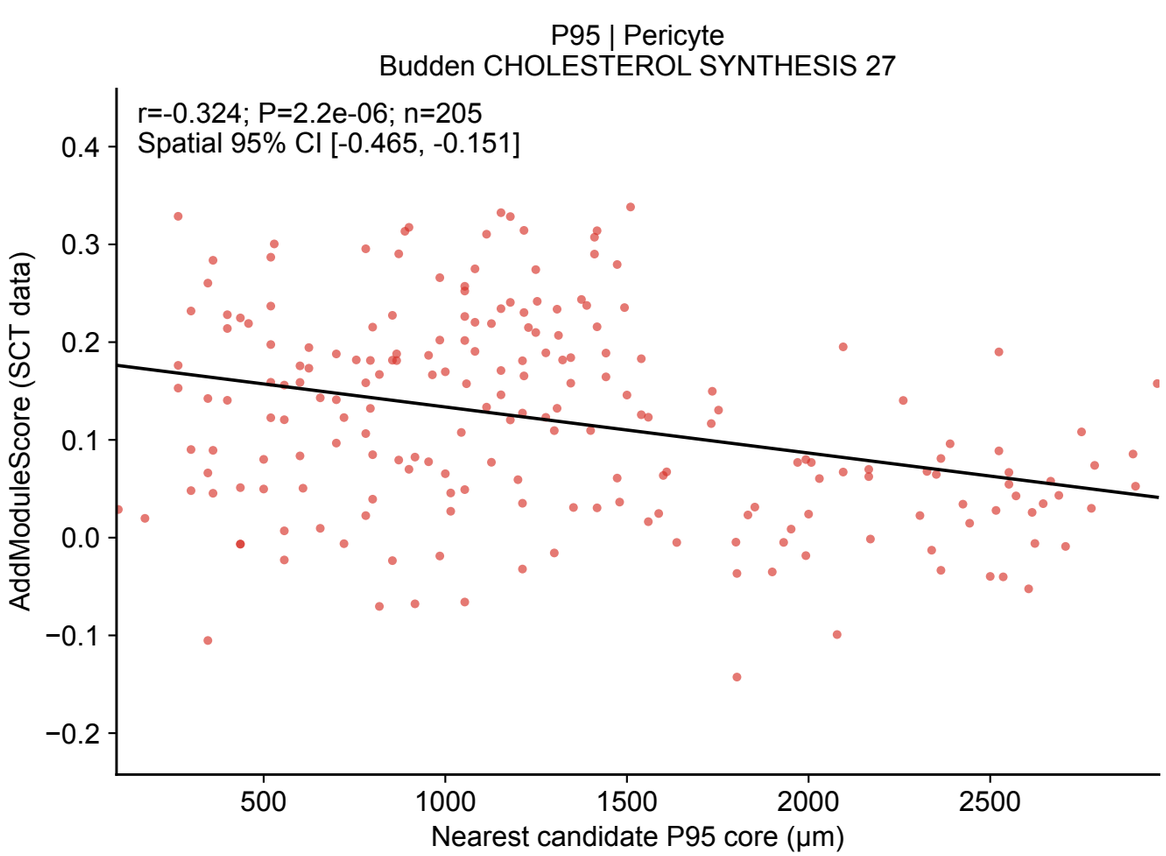
P95 minus nonP95 r | common support | grey = not estimable

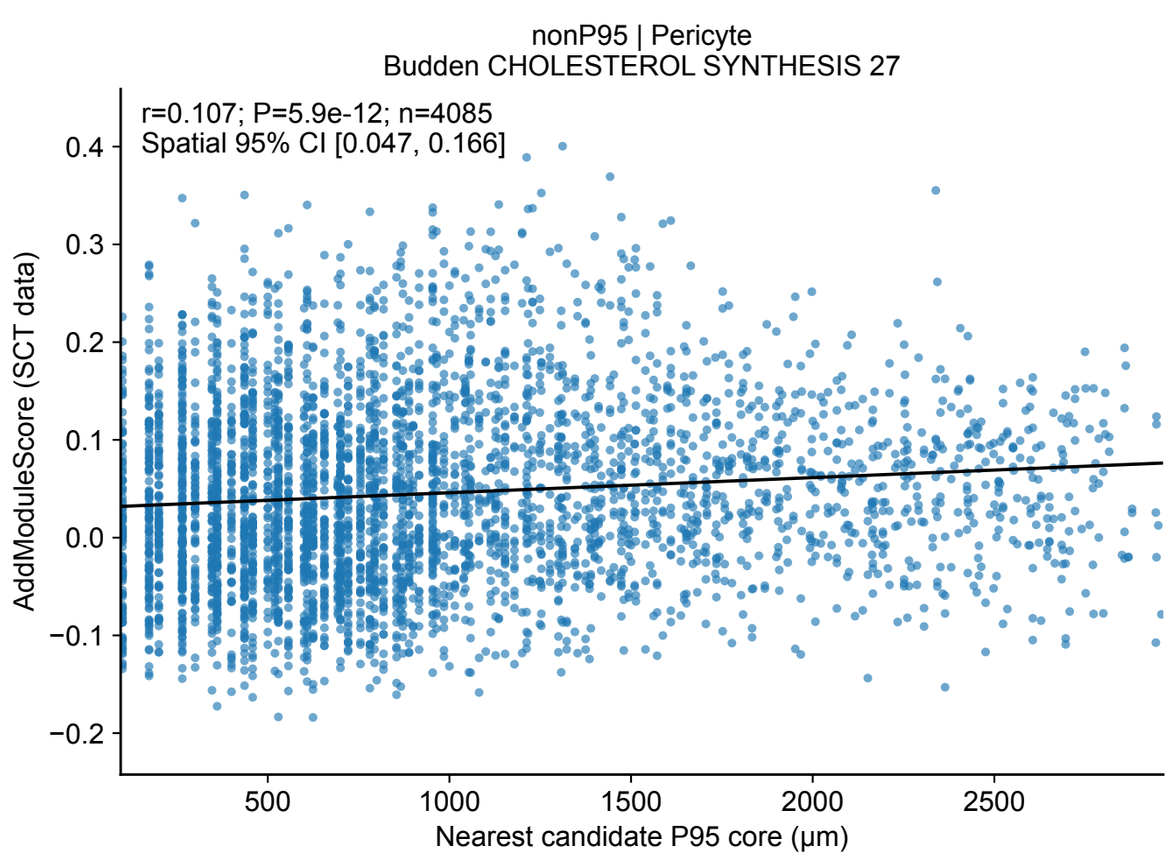


CanDis_2026_Sample2 | exploratory scatter display selected by coverage and |delta r|, not P







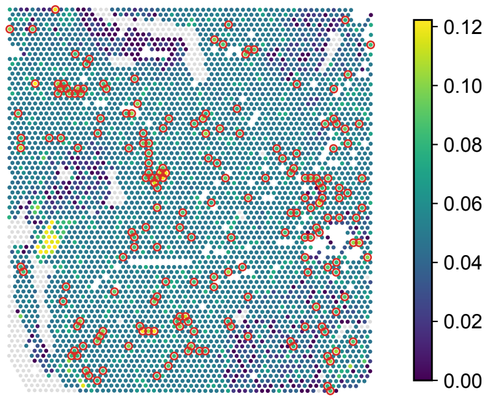


Pericyte / Budden S1P RESPONSE 18: delta r=0.497 [0.353, 0.624]; common 100–2960 µm; spatial P=6.7e-09
Pericyte / Budden CHOLESTEROL SYNTHESIS 27: delta r=-0.431 [-0.573, -0.280]; common 100–2960 µm; spatial P=1.4e-07
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

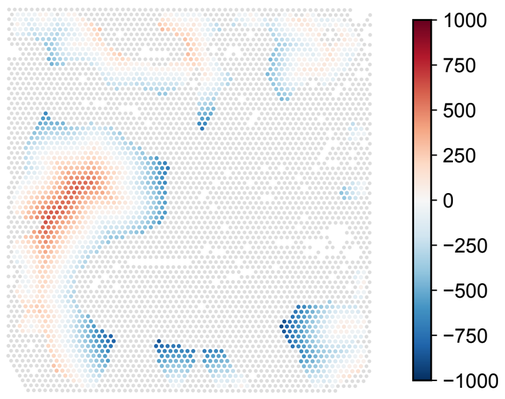
P95 nonP95 distance 0

CanDis_2026_Sample3 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

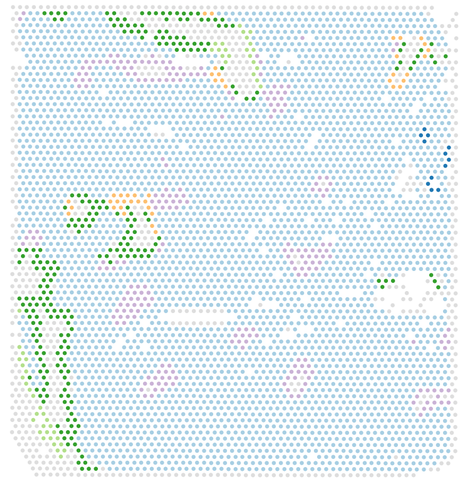
MP10 AUCell and frozen P95



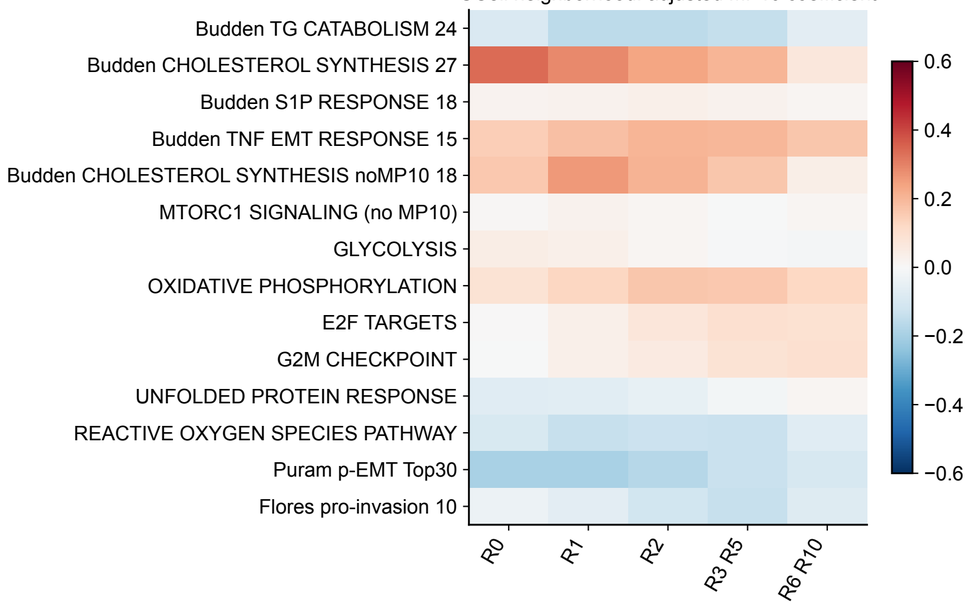
Strict v3.4 signed distance (µm)



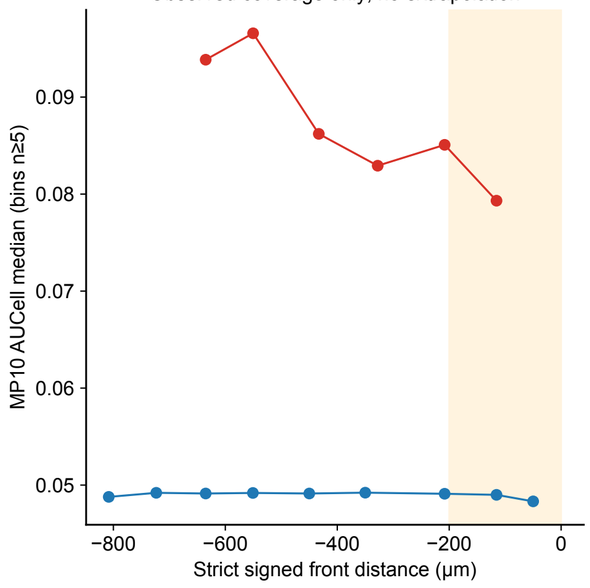
Formal R12 0–8; grey = unlabelled



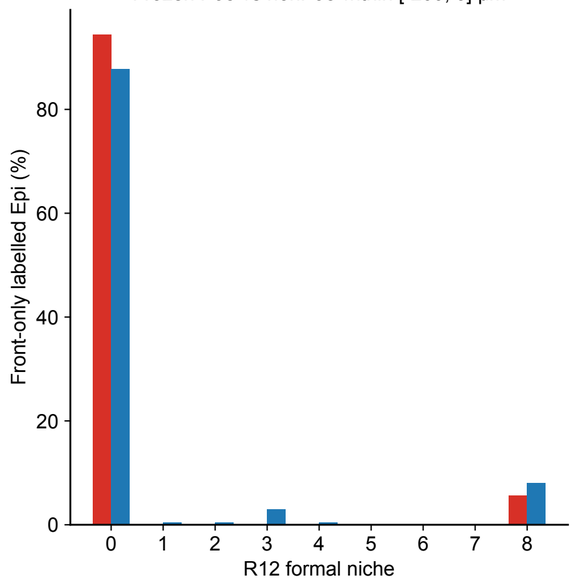
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

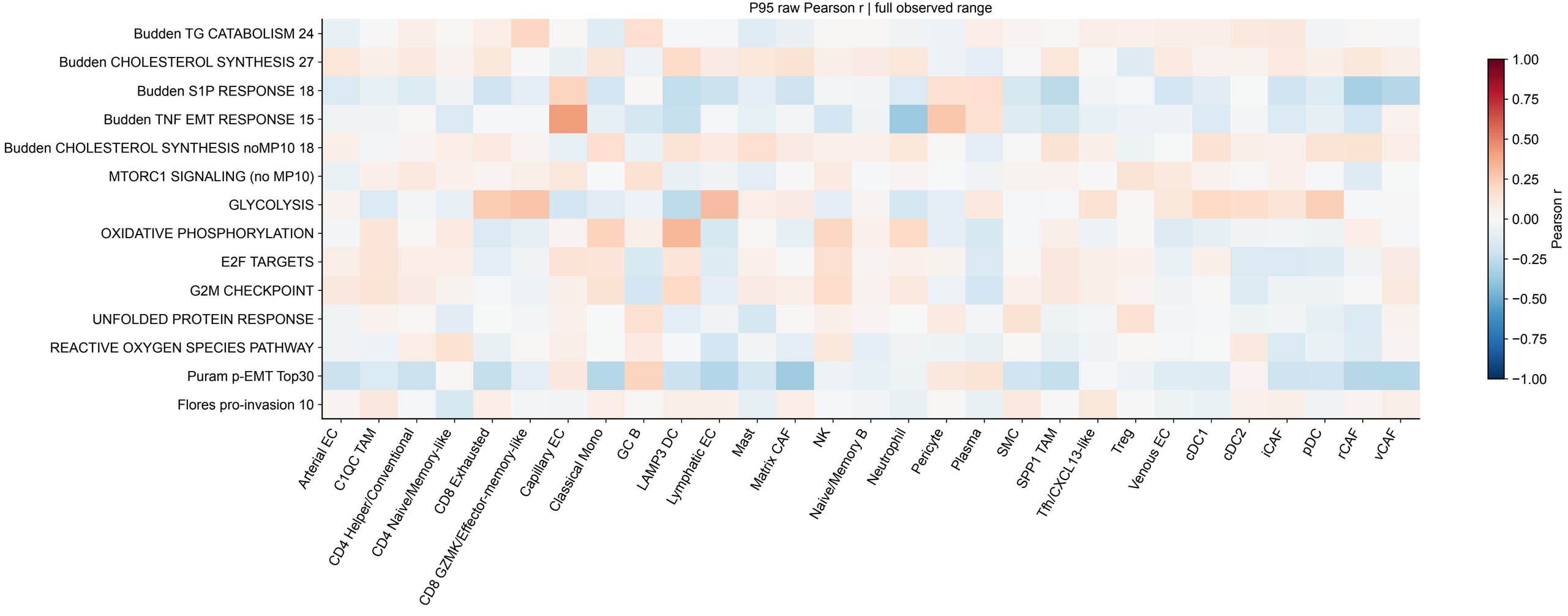


Tissue 4818 | Epi 4553 | internal 3805 | P95 191
Exact front in Epi 1607 | R12 labelled 4239

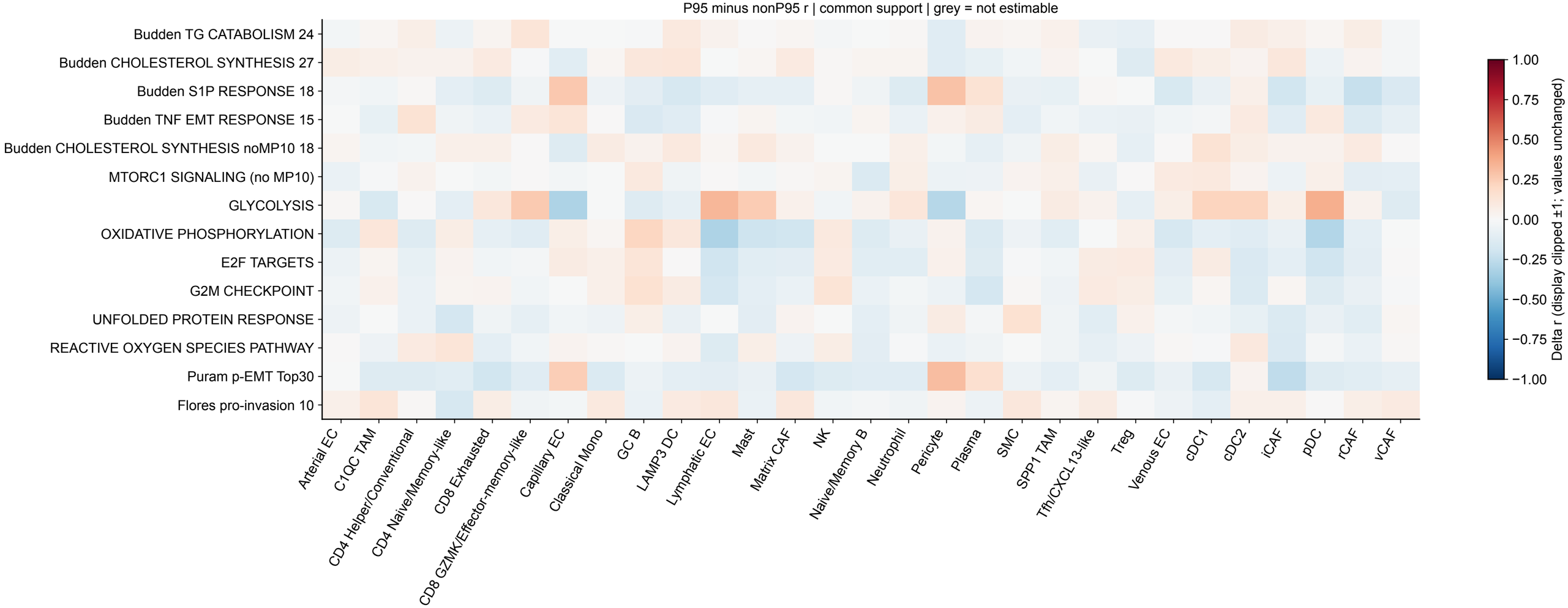
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

CanDis_2026_Sample3 | all 29 candidates; fixed core programs

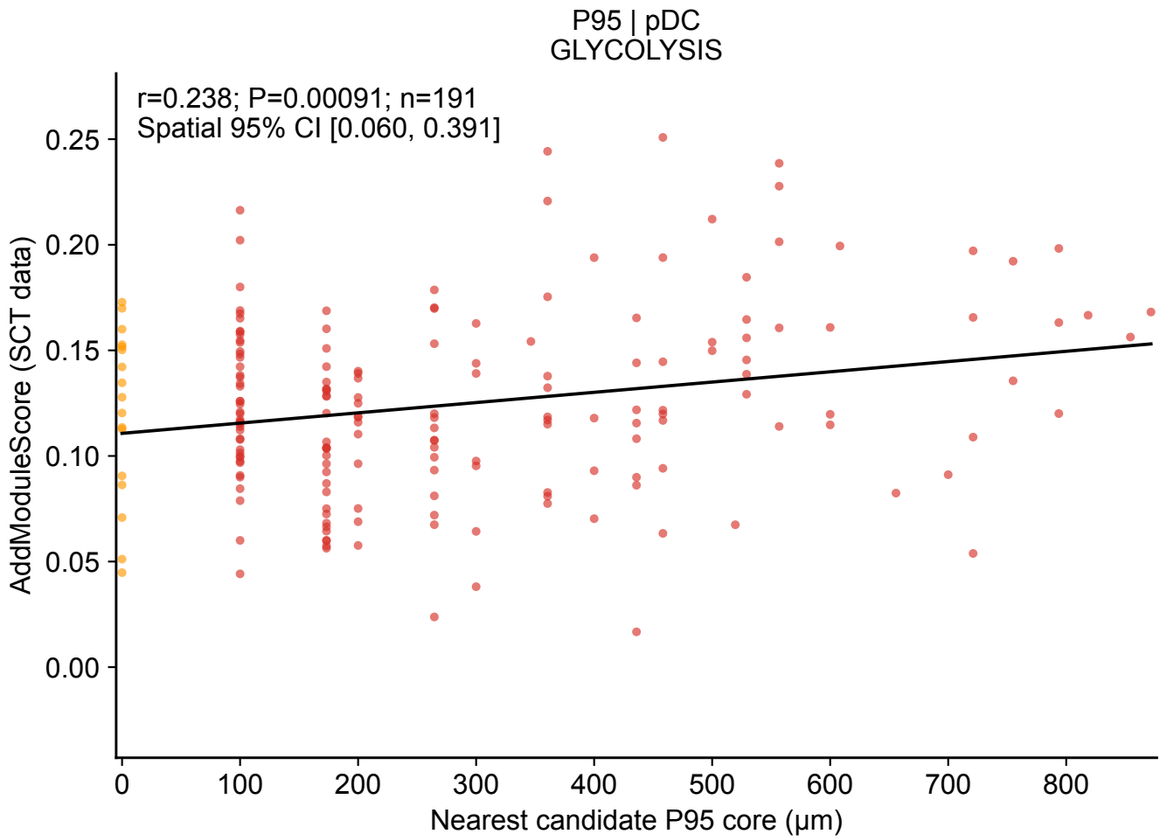
P95 raw Pearson r | full observed range

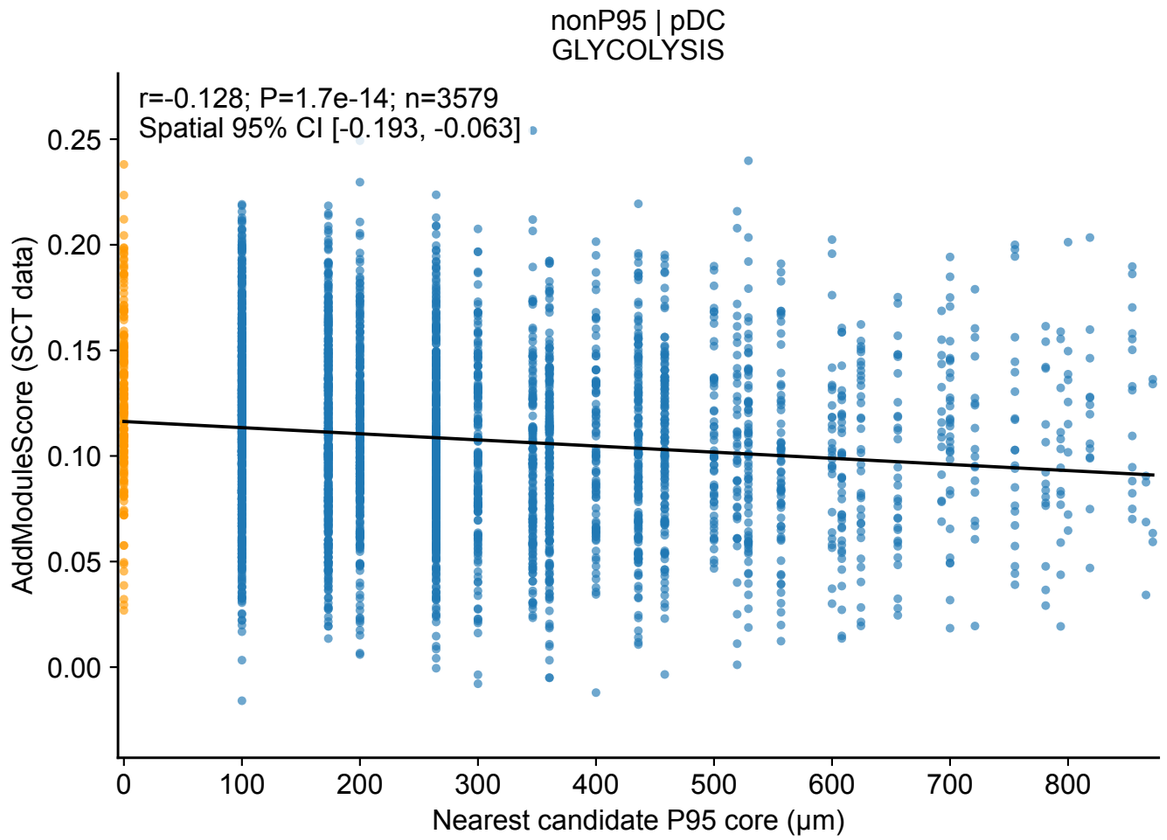


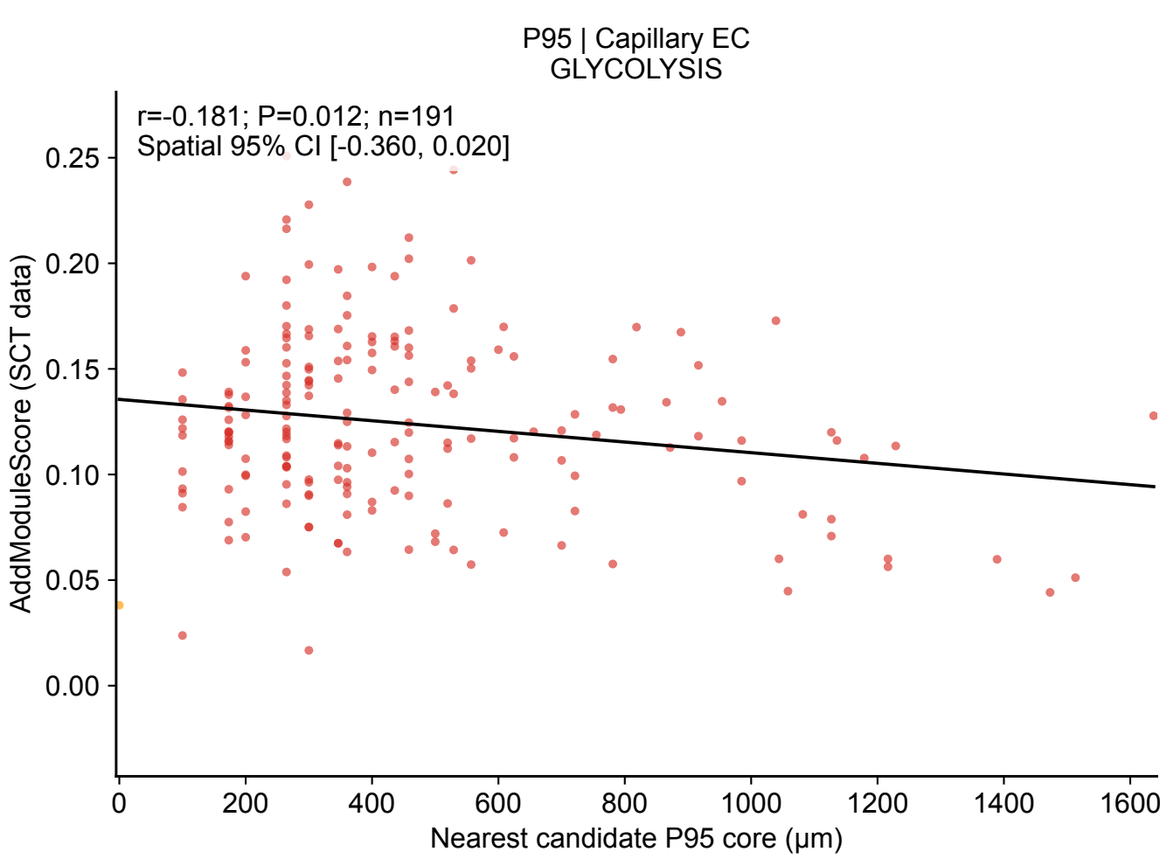
P95 minus nonP95 r | common support | grey = not estimable

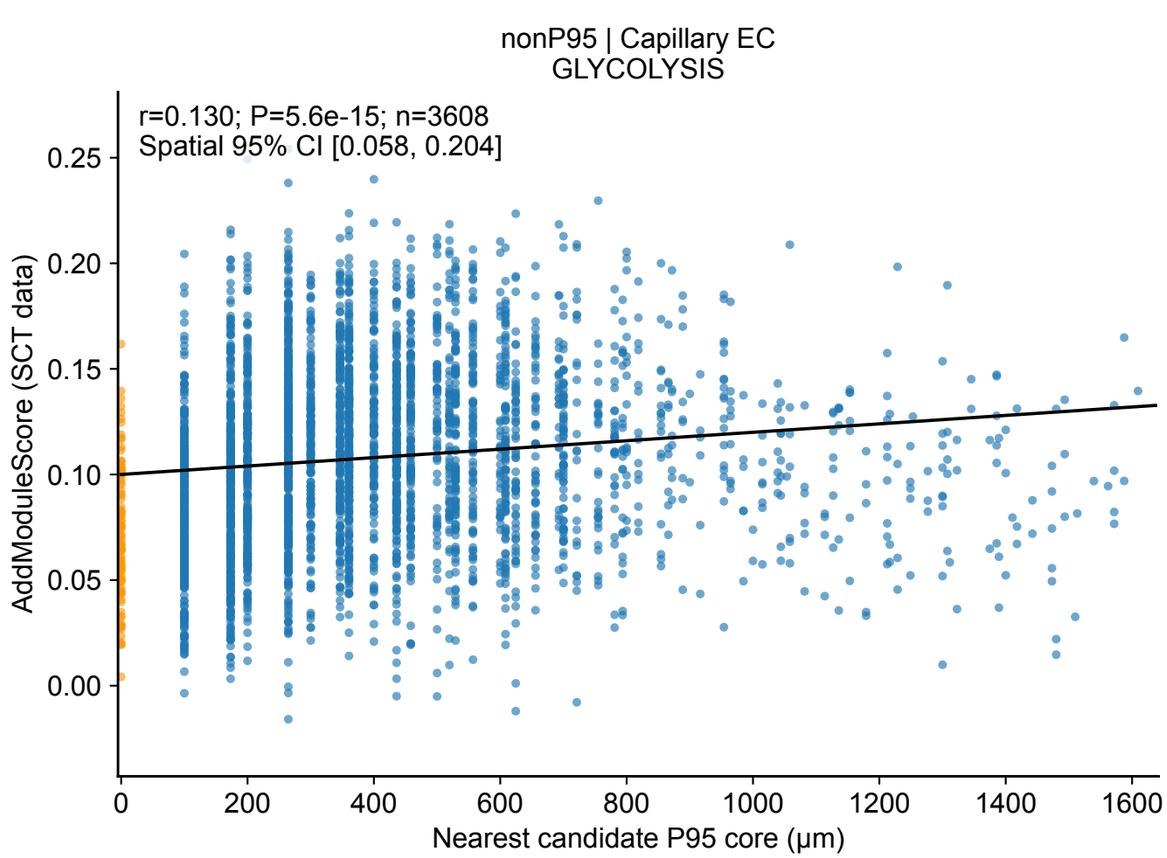


CanDis_2026_Sample3 | exploratory scatter display selected by coverage and |delta r|, not P







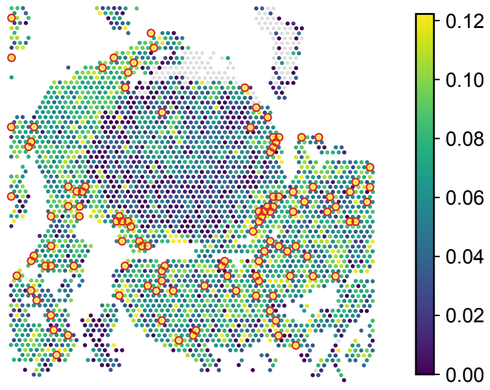


pDC / GLYCOLYSIS: delta r=0.366 [0.198, 0.521]; common 0–872 µm; spatial P=2.8e-05
Capillary EC / GLYCOLYSIS: delta r=-0.311 [-0.491, -0.114]; common 0–1637 µm; spatial P=0.0028
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

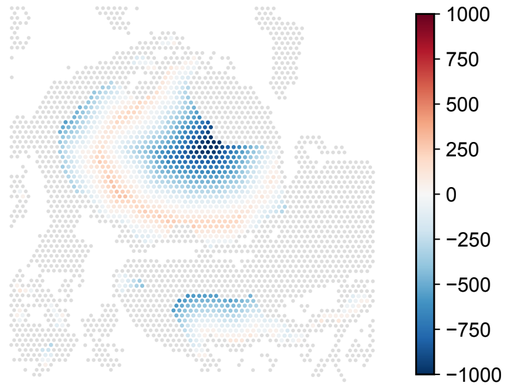
P95 nonP95 distance 0

CanDis_2026_Sample4 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

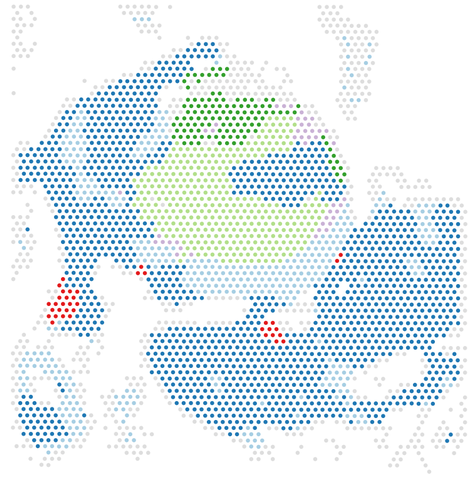
MP10 AUCell and frozen P95



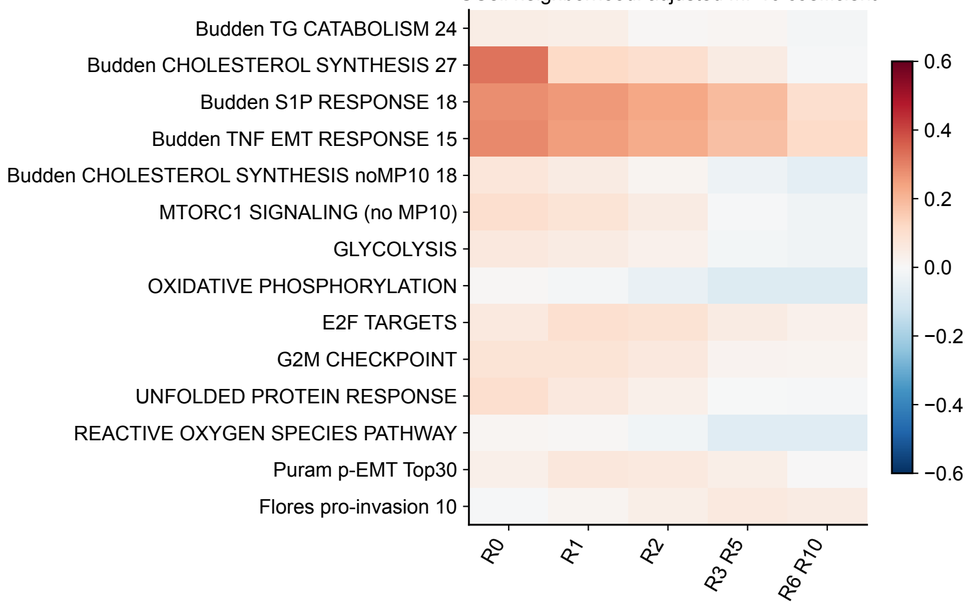
Strict v3.4 signed distance (µm)



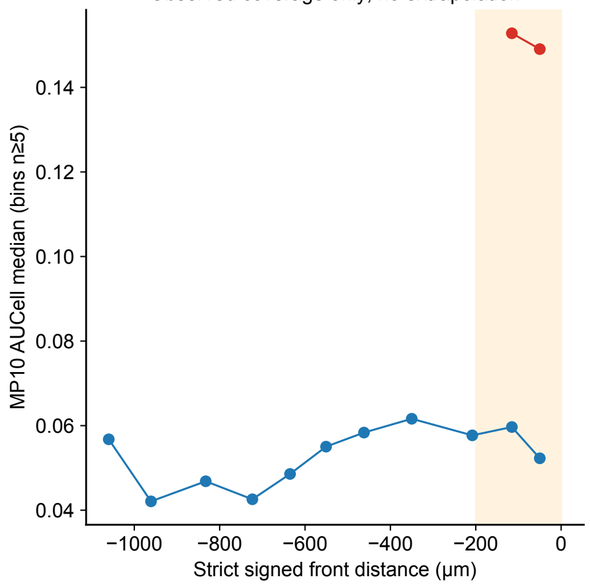
Formal R12 0–8; grey = unlabelled



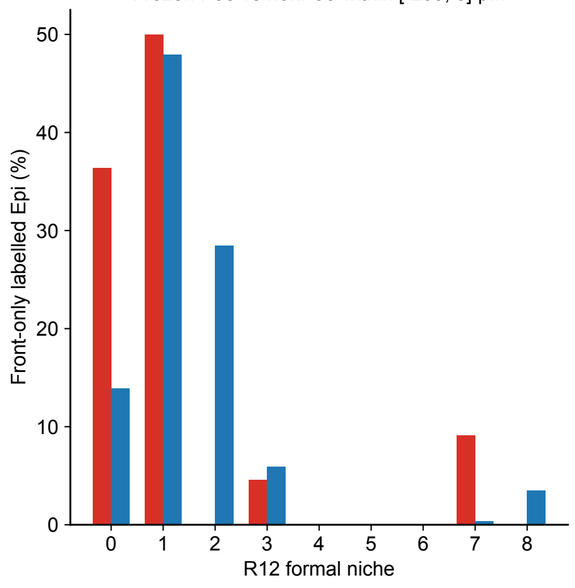
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

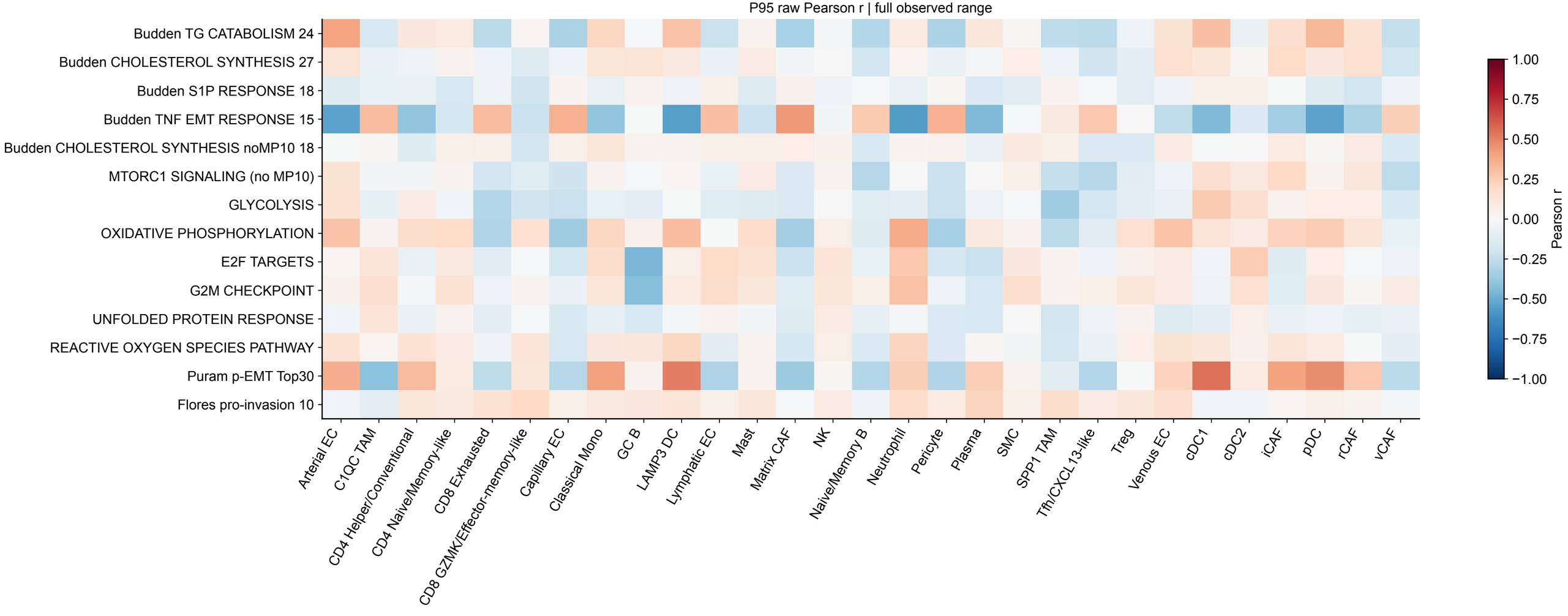


Tissue 3118 | Epi 2997 | internal 2286 | P95 115
Exact front in Epi 1114 | R12 labelled 2417

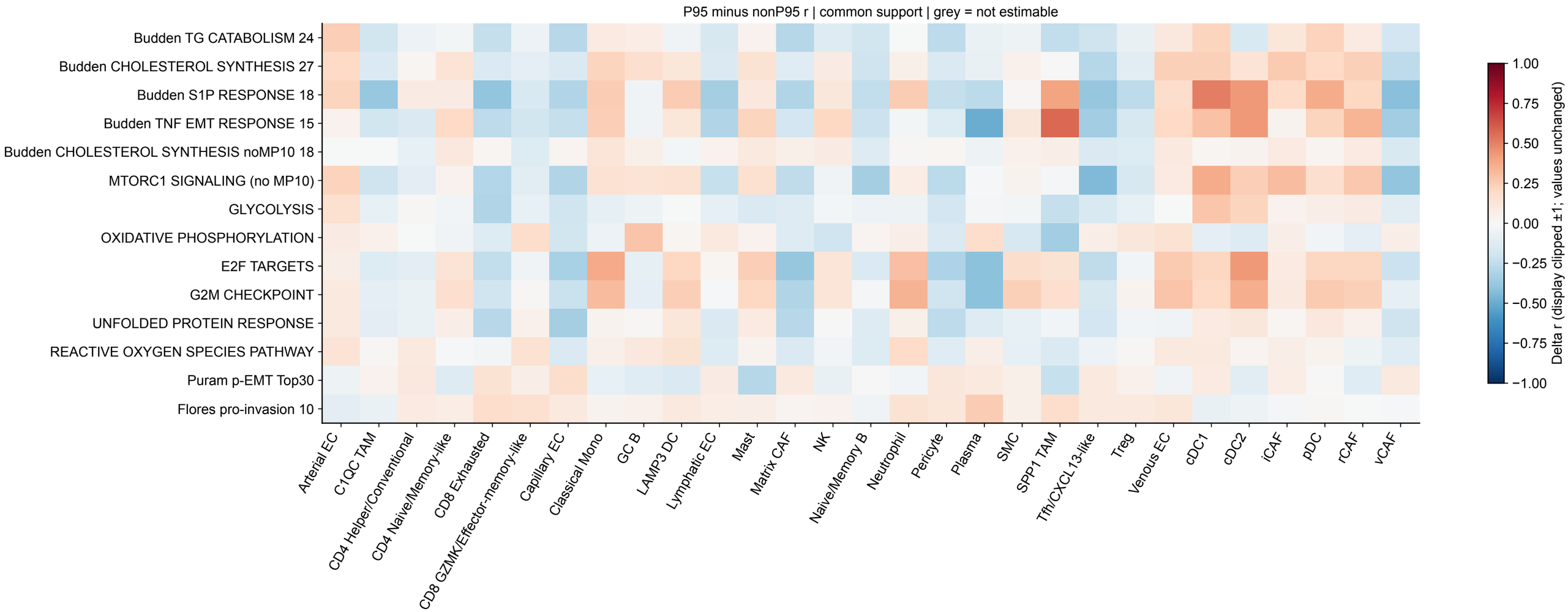
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

CanDis_2026_Sample4 | all 29 candidates; fixed core programs

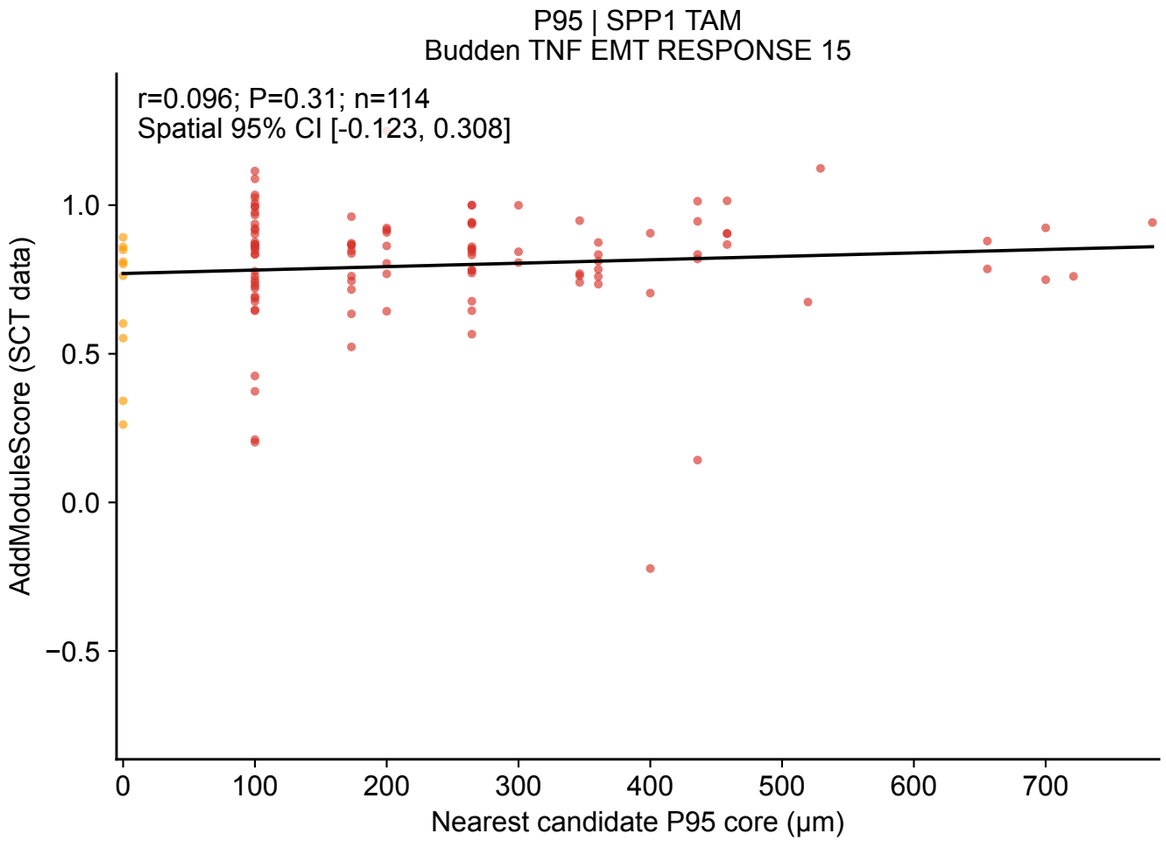
P95 raw Pearson r | full observed range

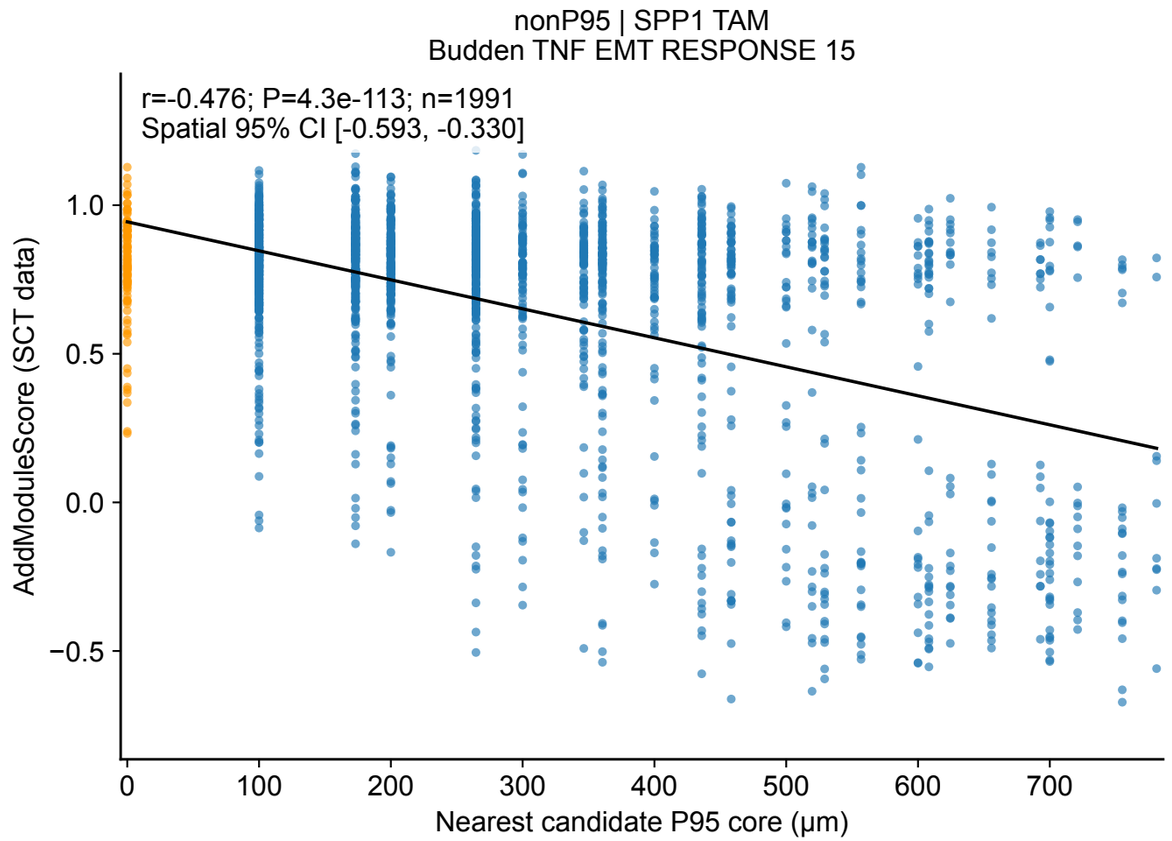


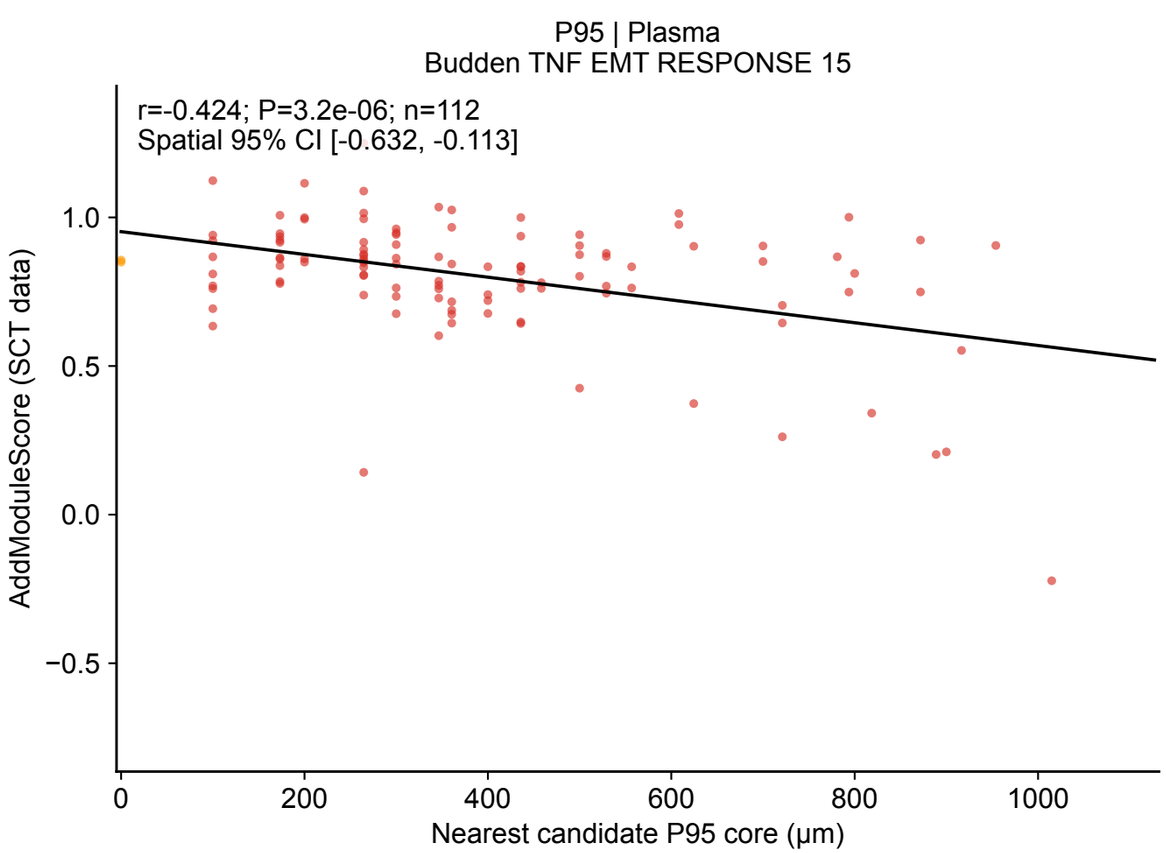
P95 minus nonP95 r | common support | grey = not estimable

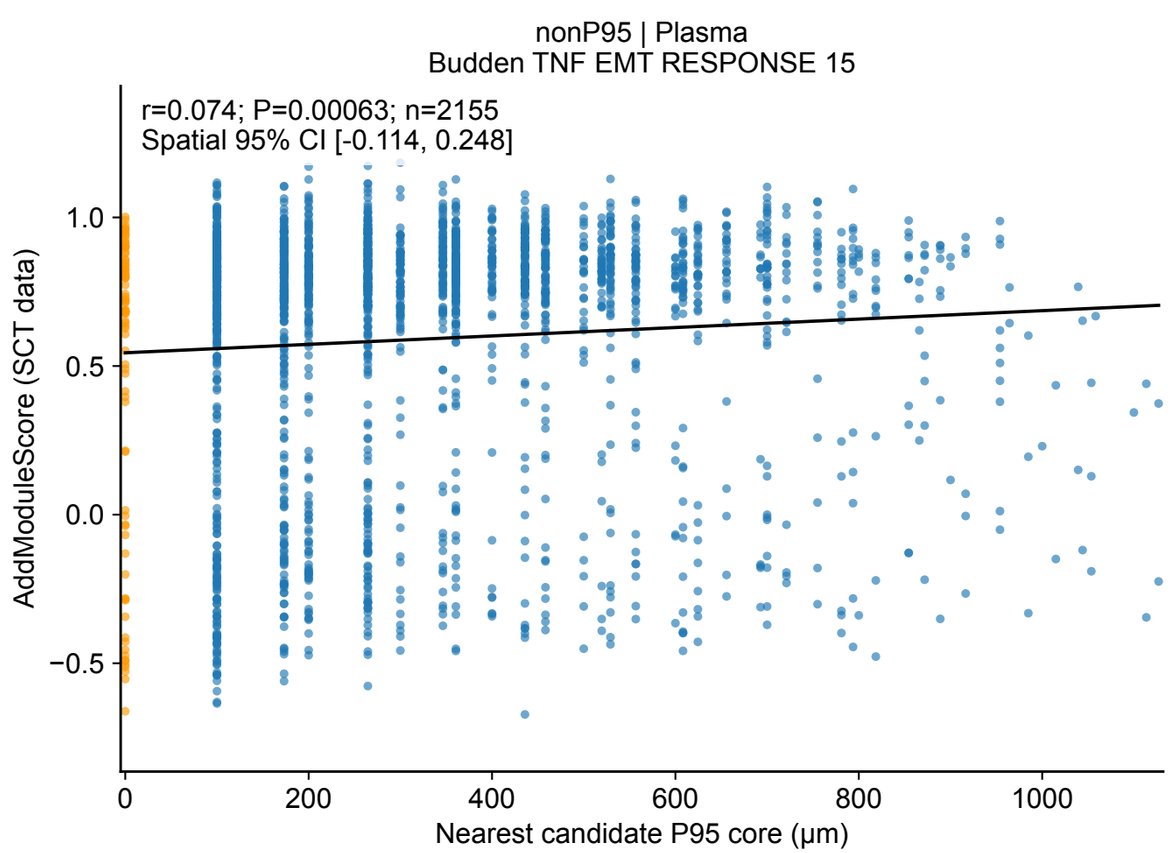


CanDis_2026_Sample4 | exploratory scatter display selected by coverage and |delta r|, not P







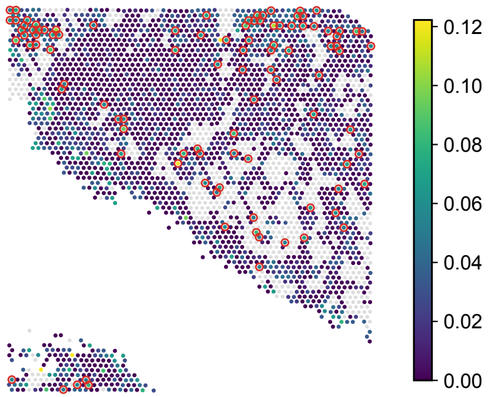


SPP1 TAM / Budden TNF EMT RESPONSE 15: delta r=0.572 [0.342, 0.796]; common 0–781 µm; spatial P=8.6e-06
Plasma / Budden TNF EMT RESPONSE 15: delta r=-0.497 [-0.716, -0.192]; common 0–1127 µm; spatial P=0.00072
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

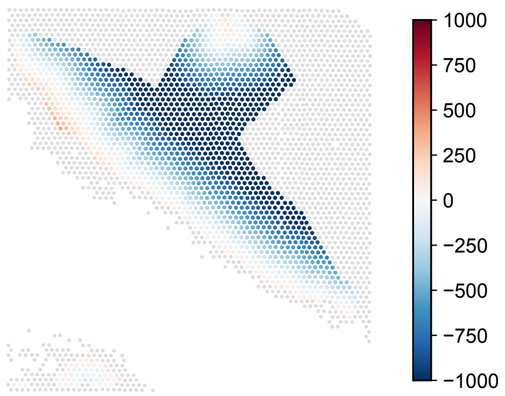
P95 nonP95 distance 0

CanDis_2026_Sample5 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

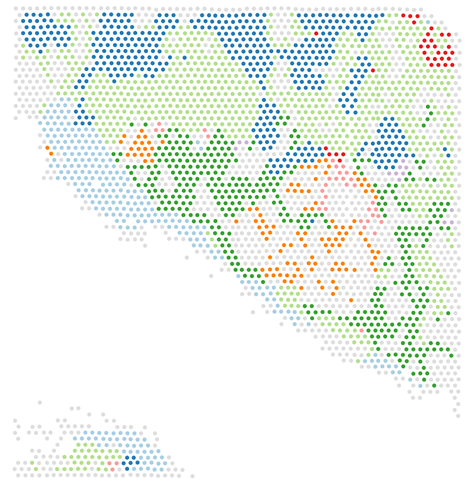
MP10 AUCell and frozen P95



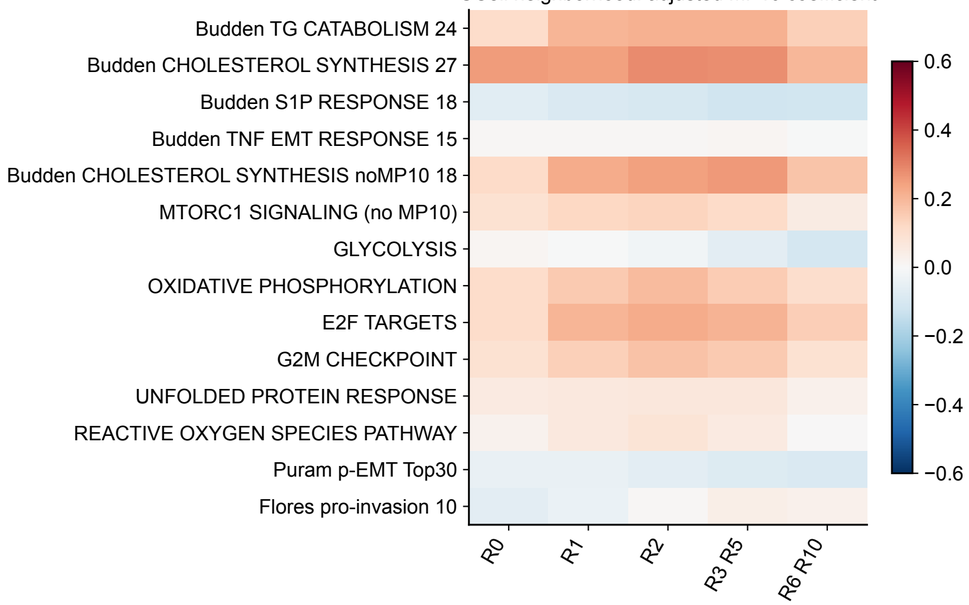
Strict v3.4 signed distance (µm)



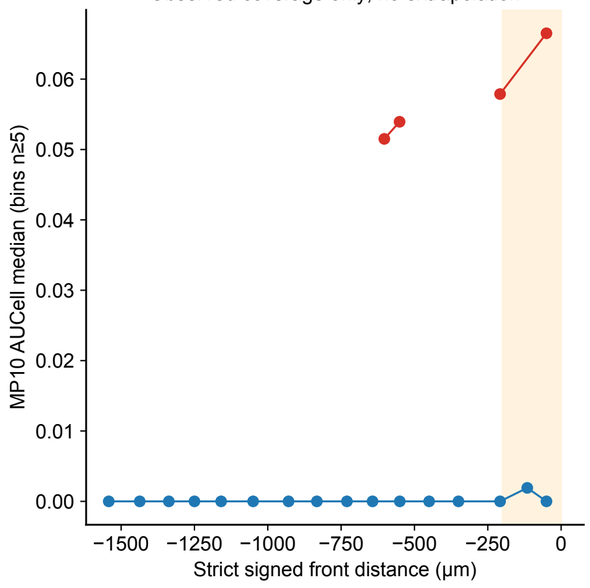
Formal R12 0–8; grey = unlabelled



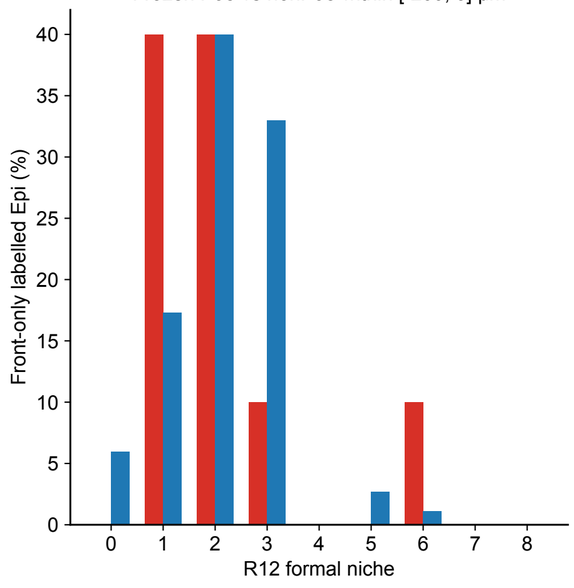
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



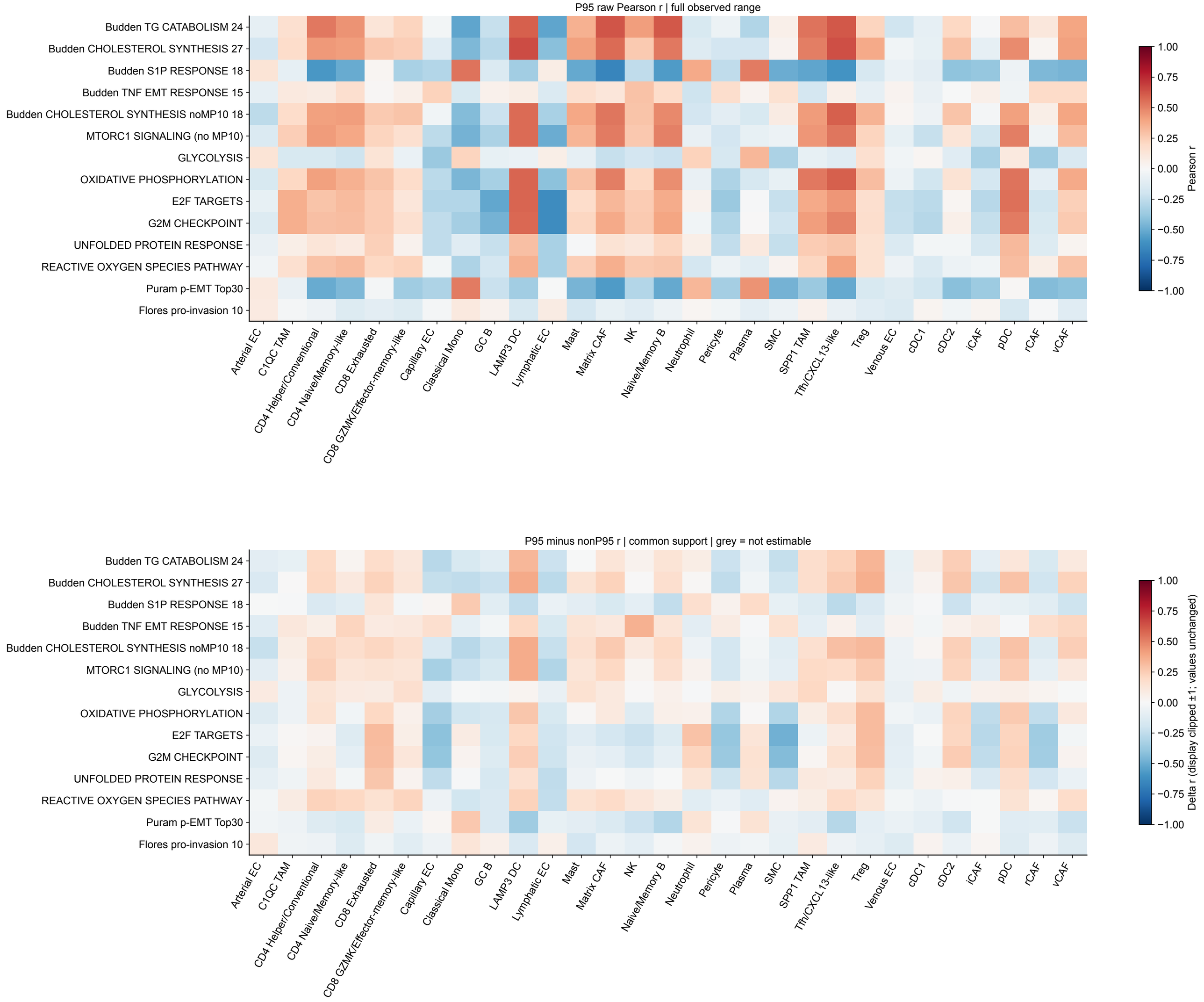
Frozen P95 vs nonP95 within [-200, 0] µm



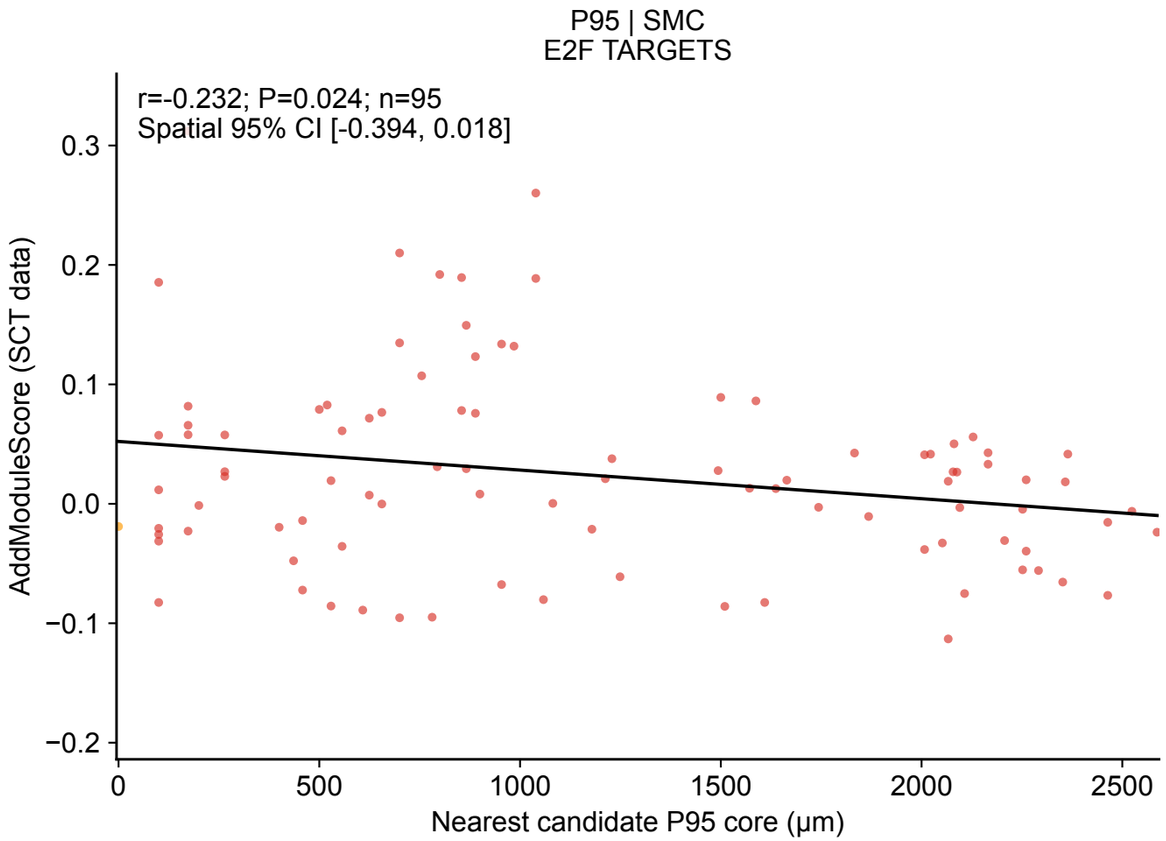
Tissue 3125 | Epi 2455 | internal 1983 | P95 100
Exact front in Epi 1186 | R12 labelled 2120

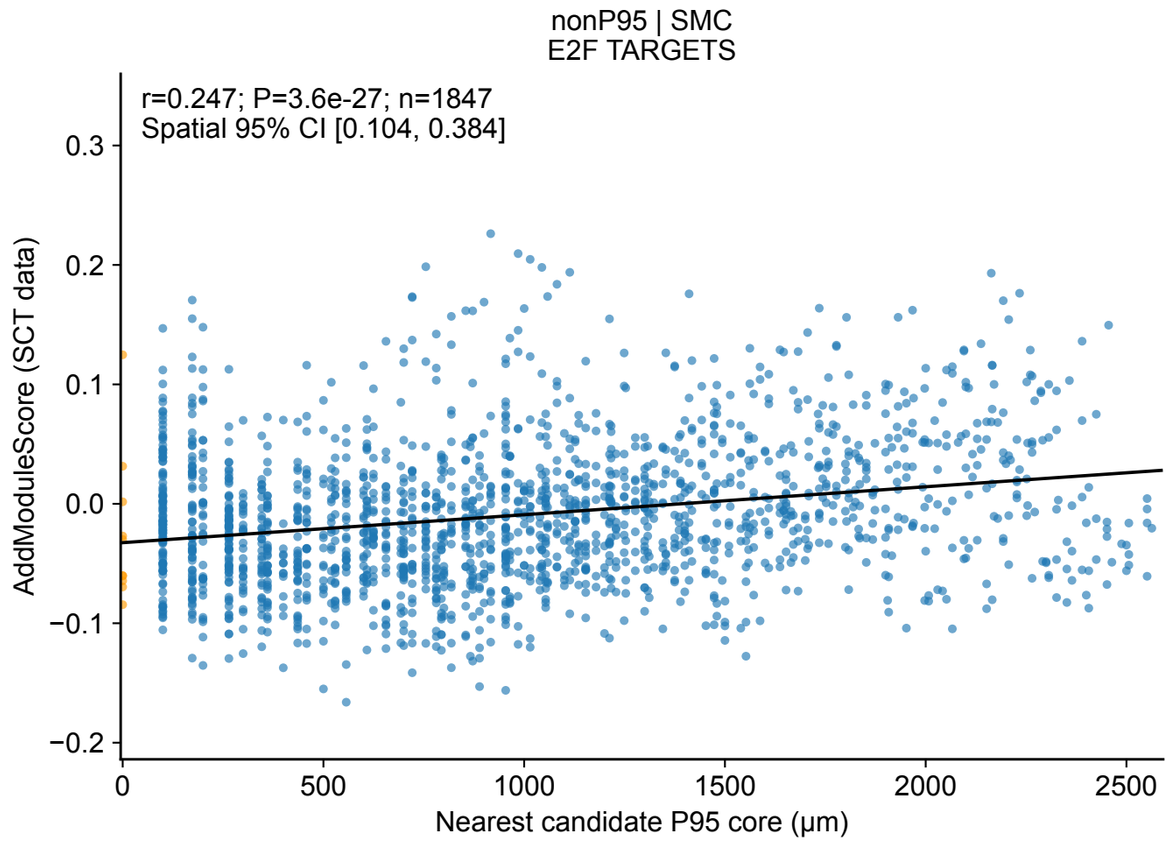
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

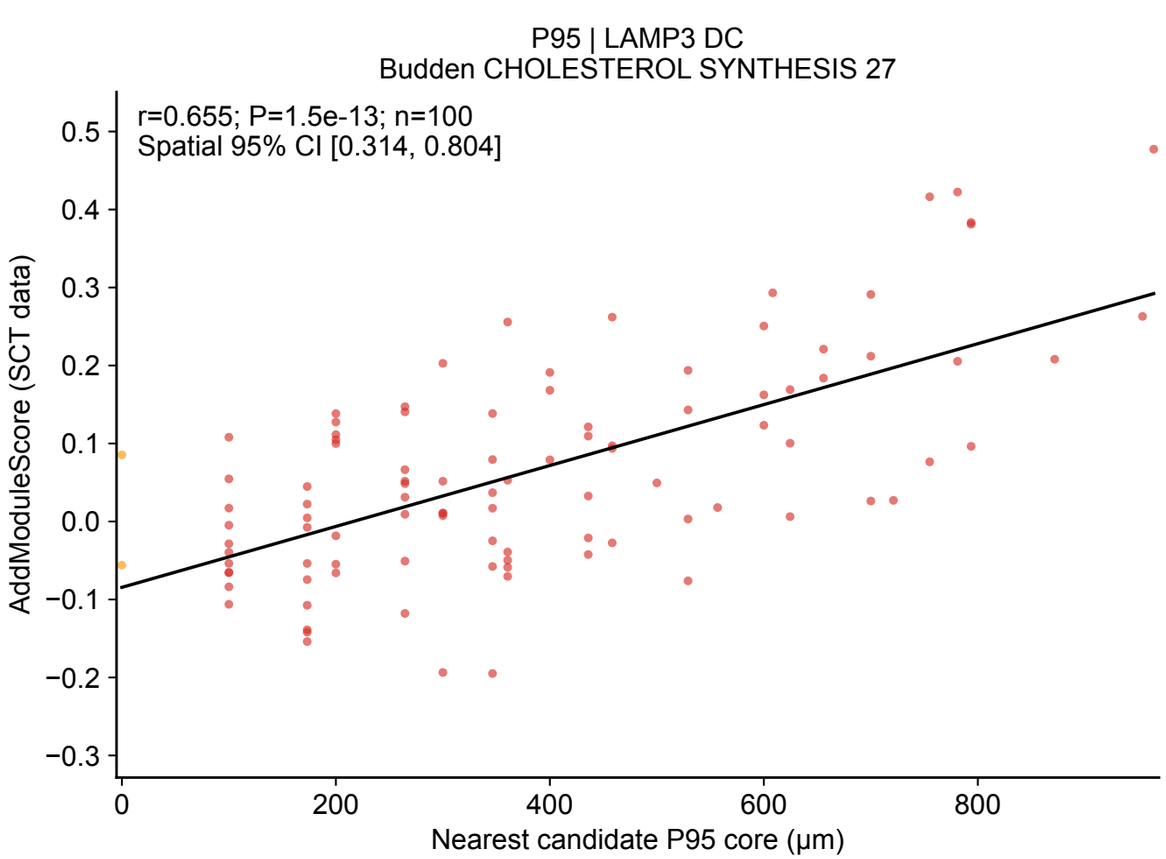
CanDis_2026_Sample5 | all 29 candidates; fixed core programs

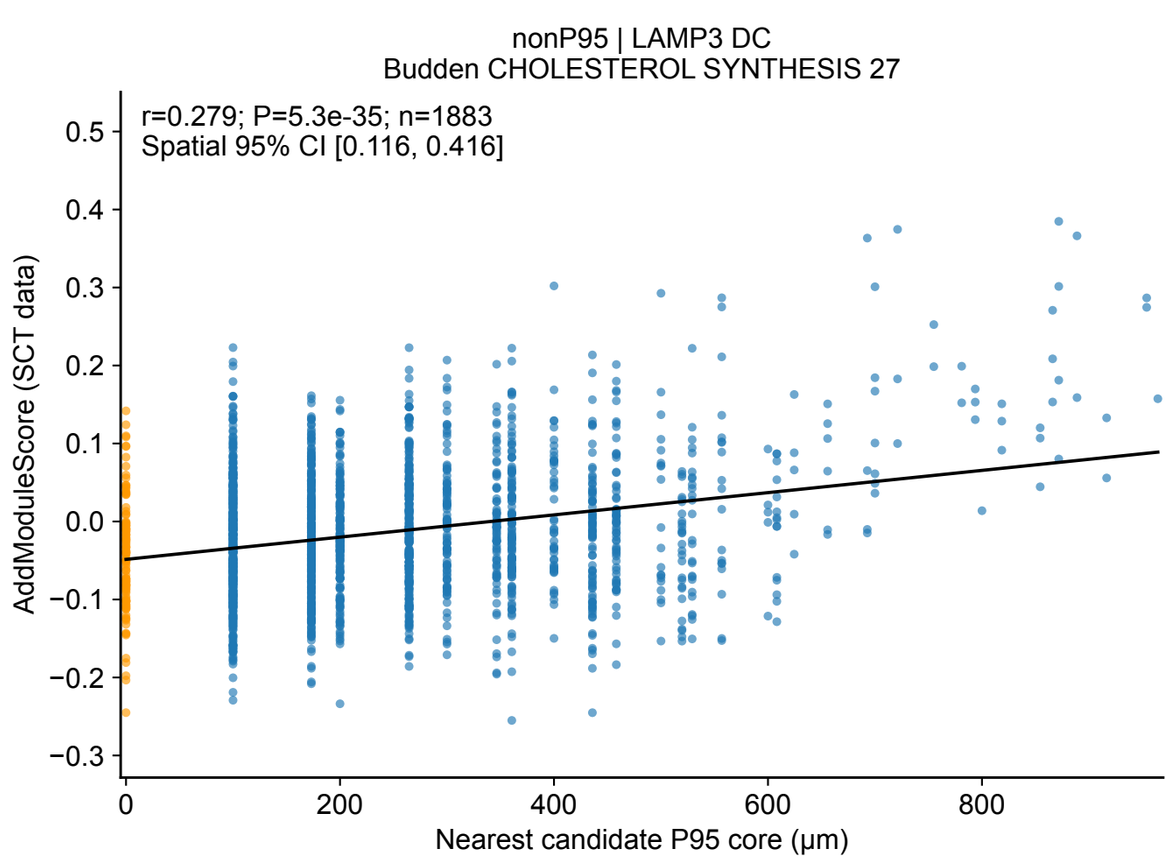


CanDis_2026_Sample5 | exploratory scatter display selected by coverage and |delta r|, not P







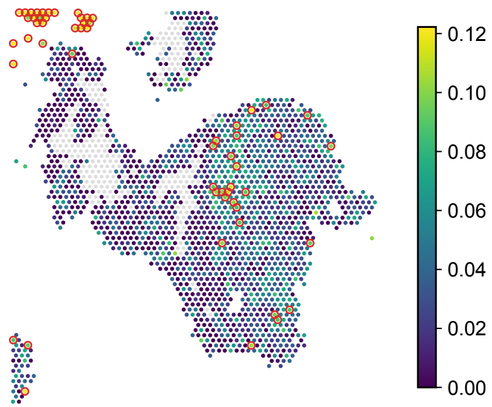


SMC / E2F TARGETS: delta r=-0.480 [-0.646, -0.234]; common 0–2587 µm; spatial P=3.9e-06
LAMP3 DC / Budden CHOLESTEROL SYNTHESIS 27: delta r=0.376 [0.135, 0.512]; common 0–964 µm; spatial P=0.0026
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

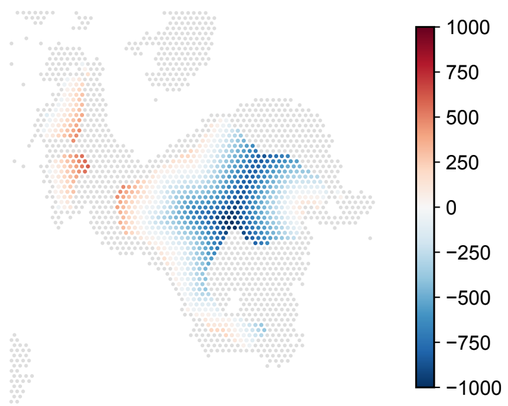
P95 nonP95 distance 0

CanDis_2026_Sample6 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

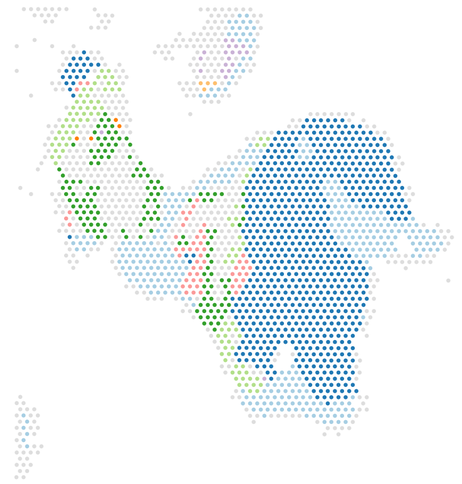
MP10 AUCell and frozen P95



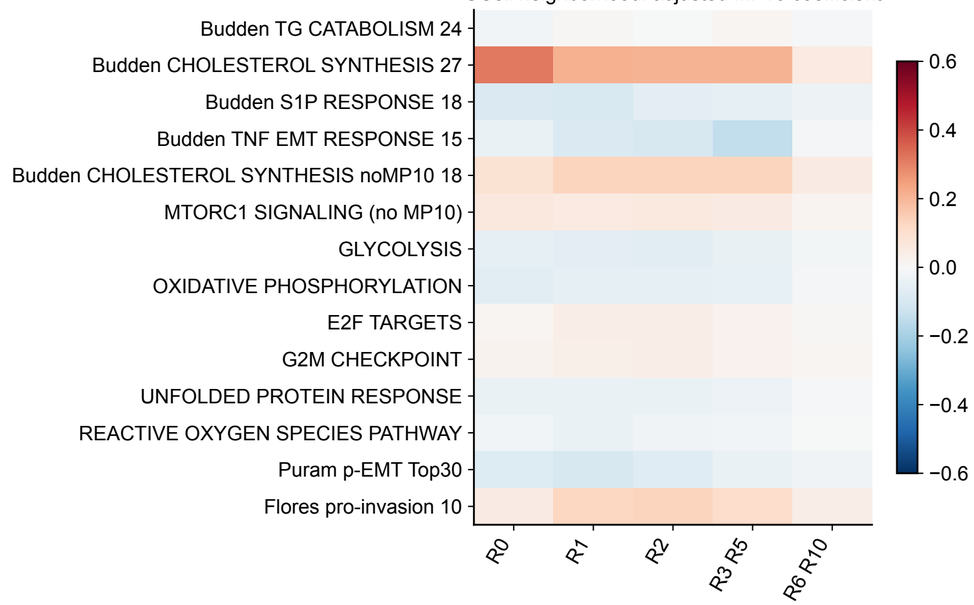
Strict v3.4 signed distance (µm)



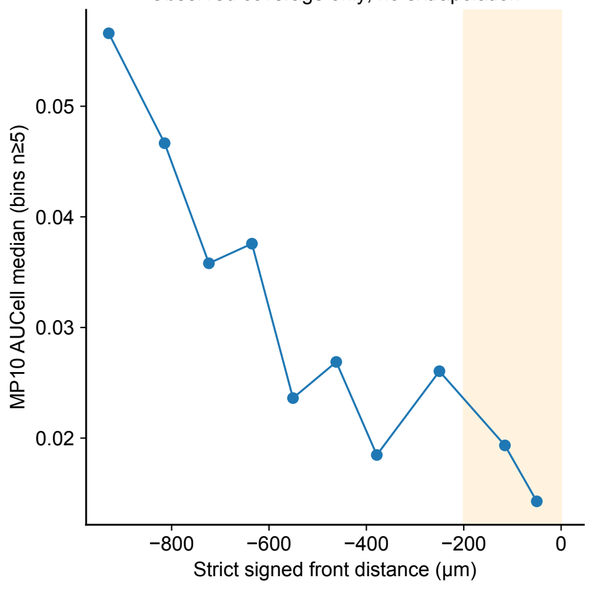
Formal R12 0–8; grey = unlabelled



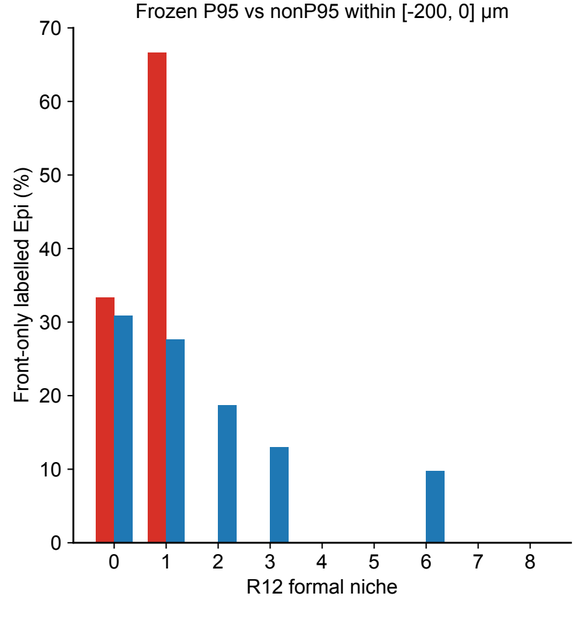
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

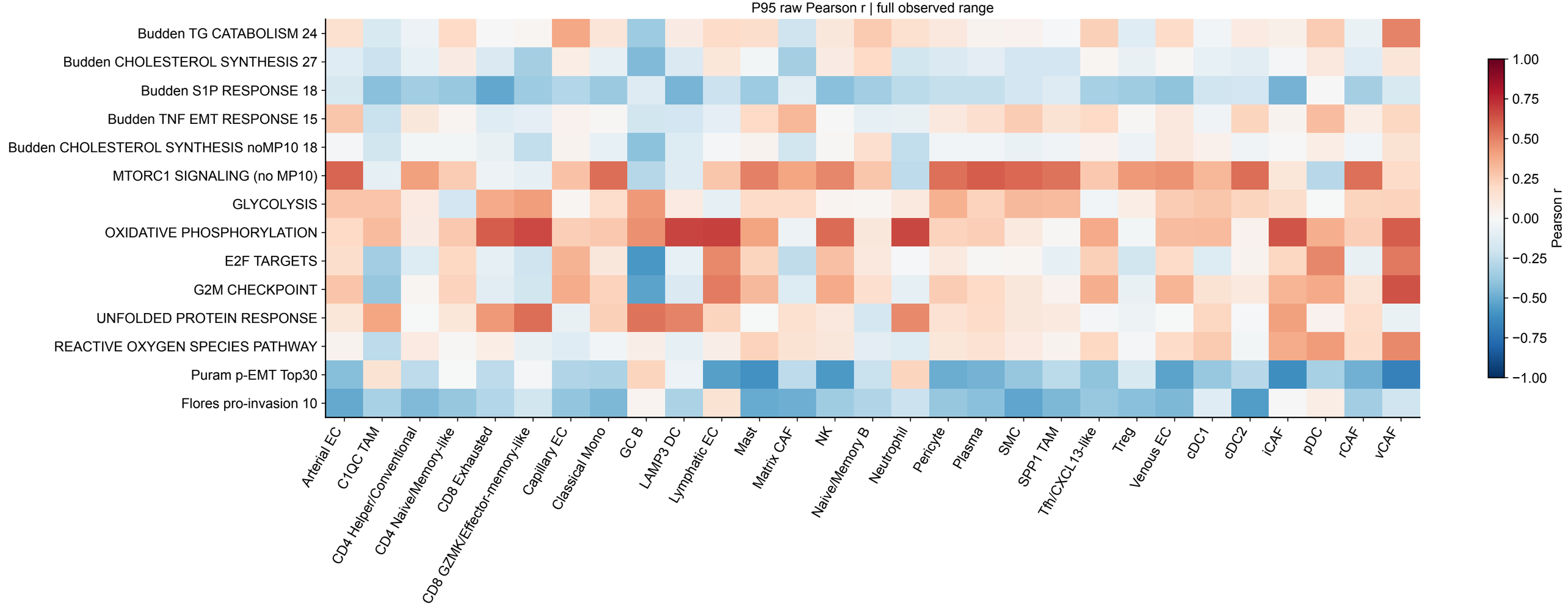


Tissue 1985 | Epi 1753 | internal 1138 | P95 57
Exact front in Epi 729 | R12 labelled 1407

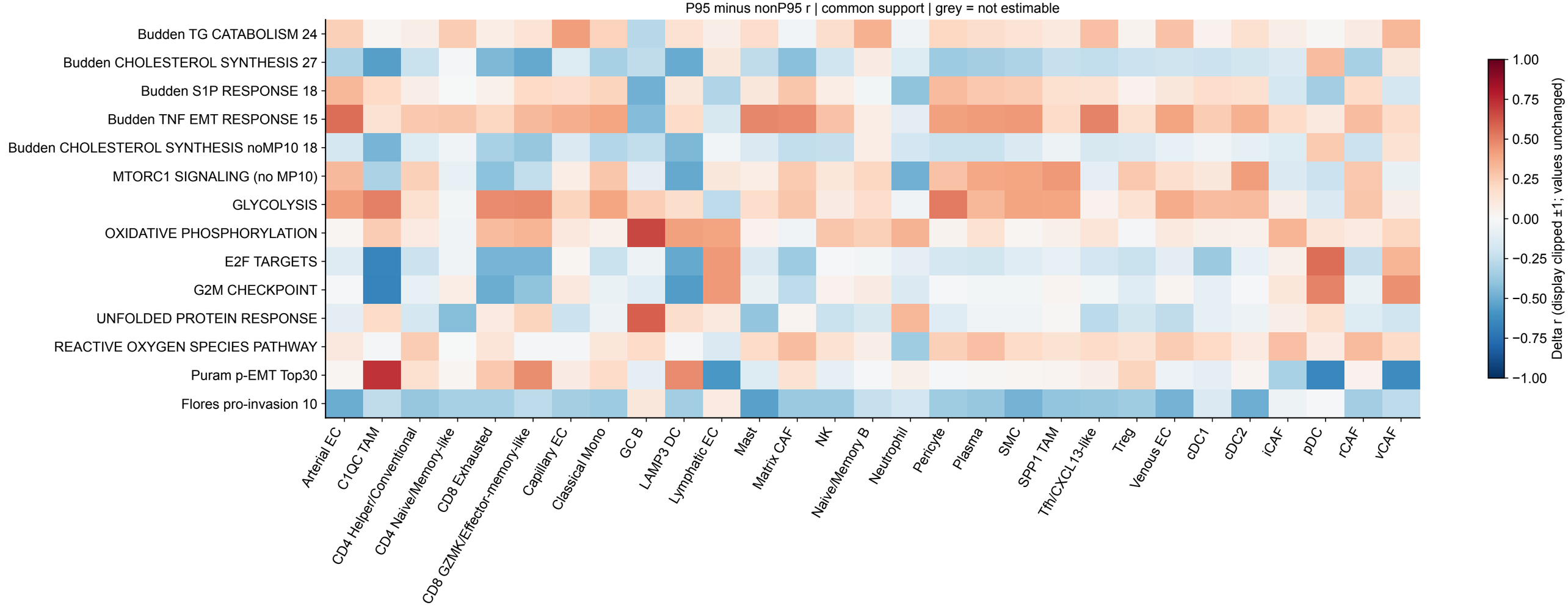
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

CanDis_2026_Sample6 | all 29 candidates; fixed core programs

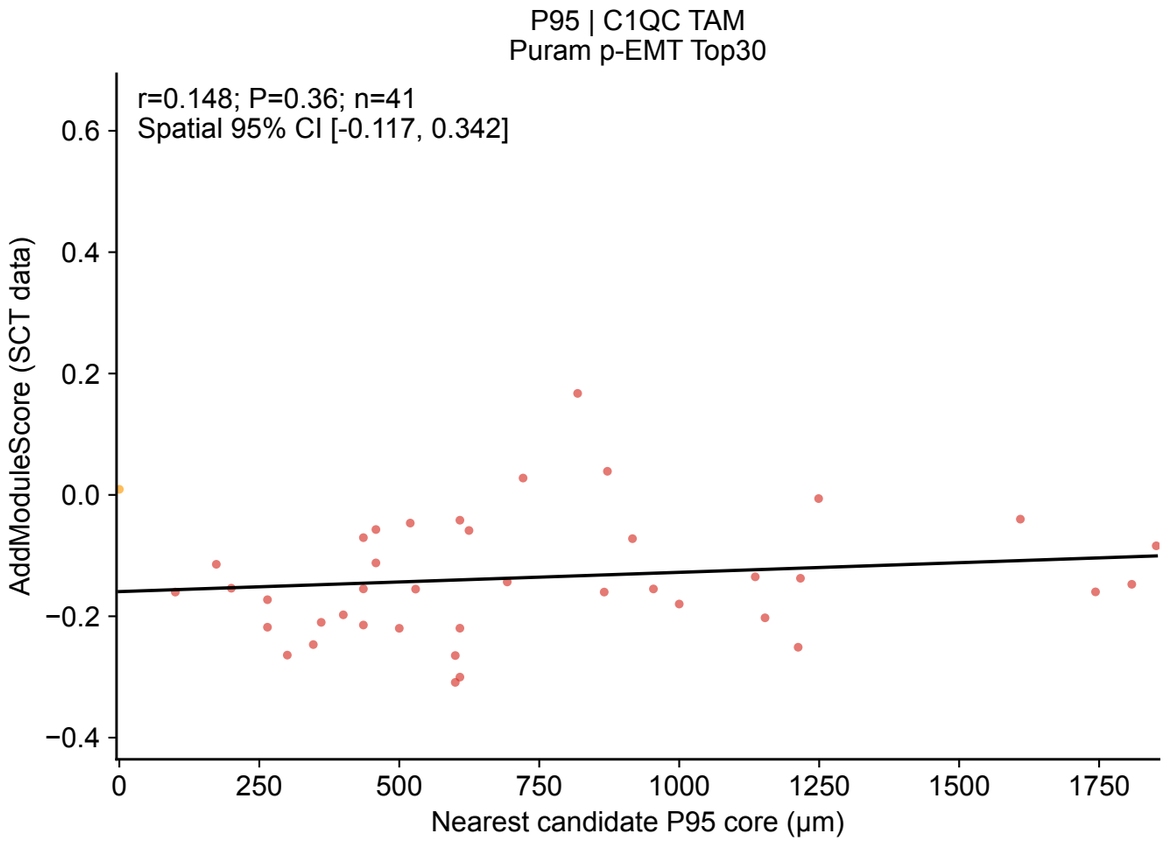
P95 raw Pearson r | full observed range

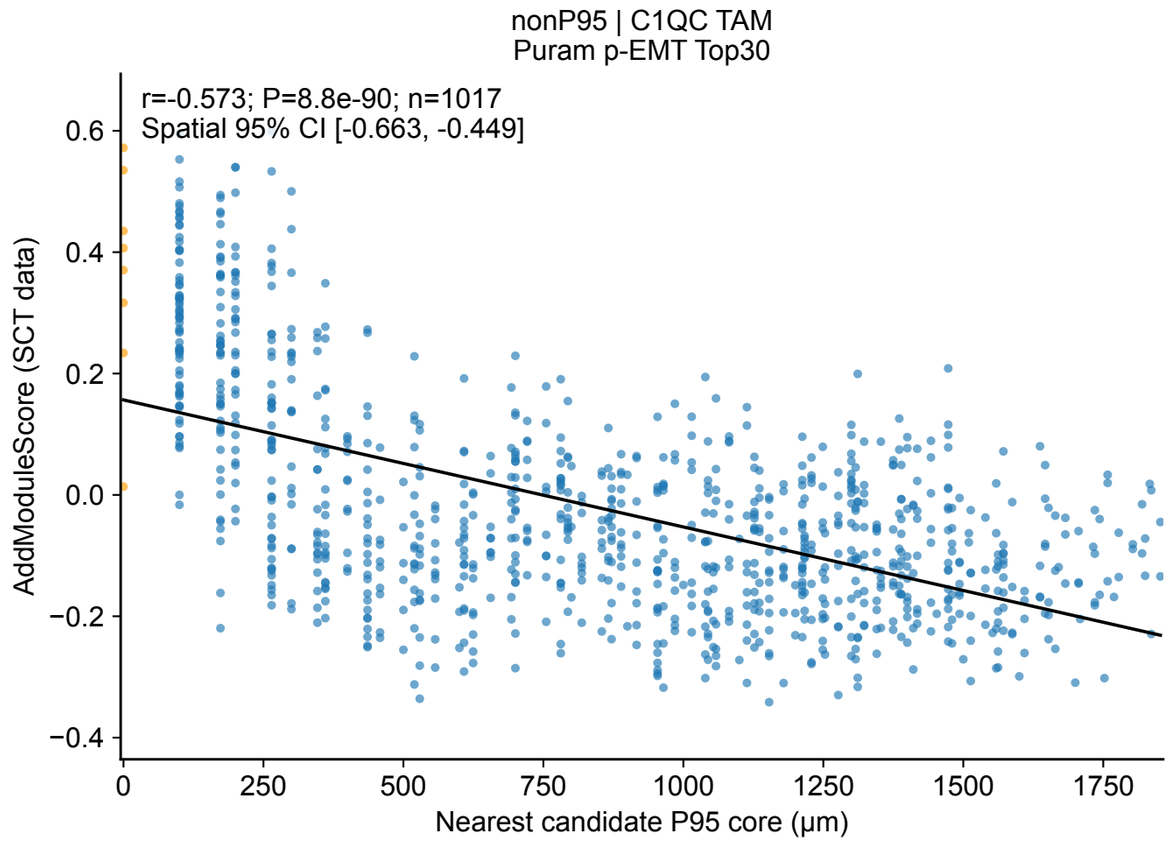


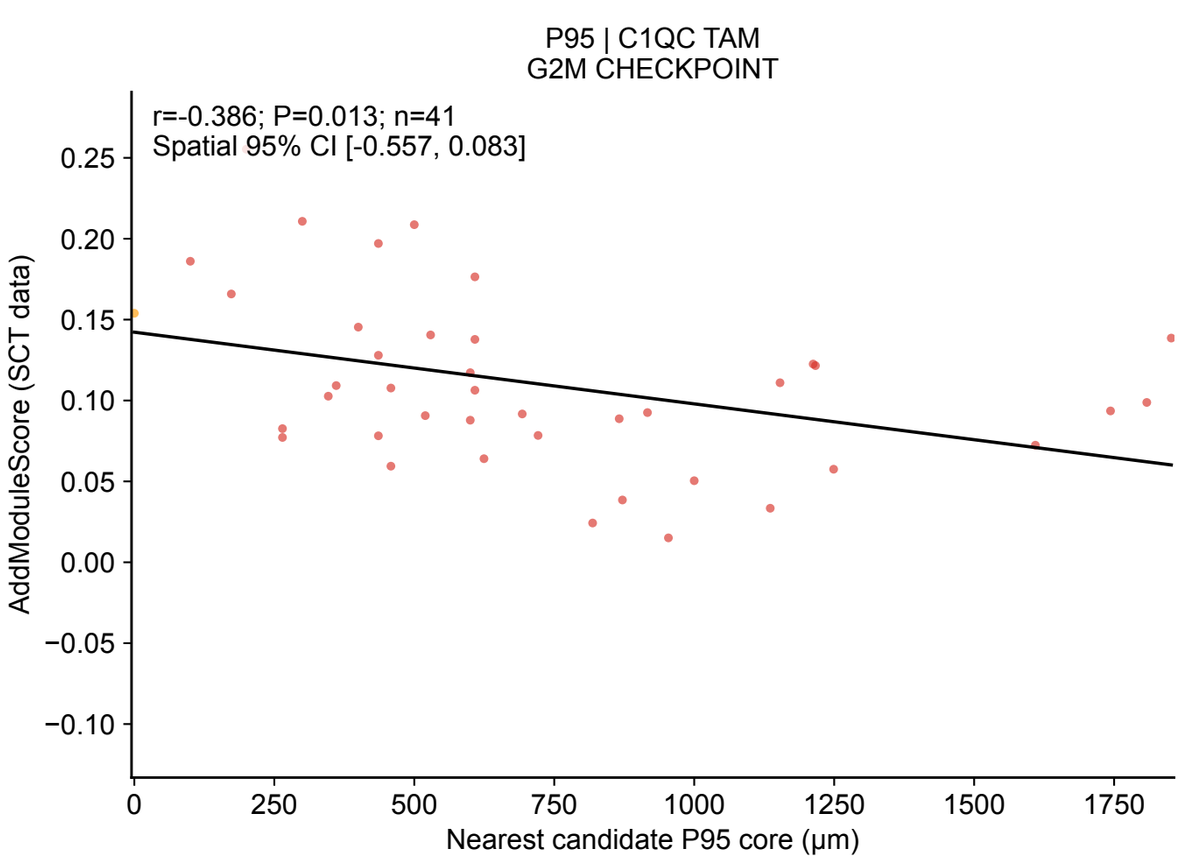
P95 minus nonP95 r | common support | grey = not estimable

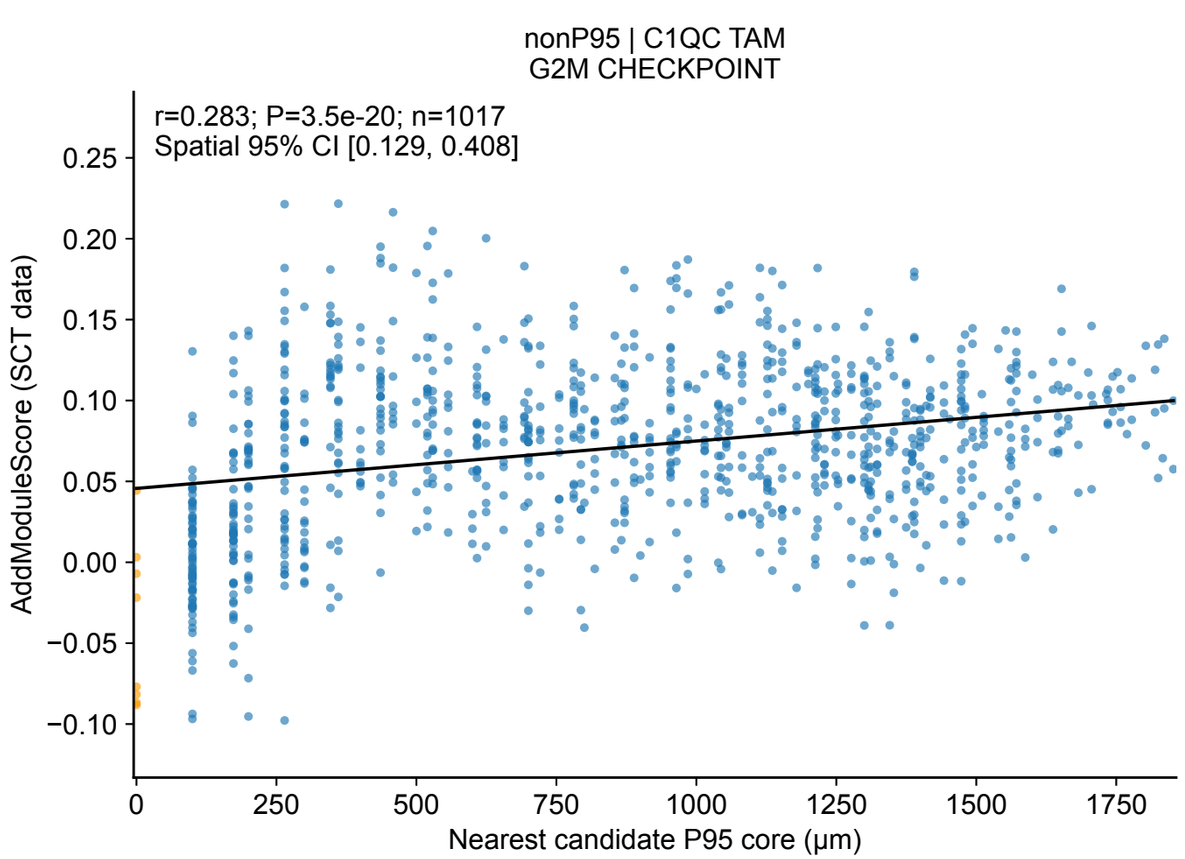


CanDis_2026_Sample6 | exploratory scatter display selected by coverage and |delta r|, not P







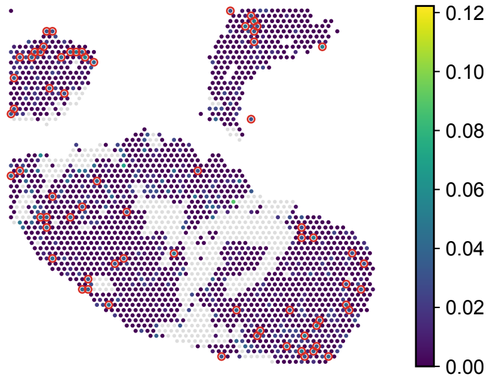


C1QC TAM / Puram p-EMT Top30: delta r=0.721 [0.458, 0.928]; common 0–1852 µm; spatial P=1.1e-09
C1QC TAM / G2M CHECKPOINT: delta r=-0.669 [-0.873, -0.219]; common 0–1852 µm; spatial P=1.3e-05
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

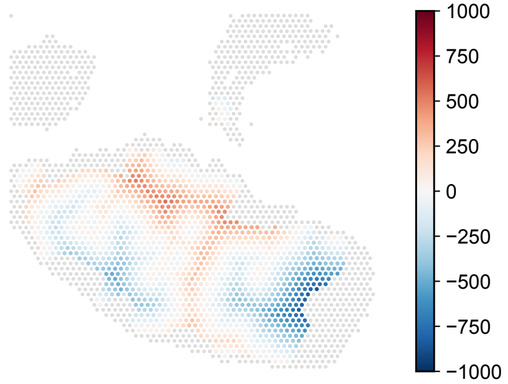
P95 nonP95 distance 0

CanDis_2026_Sample7 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

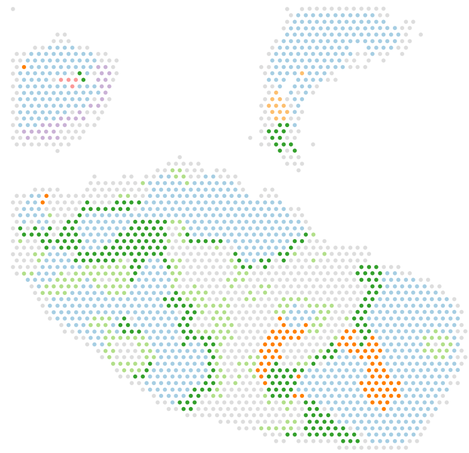
MP10 AUCell and frozen P95



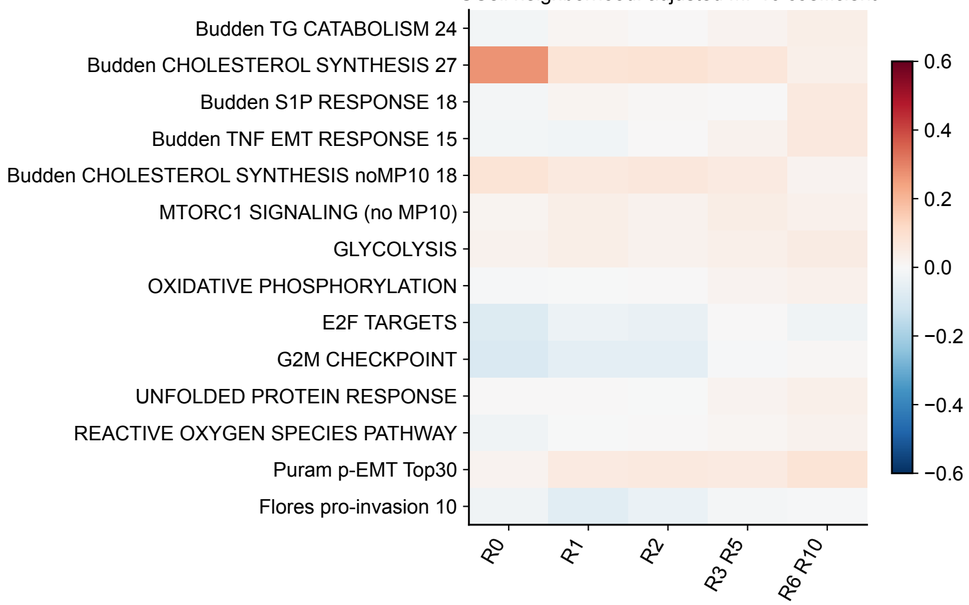
Strict v3.4 signed distance (µm)



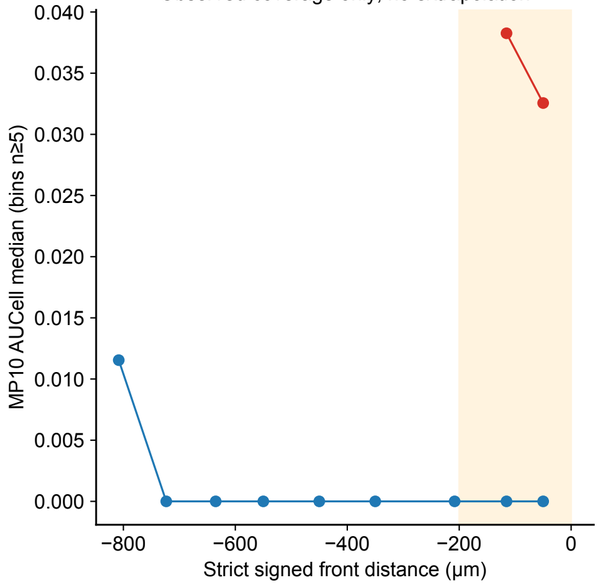
Formal R12 0–8; grey = unlabelled



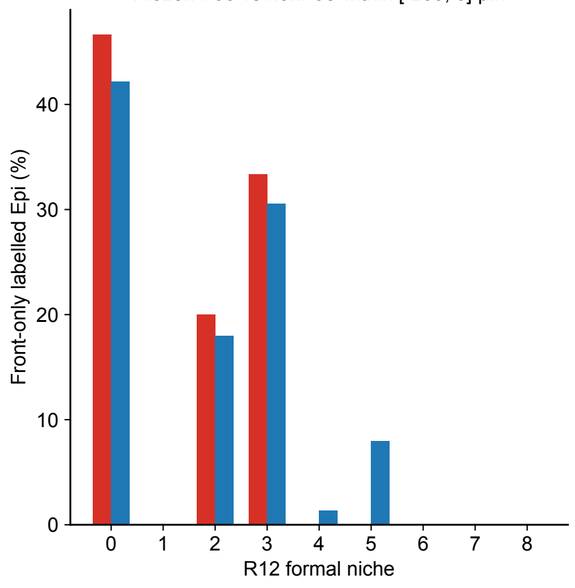
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

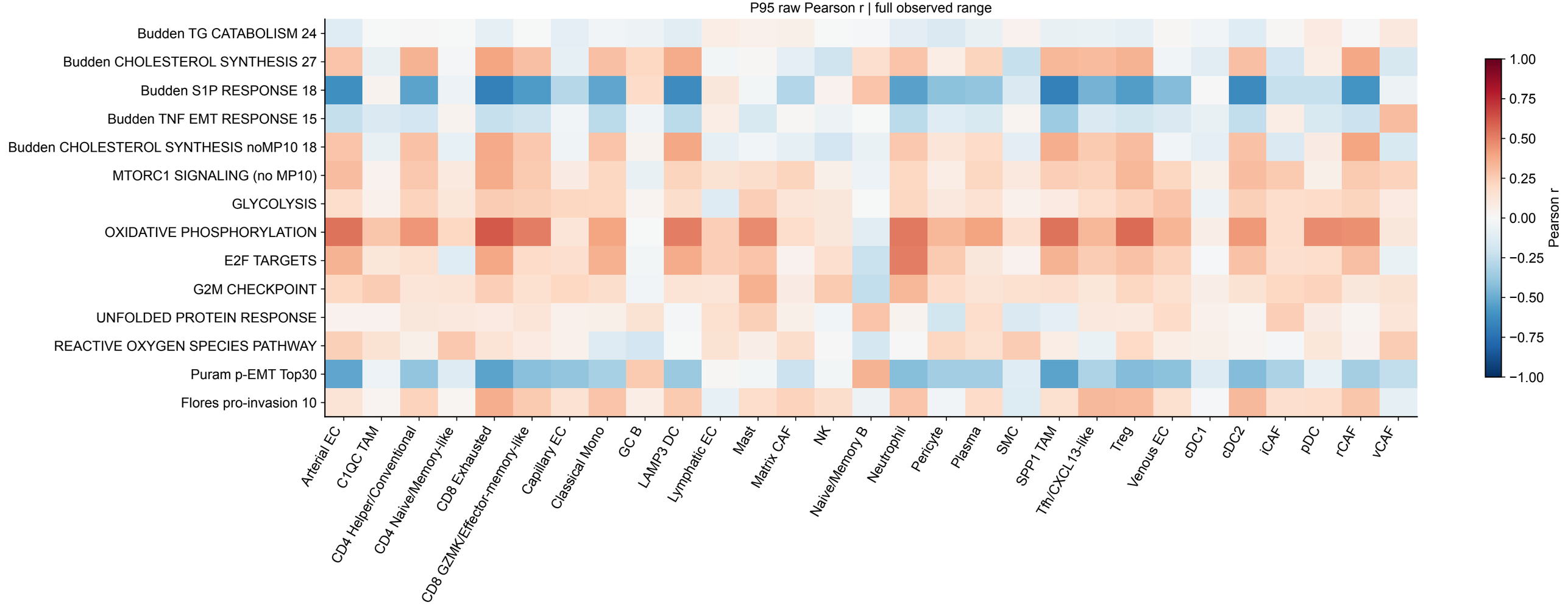


Tissue 2295 | Epi 1858 | internal 1366 | P95 69
Exact front in Epi 1033 | R12 labelled 1576

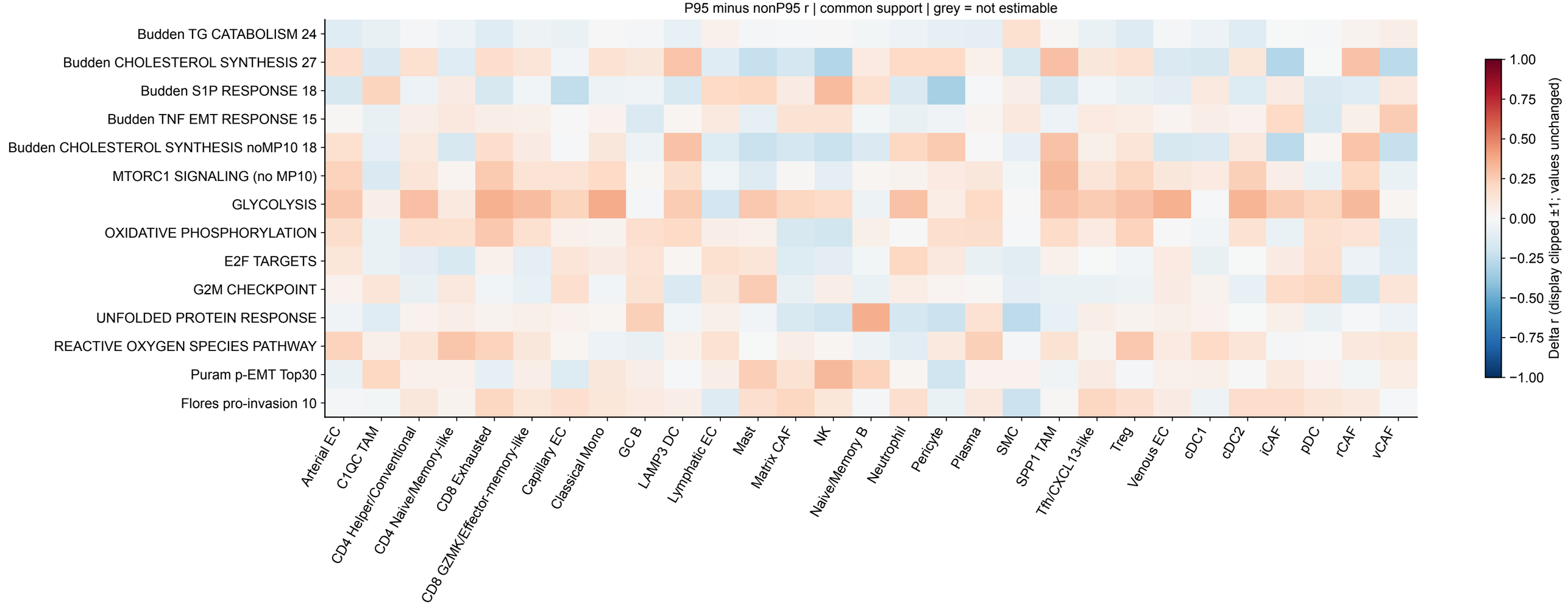
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

CanDis_2026_Sample7 | all 29 candidates; fixed core programs

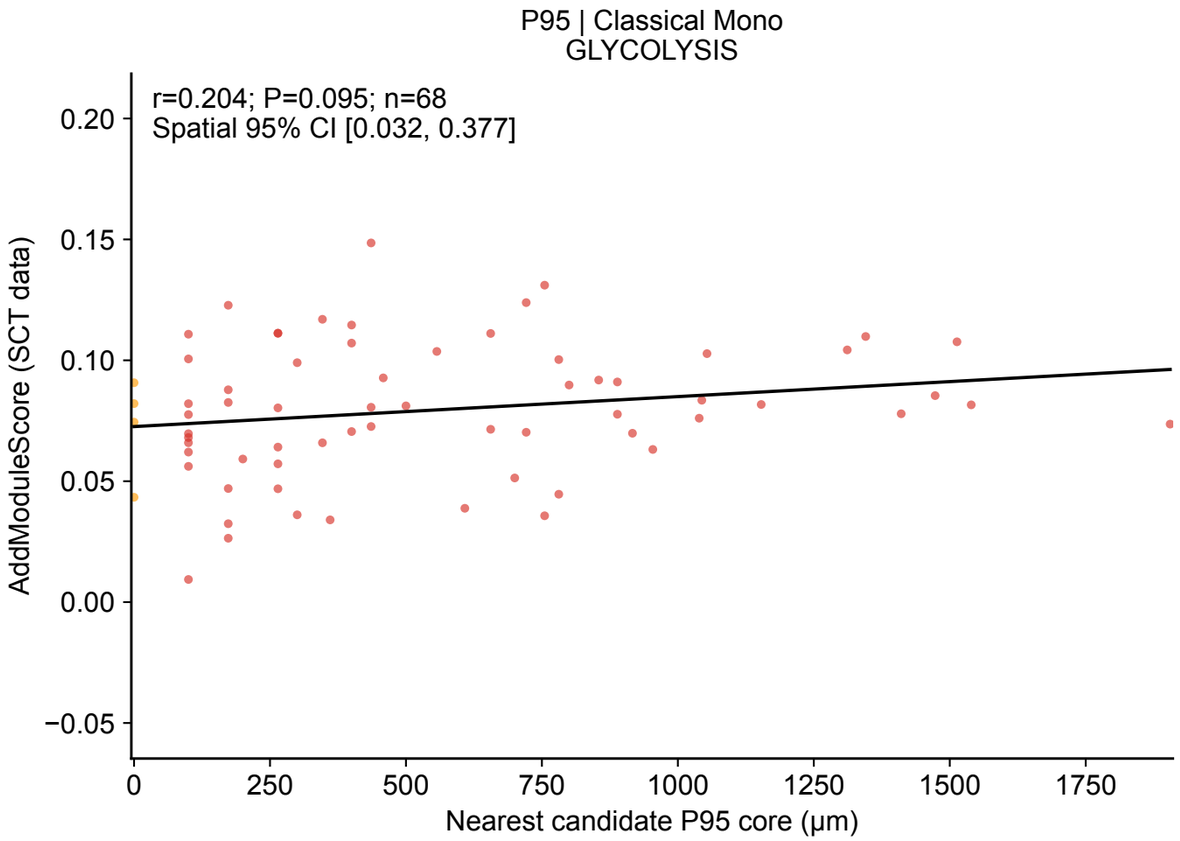
P95 raw Pearson r | full observed range

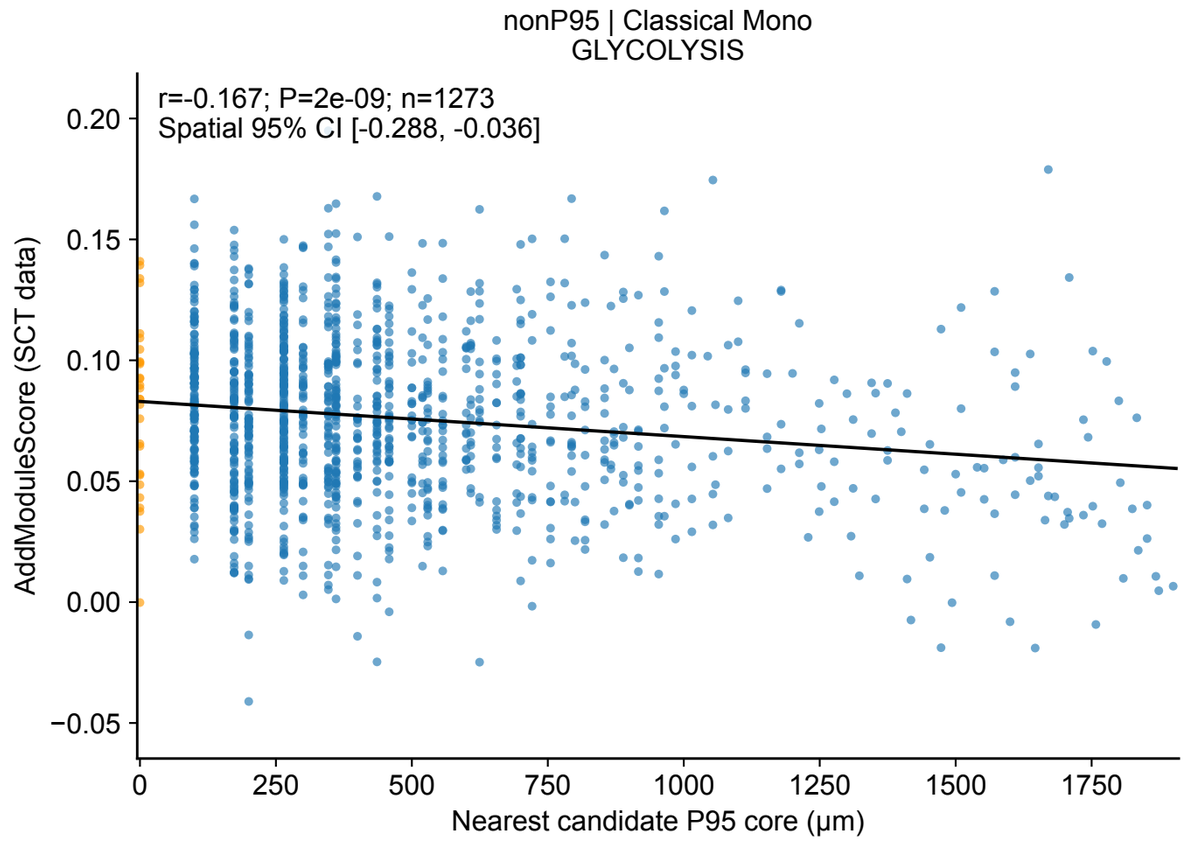


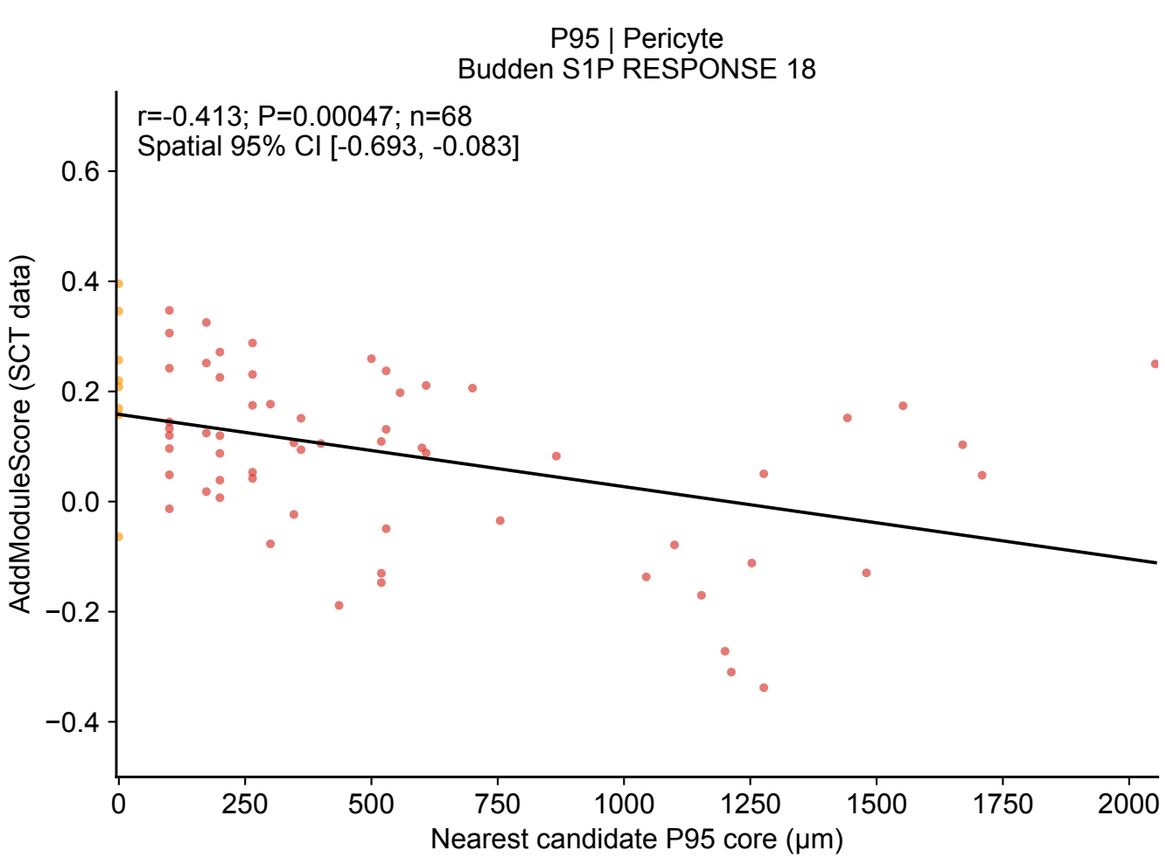
P95 minus nonP95 r | common support | grey = not estimable

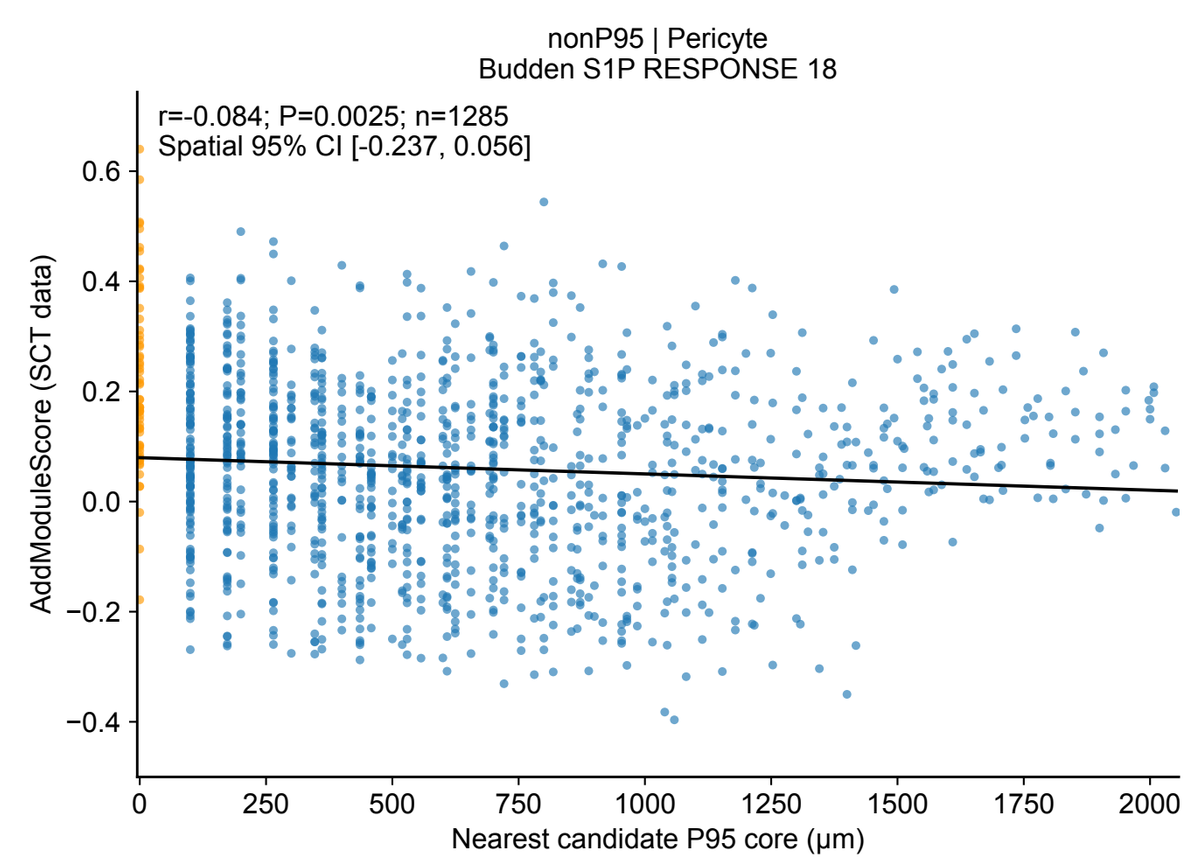


CanDis_2026_Sample7 | exploratory scatter display selected by coverage and |delta r|, not P







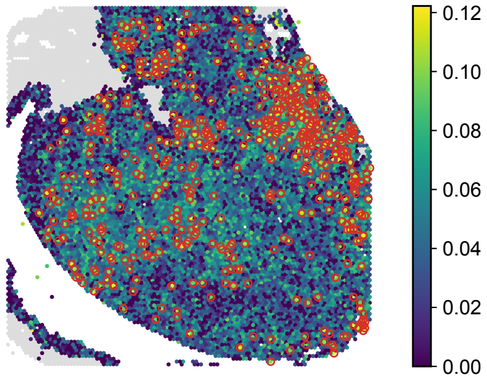


Classical Mono / GLYCOLYSIS: delta r=0.371 [0.182, 0.547]; common 0–1905 µm; spatial P=0.00016
Pericyte / Budden S1P RESPONSE 18: delta r=-0.329 [-0.574, -0.057]; common 0–2052 µm; spatial P=0.02
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

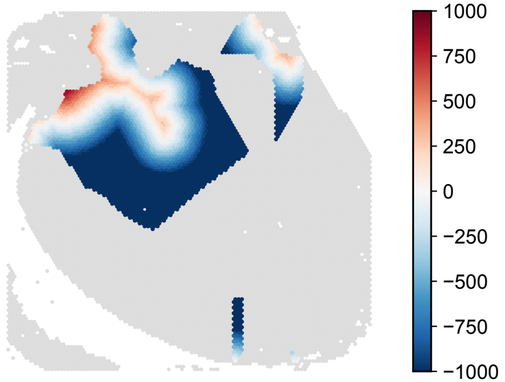
P95 nonP95 distance 0

CanDis_2026_Sample8 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

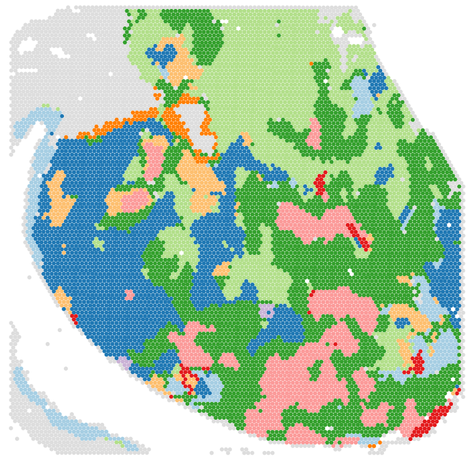
MP10 AUCell and frozen P95



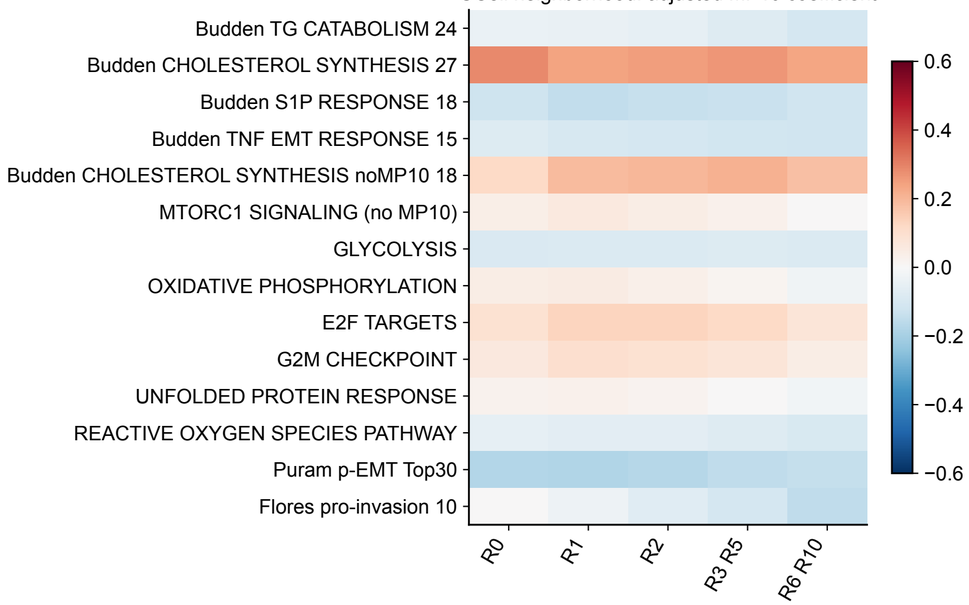
Strict v3.4 signed distance (µm)



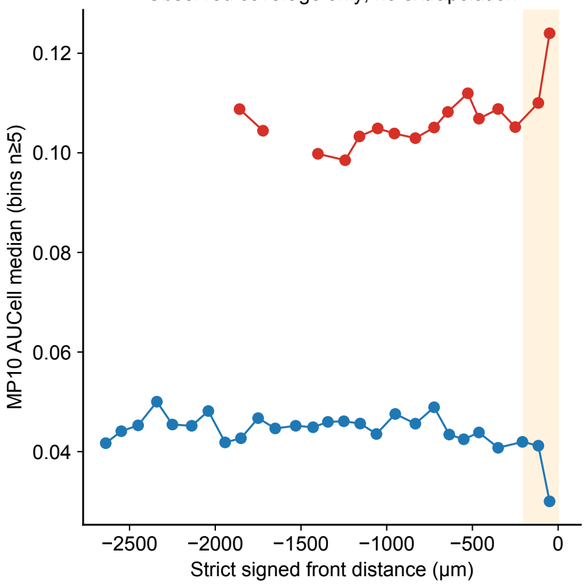
Formal R12 0–8; grey = unlabelled



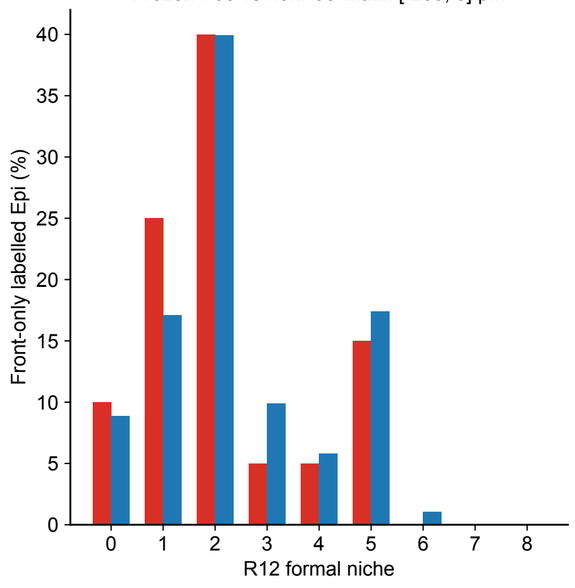
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

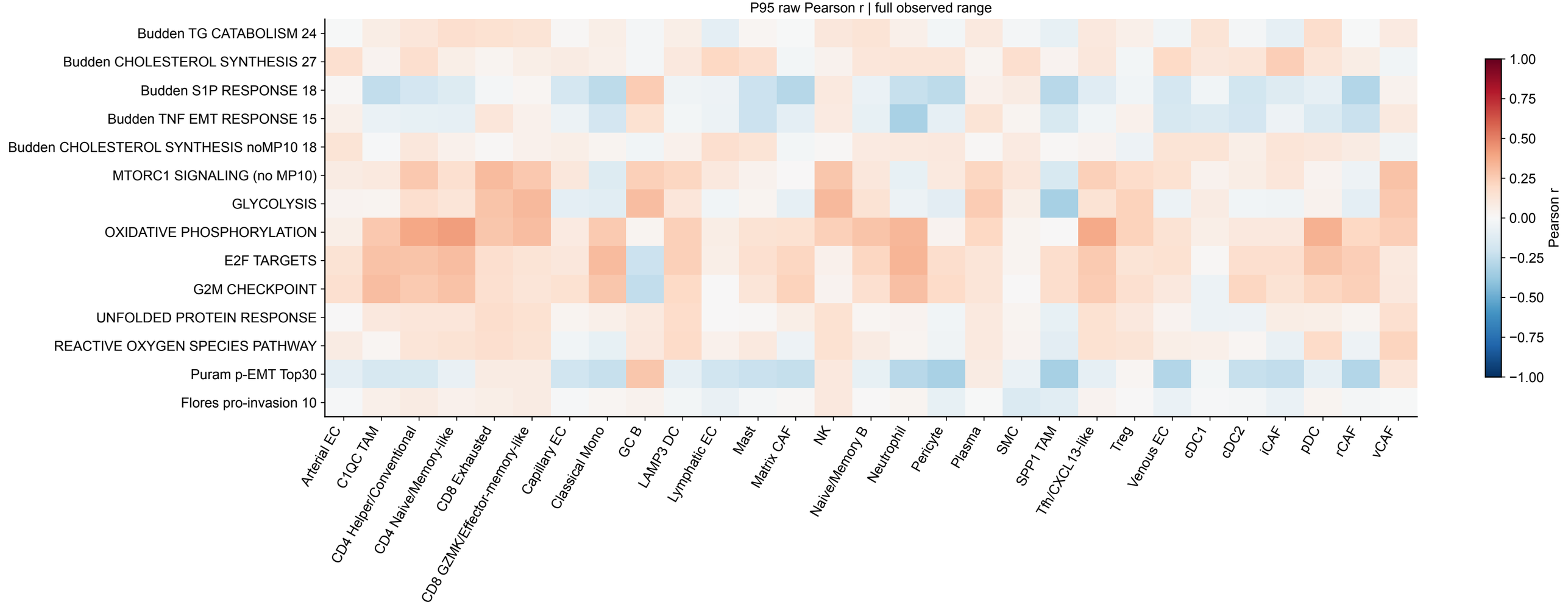


Tissue 11958 | Epi 10711 | internal 10037 | P95 502
Exact front in Epi 2755 | R12 labelled 10188

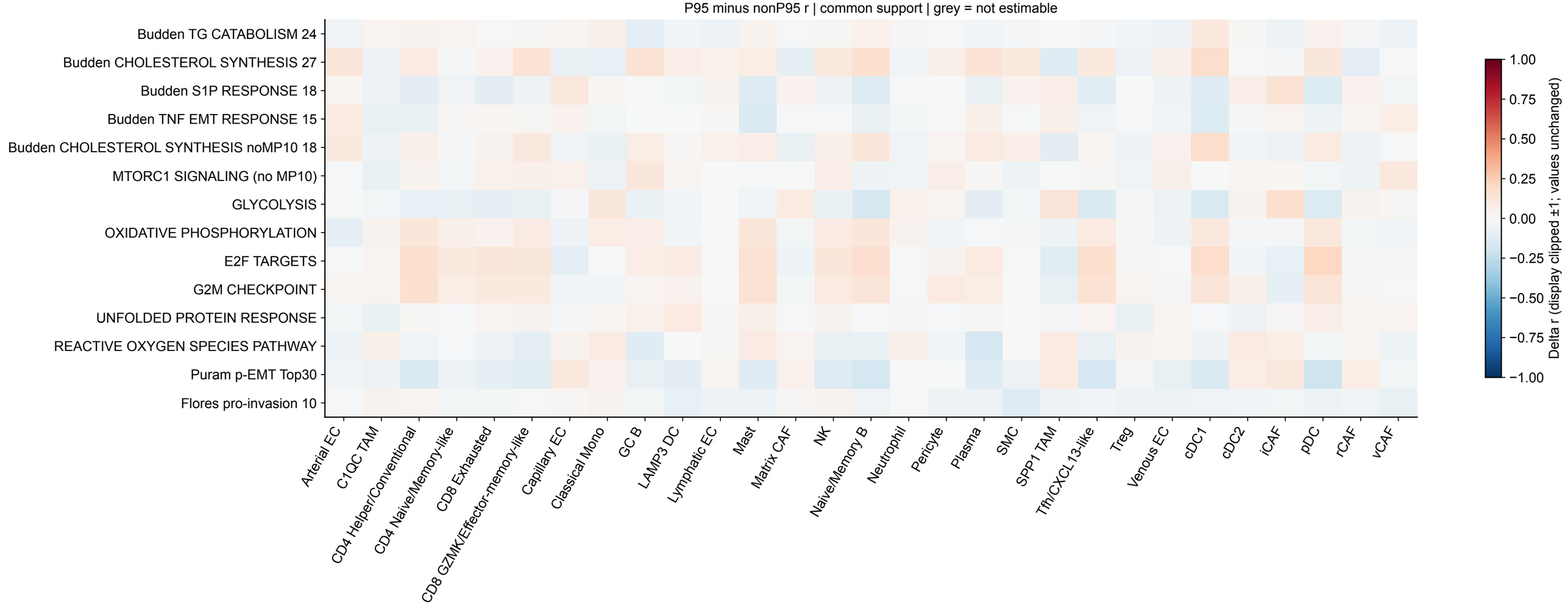
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

CanDis_2026_Sample8 | all 29 candidates; fixed core programs

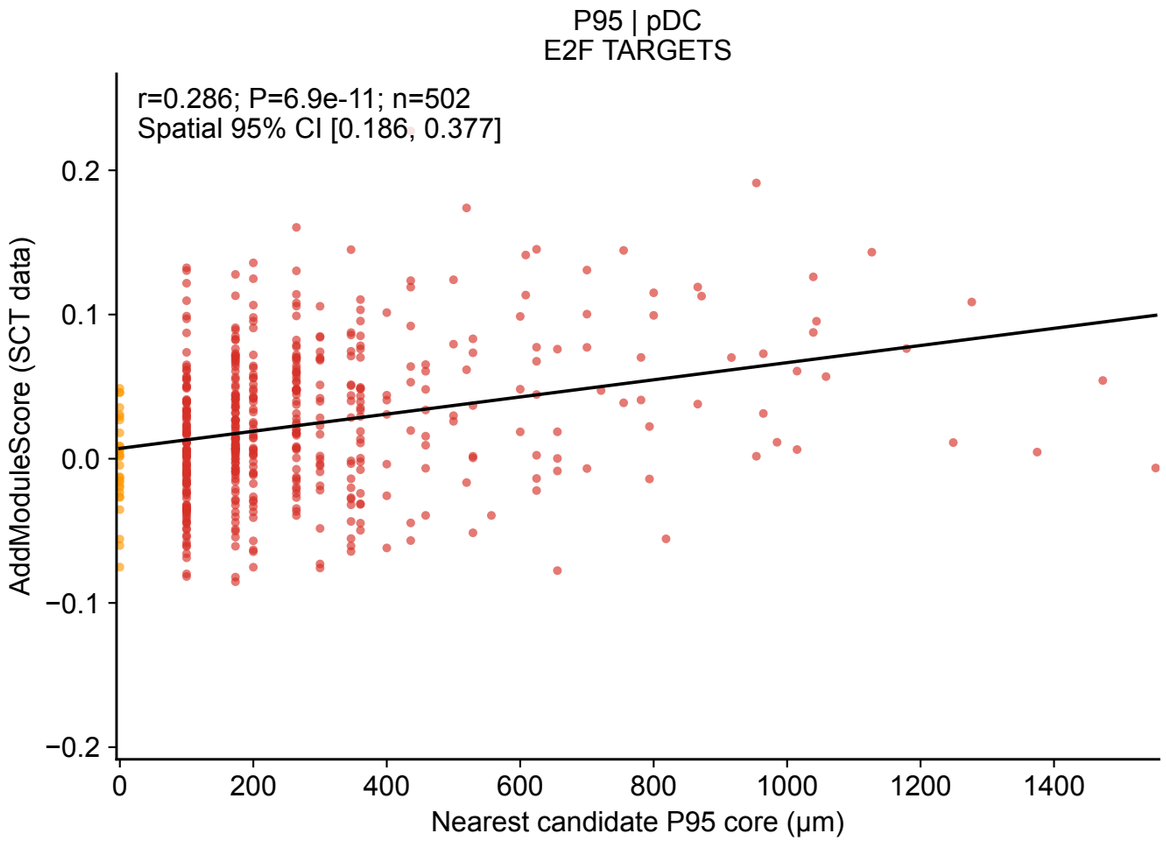
P95 raw Pearson r | full observed range

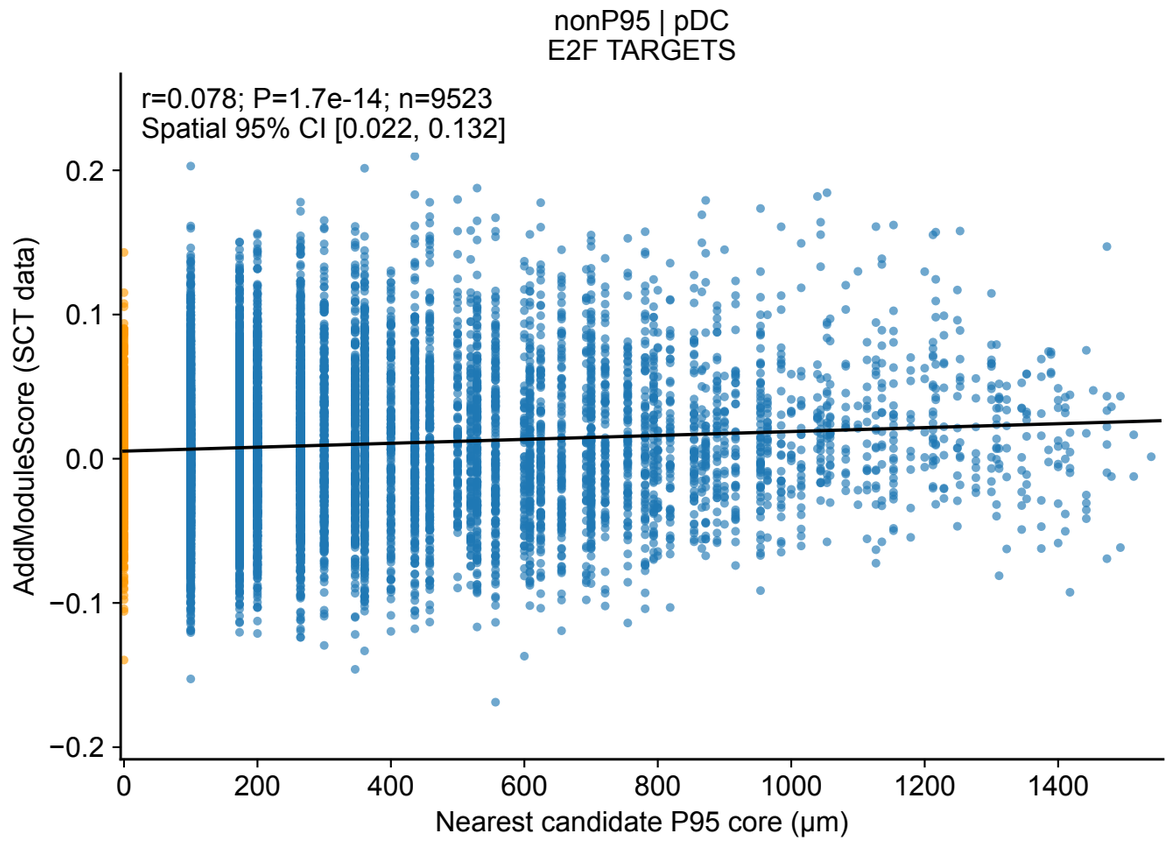


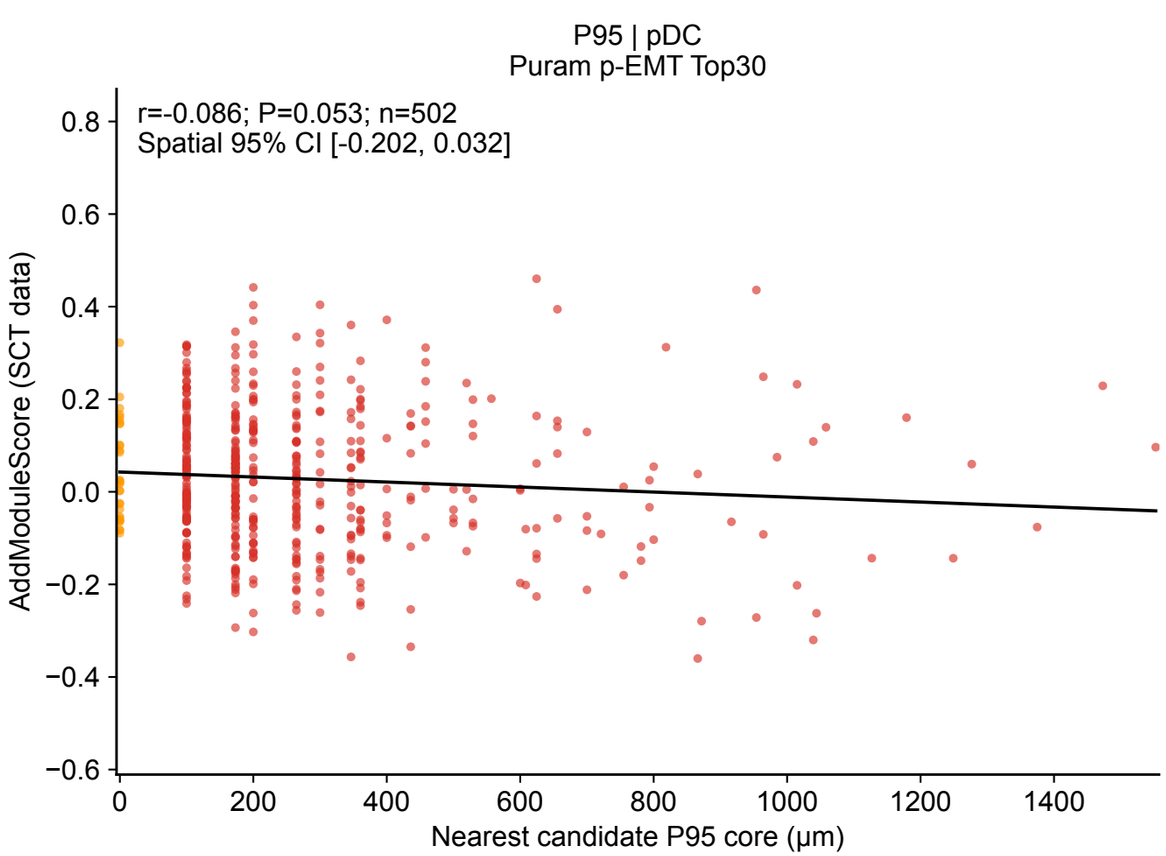
P95 minus nonP95 r | common support | grey = not estimable

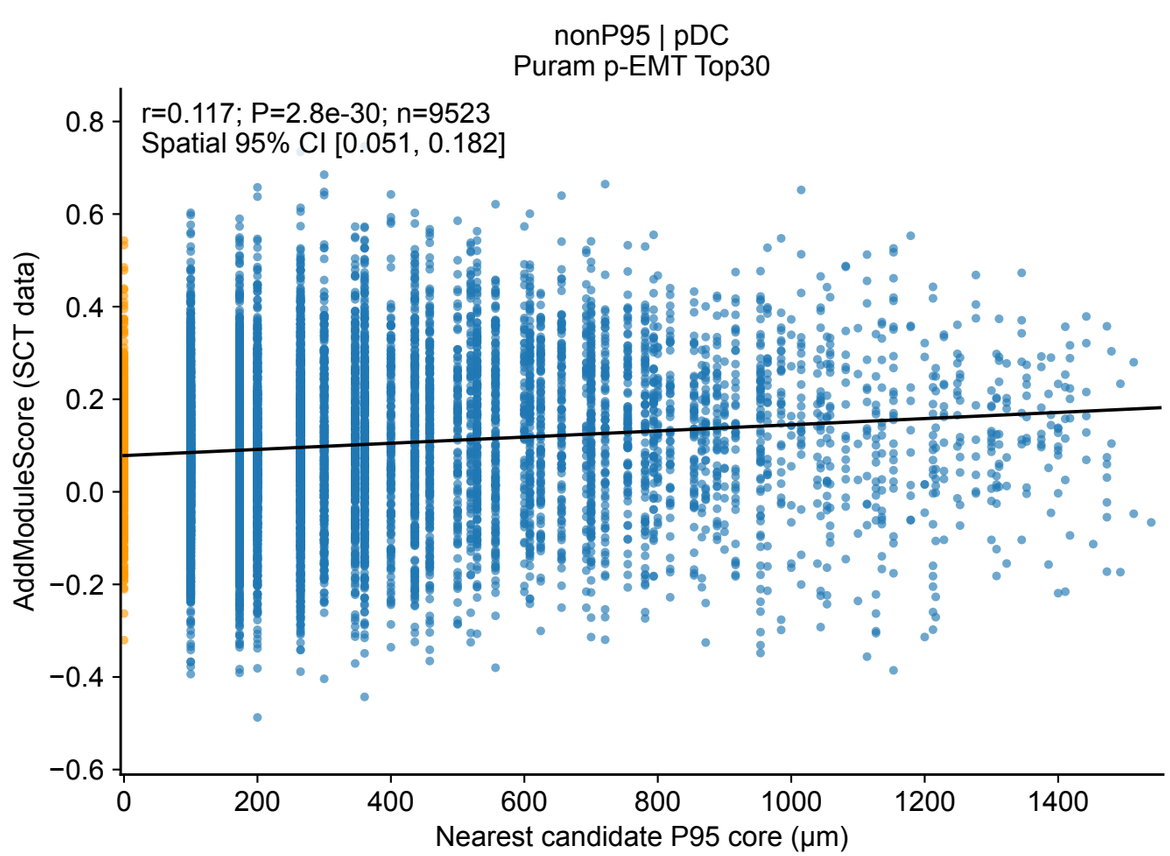


CanDis_2026_Sample8 | exploratory scatter display selected by coverage and |delta r|, not P







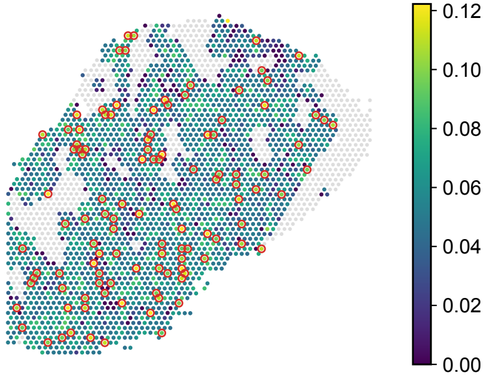


pDC / E2F TARGETS: delta r=0.207 [0.111, 0.292]; common 0–1552 µm; spatial P=1.7e-05
pDC / Puram p-EMT Top30: delta r=-0.203 [-0.307, -0.094]; common 0–1552 µm; spatial P=0.00038
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

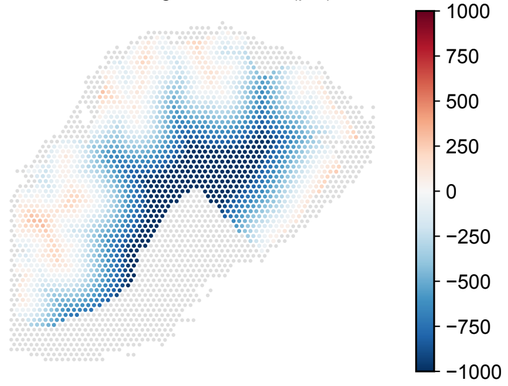
P95 nonP95 distance 0

CanDis_2026_Sample9 | tumor_study_specimen_not_region_diagnosis
Frozen mixed-spot MP10 / front / R12

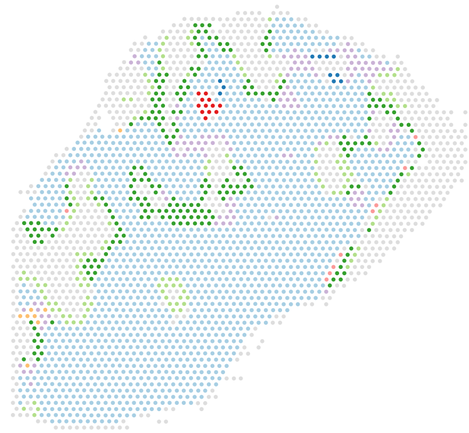
MP10 AUCell and frozen P95



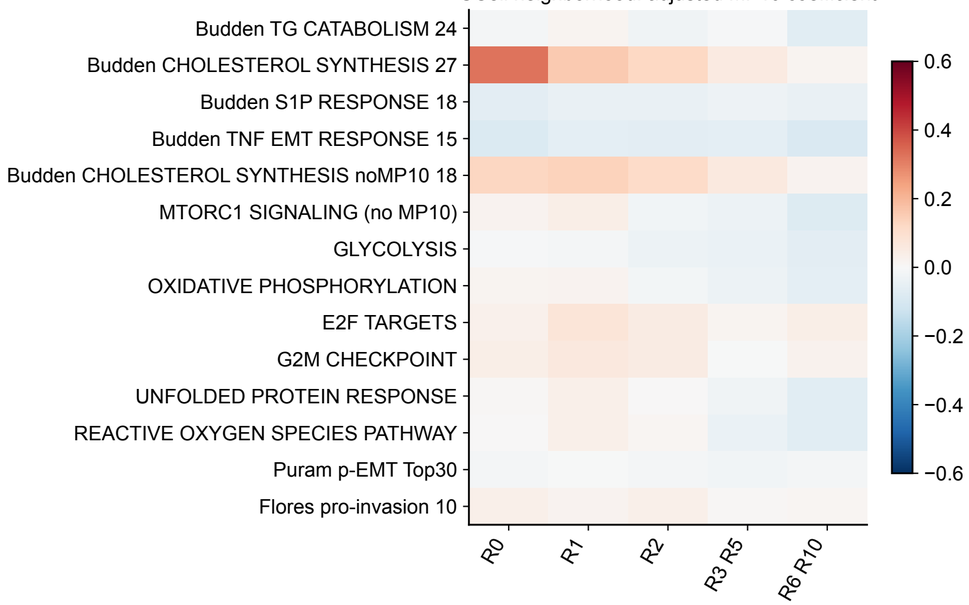
Strict v3.4 signed distance (µm)



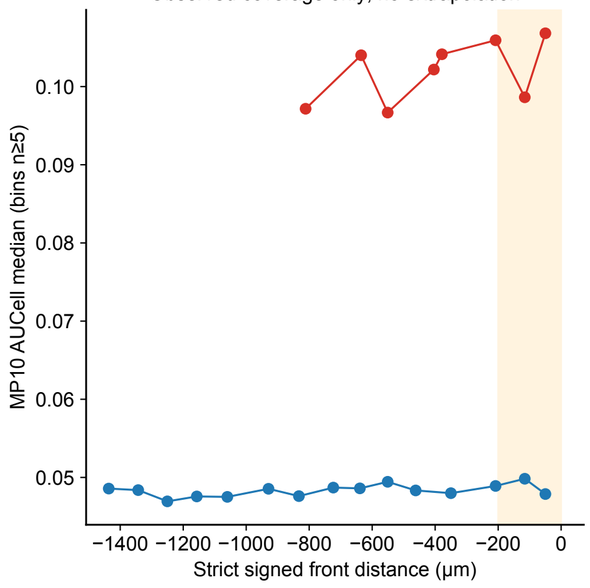
Formal R12 0–8; grey = unlabelled



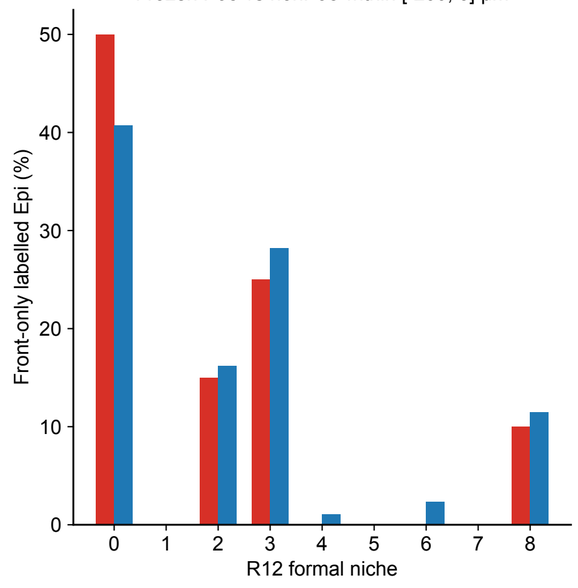
UCell neighborhood: adjusted MP10 coefficient



Observed coverage only; no extrapolation



Frozen P95 vs nonP95 within [-200, 0] µm

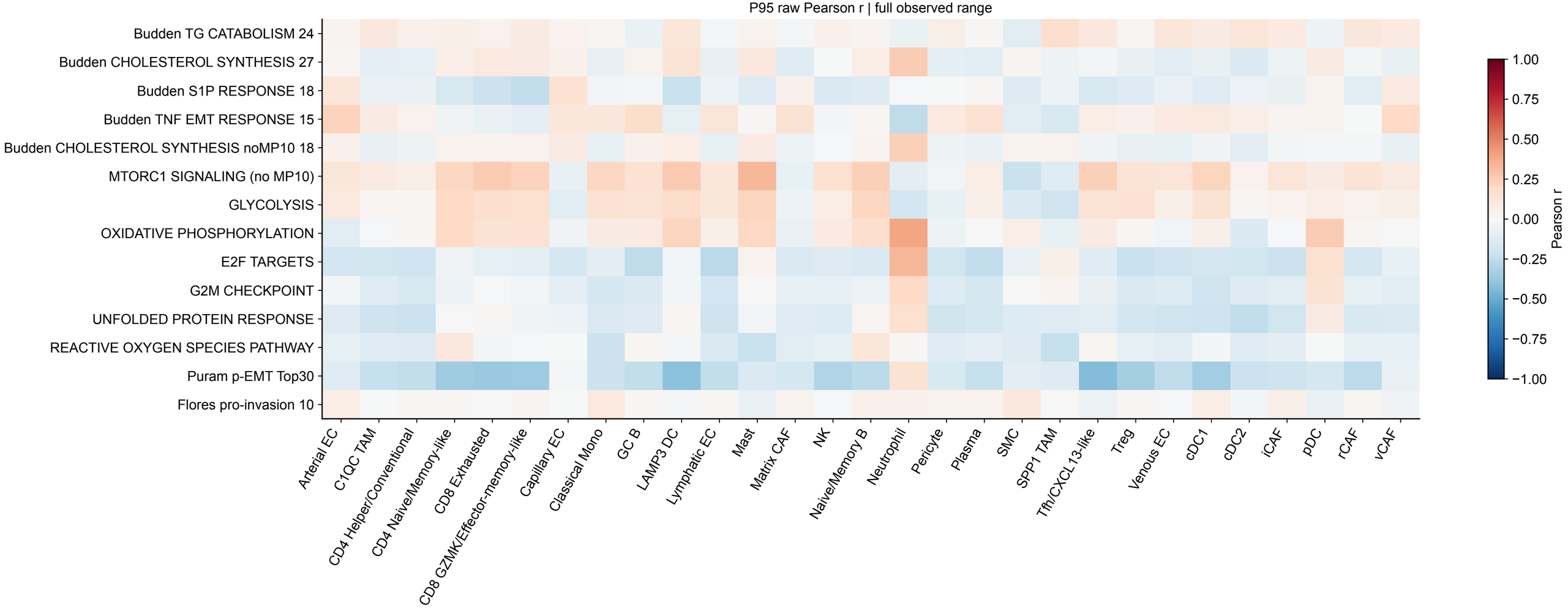


Tissue 2893 | Epi 2310 | internal 2204 | P95 111
Exact front in Epi 1588 | R12 labelled 2130

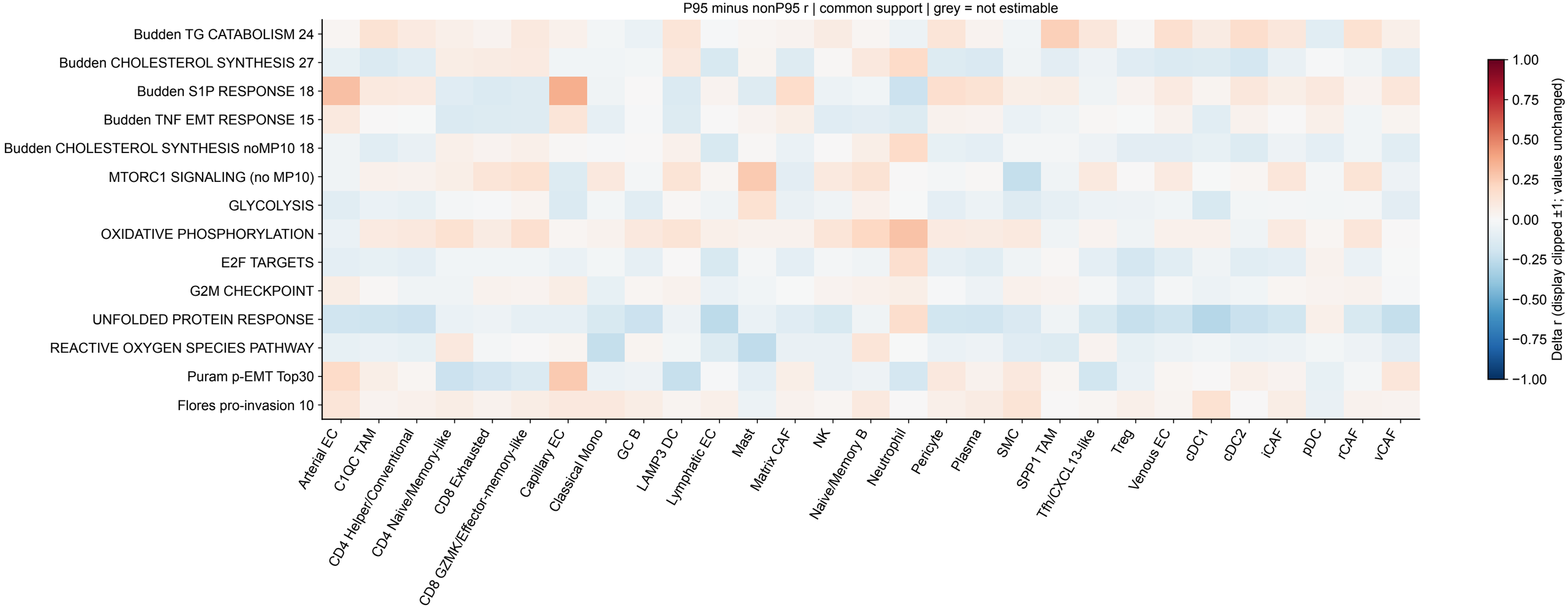
0 1 2 3 4 5 6 7 8 P95 nonP95 / all Epi

CanDis_2026_Sample9 | all 29 candidates; fixed core programs

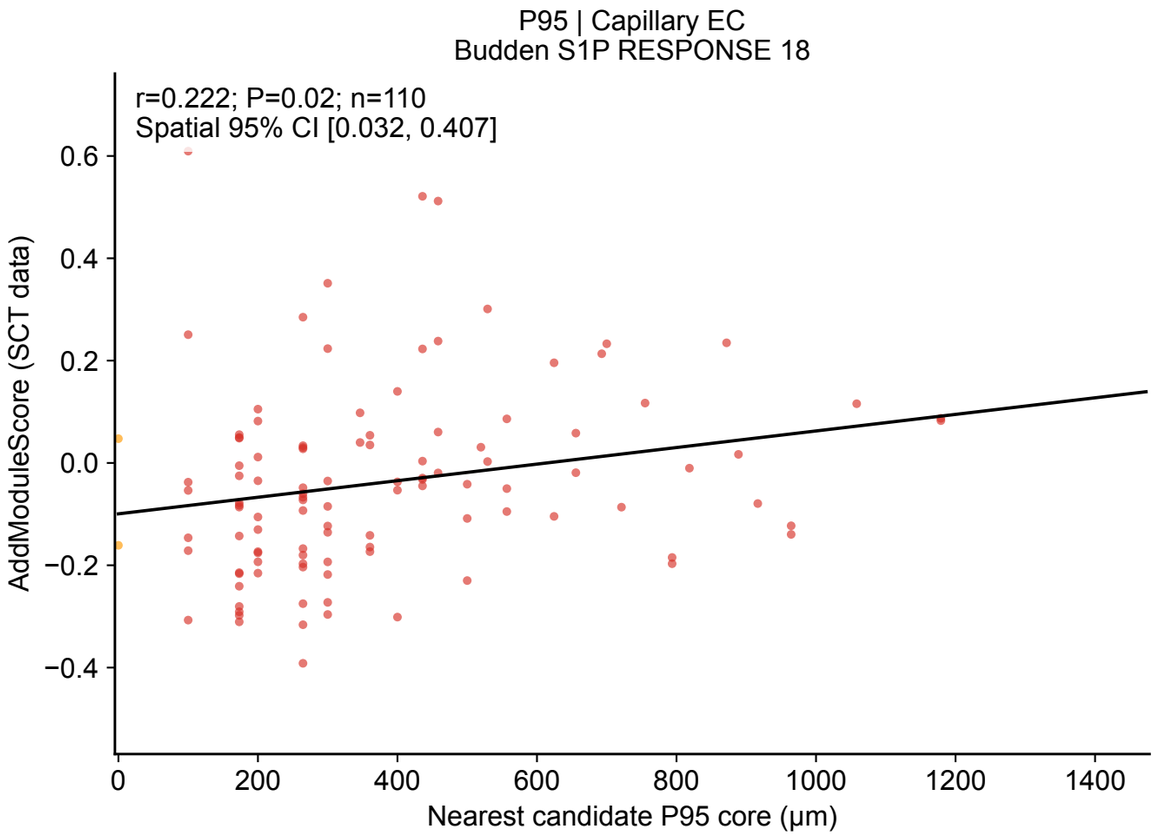
P95 raw Pearson r | full observed range

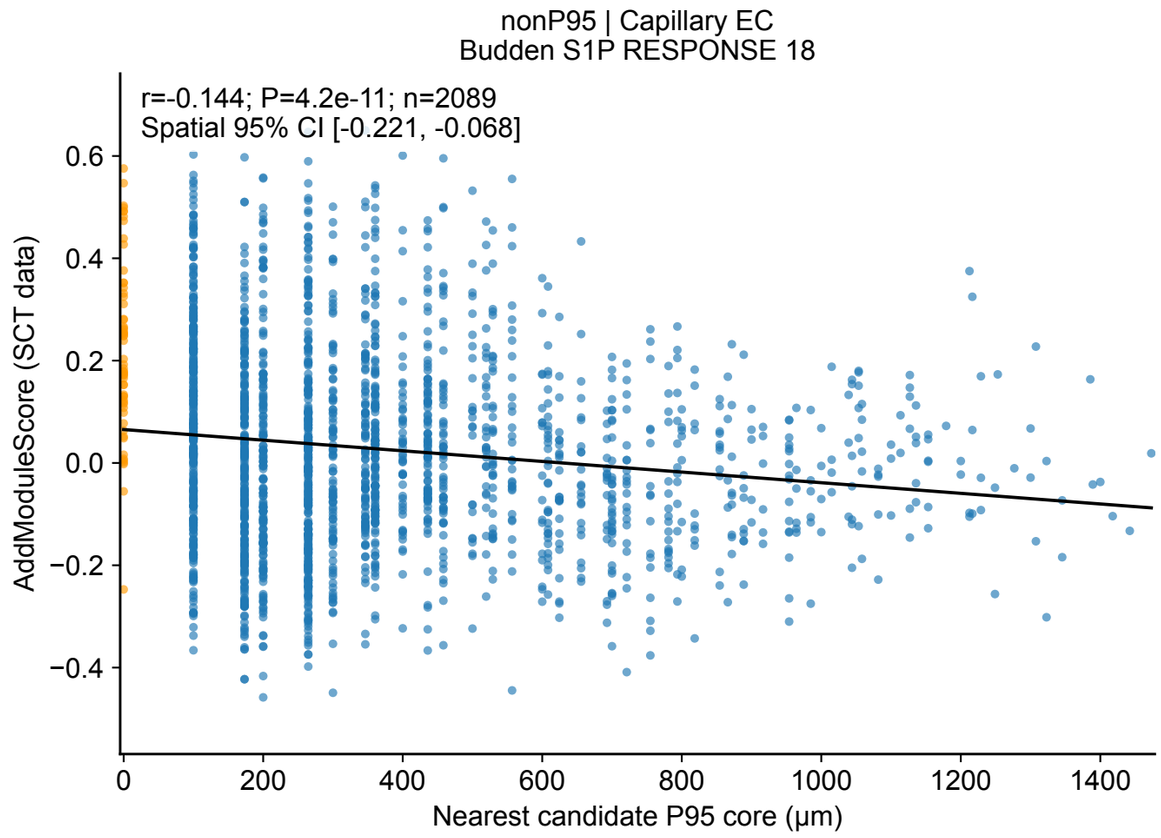


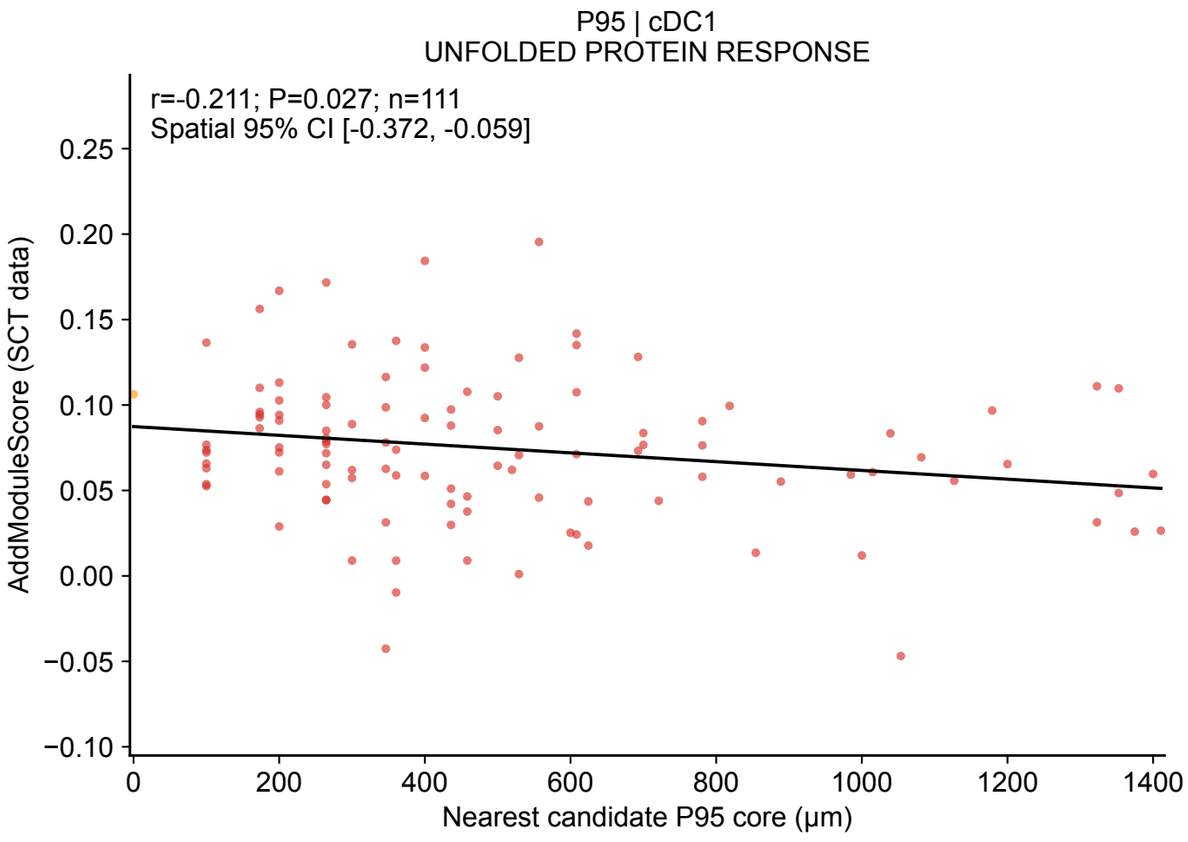
P95 minus nonP95 r | common support | grey = not estimable

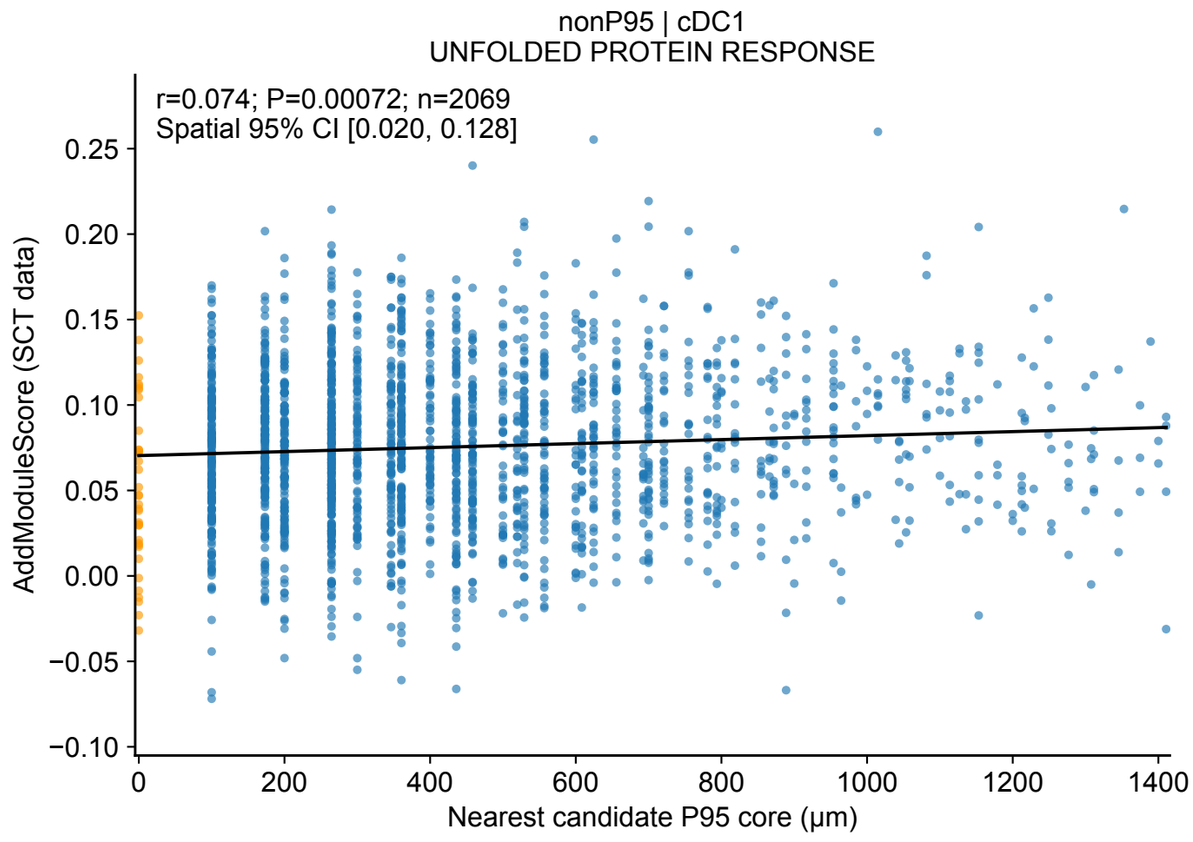


CanDis_2026_Sample9 | exploratory scatter display selected by coverage and |delta r|, not P



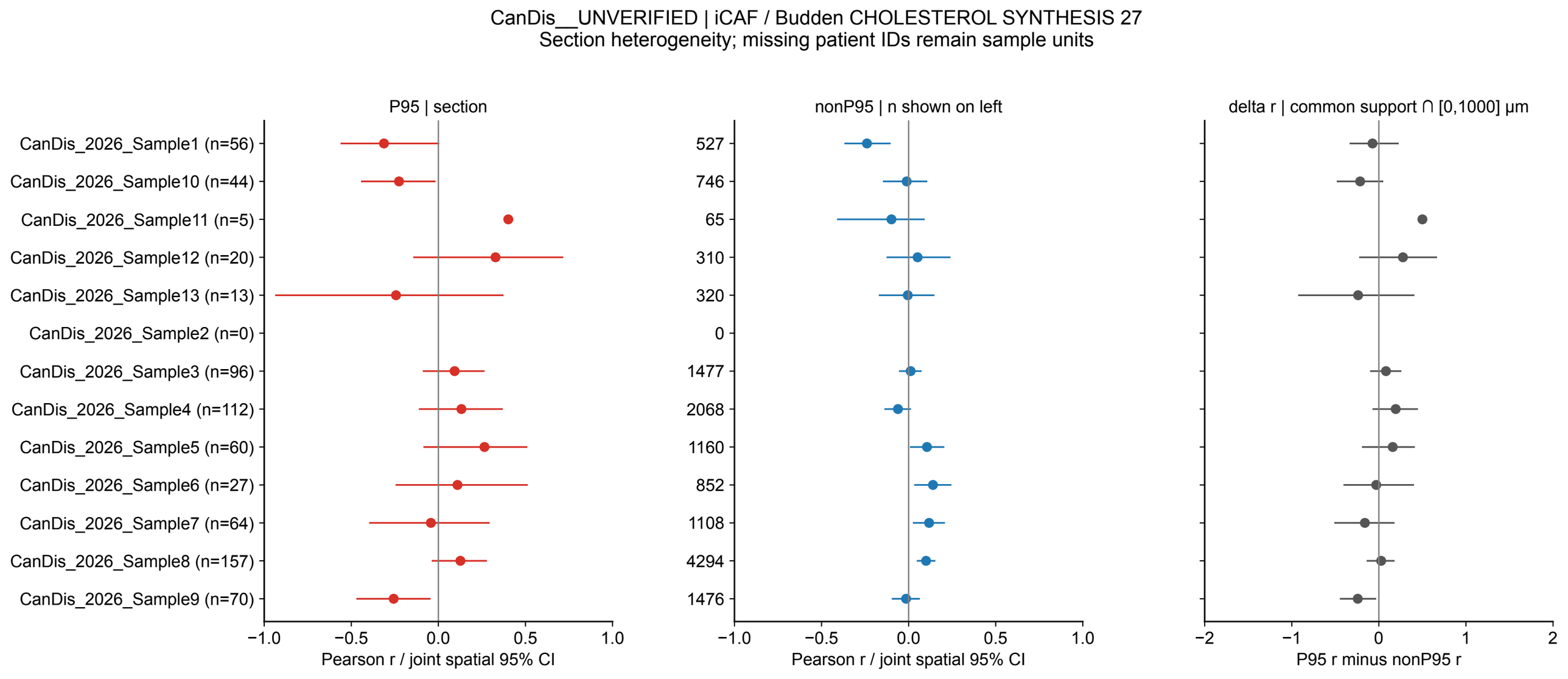


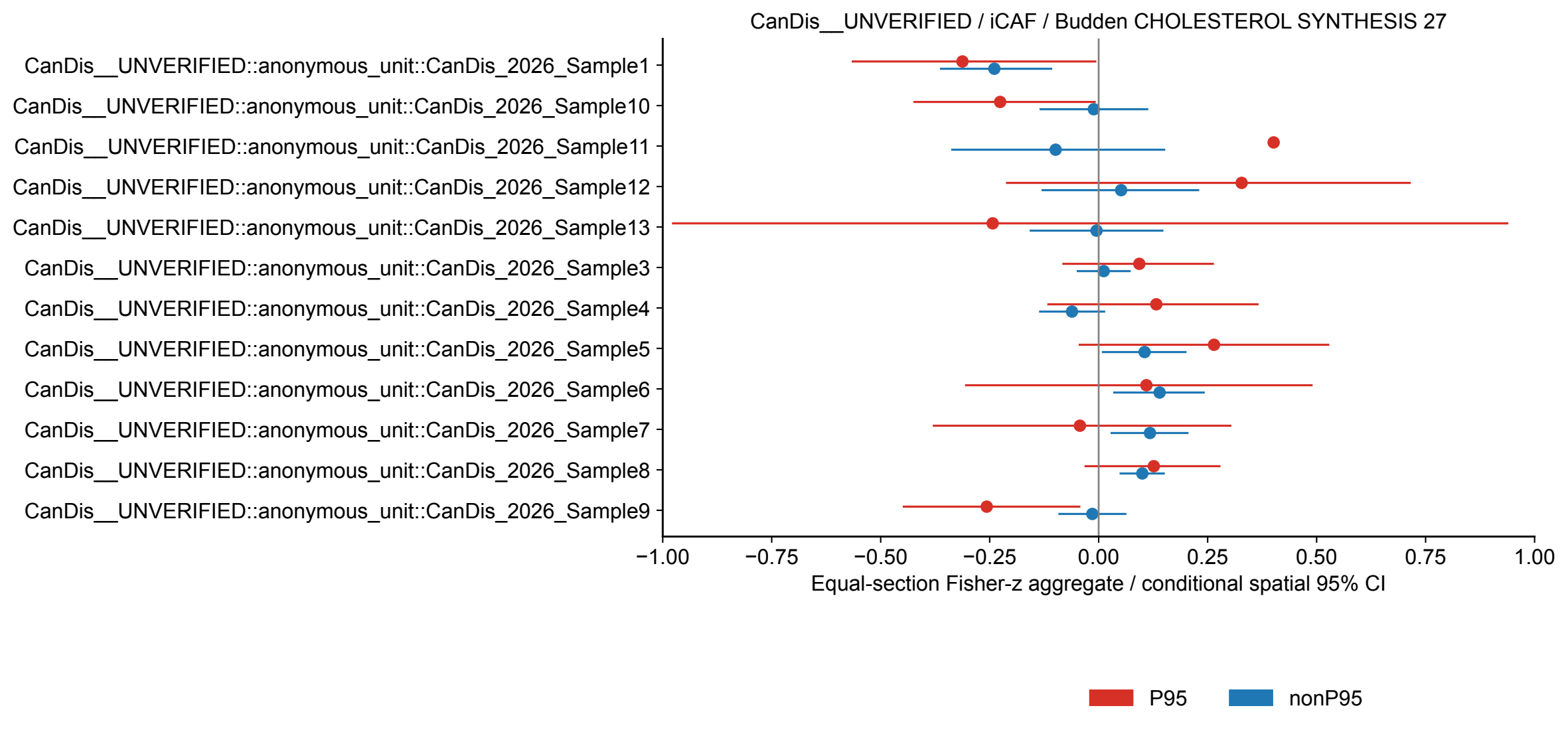


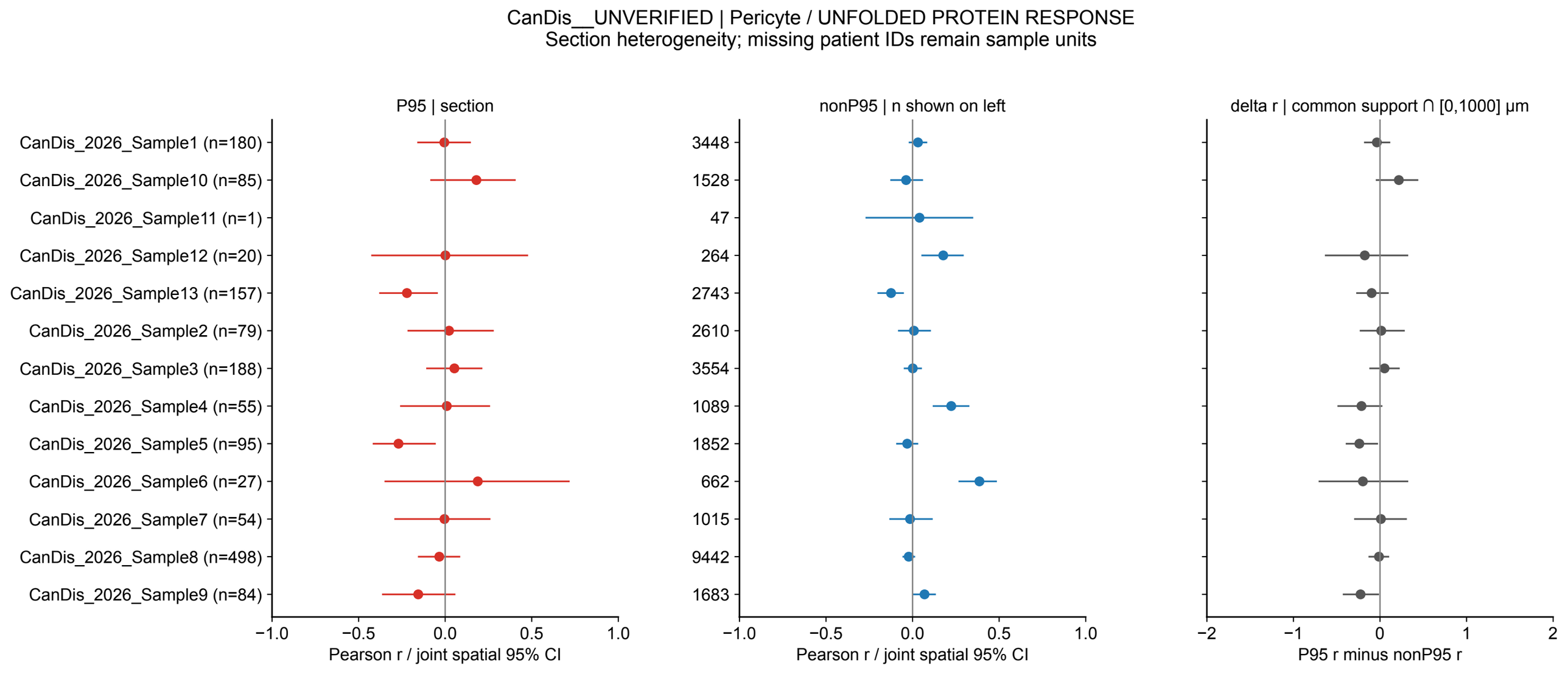


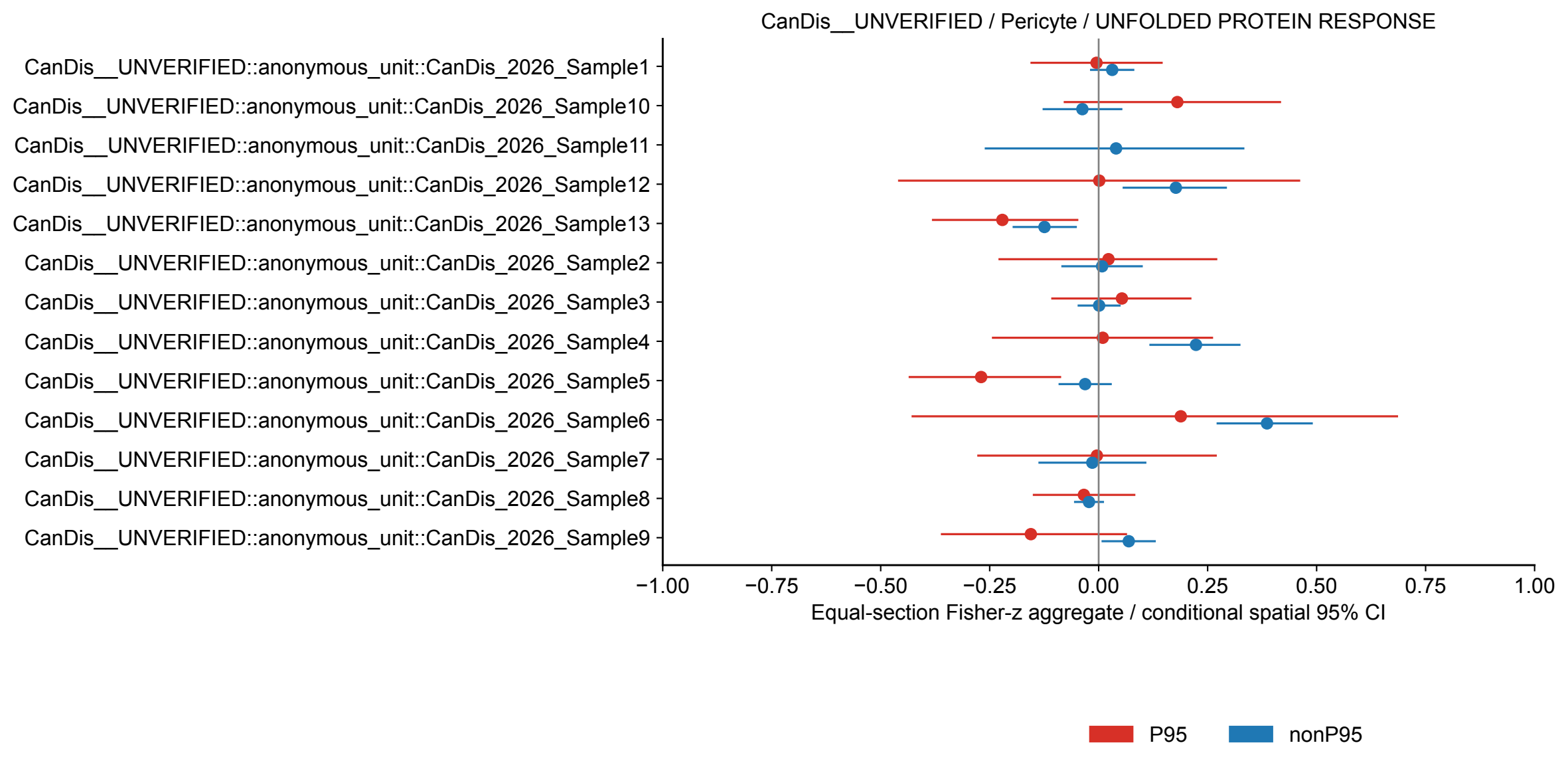
Capillary EC / Budden S1P RESPONSE 18: delta r=0.366 [0.197, 0.551]; common 0–1473 µm; spatial P=0.00013
cDC1 / UNFOLDED PROTEIN RESPONSE: delta r=-0.285 [-0.448, -0.124]; common 0–1411 µm; spatial P=0.00093
Orange points: distance = 0. Pearson P assumes independent spots; spatial intervals use 500 µm blocks.

P95 nonP95 distance 0


CanDis__UNVERIFIED | iCAF / Budden CHOLESTEROL SYNTHESIS 27
Section heterogeneity; missing patient IDs remain sample units
P95 | section
nonP95 | n shown on left
delta r | common support ∩ [0,1000] µm
CanDis_2026_Sample1 (n=56)
CanDis_2026_Sample10 (n=44)
CanDis_2026_Sample11 (n=5)
CanDis_2026_Sample12 (n=20)
CanDis_2026_Sample13 (n=13)
CanDis_2026_Sample2 (n=0)
CanDis_2026_Sample3 (n=96)
CanDis_2026_Sample4 (n=112)
CanDis_2026_Sample5 (n=60)
CanDis_2026_Sample6 (n=27)
CanDis_2026_Sample7 (n=64)
CanDis_2026_Sample8 (n=157)
CanDis_2026_Sample9 (n=70)
527
746
65
310
320
0
1477
2068
1160
852
1108
4294
1476
−1.0
−0.5
0.0
0.5
1.0
Pearson r / joint spatial 95% CI
−1.0
−0.5
0.0
0.5
1.0
Pearson r / joint spatial 95% CI
−2
−1
0
1
2
P95 r minus nonP95 r


CanDis__UNVERIFIED / iCAF / Budden CHOLESTEROL SYNTHESIS 27
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample1
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample10
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample11
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample12
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample13
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample3
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample4
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample5
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample6
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample7
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample8
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample9
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Equal-section Fisher-z aggregate / conditional spatial 95% CI
P95
nonP95


CanDis__UNVERIFIED | Pericyte / UNFOLDED PROTEIN RESPONSE
Section heterogeneity; missing patient IDs remain sample units
P95 | section
nonP95 | n shown on left
delta r | common support ∩ [0,1000] µm
CanDis_2026_Sample1 (n=180)
CanDis_2026_Sample10 (n=85)
CanDis_2026_Sample11 (n=1)
CanDis_2026_Sample12 (n=20)
CanDis_2026_Sample13 (n=157)
CanDis_2026_Sample2 (n=79)
CanDis_2026_Sample3 (n=188)
CanDis_2026_Sample4 (n=55)
CanDis_2026_Sample5 (n=95)
CanDis_2026_Sample6 (n=27)
CanDis_2026_Sample7 (n=54)
CanDis_2026_Sample8 (n=498)
CanDis_2026_Sample9 (n=84)
3448
1528
47
264
2743
2610
3554
1089
1852
662
1015
9442
1683
−1.0
−0.5
0.0
0.5
1.0
Pearson r / joint spatial 95% CI
−2
−1
0
1
2
P95 r minus nonP95 r


CanDis__UNVERIFIED / Pericyte / UNFOLDED PROTEIN RESPONSE
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample1
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample10
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample11
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample12
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample13
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample2
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample3
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample4
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample5
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample6
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample7
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample8
CanDis__UNVERIFIED::anonymous_unit::CanDis_2026_Sample9
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Equal-section Fisher-z aggregate / conditional spatial 95% CI
P95
nonP95